

A PROJECT REPORT ON
DUAL STAGE ANN- RNN HYBRID MODEL FOR AUTOMATED
DIABETIC RETINOPATHY DETECTION

Submitted in partial fulfillment of the requirements for the award of the degree

of

BACHELOR OF TECHNOLOGY

in

CSE (ARTIFICIAL INTELLIGENCE)

Under the Esteemed Guidance of

Ms. M. Deepika

Assistant Professor



BY

SAVARALA THANMAYEE
P KEERTHI
K YOHITHA
SAMUDRALA KOKILA

22751A3158
22751A3144
22751A3125
22751A3157

SREENIVASA INSTITUTE OF TECHNOLOGY AND MANAGEMENT STUDIES
(Autonomous)

(Approved by AICTE, New Delhi & Affiliated to JNTUA, Ananthapuramu),

Chittoor, A.P-517127.

DEPARTMENT OF CSE (ARTIFICIAL INTELLIGENCE)

(2025 -26)

SREENIVASA INSTITUTE OF TECHNOLOGY AND MANAGEMENT STUDIES



(Autonomous)

**(Approved by AICTE, New Delhi & Affiliated to JNTUA,
Ananthapuramu)**

Chittoor, A.P-517127.

DEPARTMENT OF CSE (ARTIFICIAL INTELLIGENCE)

CERTIFICATE

This is to certify that the project work entitled “**DUAL STAGE ANN-RNN HYBRID MODEL
FOR AUTOMATED DIABETIC RETINOPATHY DETECTION**” is a genuine work of

SAVARALA THANMAYEE 22751A3158

P KEERTHI 22751A3144

K YOHITHA 22751A3125

SAMUDRALA KOKILA 22751A3157

submitted to the department of Computer Science and Engineering (Artificial Intelligence), in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in CSE (ARTIFICIAL INTELLIGENCE)** during the academic year 2025-26.

Signature of the Supervisor

Ms. M. DEEPIKA,
Assistant Professor & Supervisor.

Signature of the Head of the Department

Prof. M.E. PALANIVEL.,
Professor & Head of the Department.

Submitted for University Examination (Viva-Voice) held on.....

INTERNAL EXAMINER

EXTERNAL EXAMINER



SREENIVASA INSTITUTE OF TECHNOLOGY AND MANAGEMENT STUDIES

(Autonomous)

(Approved by AICTE, New Delhi & Affiliated to JNTUA, Ananthapuramu)

Chittoor, A.P-517127.

DEPARTMENT OF CSE (ARTIFICIAL INTELLIGENCE)

Institute Vision

To emerge as a Centre of Excellence for Learning and Research in the domains of engineering, computing and management.

Institute Mission

M1: Provide congenial academic ambience with state-art of resources for learning and research.

M2: Ignite the students to acquire self-reliance in the latest technologies.

M3: Unleash and encourage the innate potential and creativity of students.

M4: Inculcate confidence to face and experience new challenges.

M5: Foster enterprising spirit among students.

Department Vision

To contribute for the society through excellence in Computer Science and Engineering with a deep passion for wisdom, culture and values.

Department Mission

- Provide congenial academic ambience with necessary infrastructure and learning resources.
- Inculcate confidence to face and experience new challenges from industry and society.

SREENIVASA INSTITUTE OF TECHNOLOGY AND MANAGEMENT STUDIES



(Autonomous)

- Ignite the students to acquire self-reliance in the latest technologies.
- Foster Enterprising spirit among students.

(Approved by AICTE, New Delhi & Affiliated to JNTUA, Ananthapuramu) Chittoor,
A.P-517127.

DEPARTMENT OF CSE (ARTIFICIAL INTELLIGENCE)

PROGRAMME EDUCATIONAL OBJECTIVES(PEO's):

PEO1: Expertise with computer science and Engineering, artificial intelligence and machine learning disciplines through quality studies, enabling success in IT industries.

(Professional Competency)

PEO2: Establish start-up companies or employed in reputed computing industries or government sectors or pursue higher studies in the domain of CSE (AI) **(Successful Career Goals)**

PEO3: Enhance knowledge by updating advanced technological concepts for facing the rapidly changing world and contribute to society through innovation and creativity **(Continuing Education and Contribution to Society).**

PROGRAM SPECIFIC OUTCOMES (PSO's):

PSO1: Demonstrate and analyze the mathematical methodologies to develop computer programs using networking, web design, big data analytics, machine learning algorithms, data science, neural networks and deep learning techniques.

PSO2: Deploy the artificial intelligence techniques to perform human intelligence in creating innovative products and solutions.

PROGRAMME OUTCOMES (PO's):

PO1-Engineeringknowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2 - Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3 - Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.



SREENIVASA INSTITUTE OF TECHNOLOGY AND MANAGEMENT STUDIES

(Autonomous)

(Approved by AICTE, New Delhi & Affiliated to JNTUA, Ananthapuramu)

Chittoor, A.P-517127.

DEPARTMENT OF CSE (ARTIFICIAL INTELLIGENCE)

PO4 - Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions. **PO5 - Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6 - The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7 – Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8 -Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9 - Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10-Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11 - Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12 - Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.



SREENIVASA INSTITUTE OF TECHNOLOGY AND MANAGEMENT STUDIES
(Autonomous)

(Approved by AICTE, New Delhi & Affiliated to JNTUA, Ananthapuramu)
Chittoor, A.P-517127.

DEPARTMENT OF CSE (ARTIFICIAL INTELLIGENCE)

Course Out Comes for Project Work

On completion of project work the student will be able to

- CO1.** Demonstrate in-depth knowledge on the project topic.
- CO2.** Identify, analyze and formulate complex problem chosen for project work to attain substantiated conclusions.
- CO3.** Design solutions to the chosen project problem.
- CO4.** Undertake investigation of project problem to provide valid conclusions.
- CO5.** Use the appropriate techniques, resources and modern engineering tools necessary for project work.
- CO6.** Apply project results for sustainable development of the society.
- CO7.** Understand the impact of project results in the context of environmental sustainability.
- CO8.** Understand professional and ethical responsibilities while executing the project work.
- CO9.** Function effectively as individual and a member in the project team.
- CO10.** Develop communication skills, both oral and written for preparing and presenting project report.
- CO11.** Demonstrate knowledge and understanding of cost and time analysis required for carrying out the project.
- CO12.** Engage in lifelong learning to improve knowledge and competence in the chosen area of the project.



(Autonomous)

(Approved by AICTE, New Delhi & Affiliated to JNTUA, Ananthapuramu) Chittoor,
A.P-517127.

DEPARTMENT OF CSE (ARTIFICIAL INTELLIGENCE)

Evaluation Rubrics for Project Work

Rubric(CO)	Excellent(wt=3)	Good(wt=2)	Fair(wt=1)
Selection of Topic (CO1)	Selected a latest topic through complete knowledge of facts and Concepts	Selected a topic through partial knowledge off acts and concepts	Selected a topic through improper knowledge of facts and concepts
Analysis and Synthesis (CO2)	Thorough Comprehension through analysis/ synthesis	Reasonable Comprehension through analysis/ synthesis	Improper Comprehension through analysis/ synthesis
Problem Solving (CO3)	Thorough comprehension about What is proposed in the literature papers	Reasonable comprehension about What is proposed in the literature papers	Improper comprehension about What is proposed in the literature
Literature Survey (CO4)	Extensive literature Survey with standard References	Considerable literature Survey with standard References	Incomplete literature Survey with substandard References
Usage of Techniques & Tools (CO5)	Clearly identified and has complete knowledge Of techniques & tools used in the project work	Identified and has sufficient knowledge of techniques & tools used in the project work	Identified and has in adequate knowledge of techniques & tools used in project work
Project work impact on Society (CO6)	Conclusion of project work has strong impact on society	Conclusion of project work has considerable impact on society	Conclusion of project work has feeble impact on society
Project work impact on Environment (CO7)	Conclusion of project work has strong impact on Environment	Conclusion of project work has considerable impact on environment	Conclusion of project work has feeble impact on environment
Ethical attitude (CO8)	Clearly understands ethical and social practices.	Moderate understanding of ethical and social practices.	Insufficient understanding of ethical and social practices.
Independent earning (CO9)	Did literature survey and selected topic with little Guidance	Did literature survey and selected topic with considerable guidance	Selected a topic suggested by the Supervisor
Oral Presentation (CO10)	Presentation in logical sequence with key points, clear conclusion and excellent language	Presentation with key points, conclusion and good language	Presentation with insufficient key points and improper Conclusion
Report Writing (CO10)	Status report with clear and logical sequence of Chapters using excellent language	Status report with logical sequence of chapters using understandable language	Status report not properly organized
Time and Cost Analysis (CO11)	Comprehensive time and cost analysis	Moderate time and cost analysis	Reasonable time and cost analysis
Continuous learning (CO12)	Highly enthusiastic Towards continuous Learning	Interested in continuous learning	Inadequate interest in continuous learning

ACKNOWLEDGEMENT

A Project of this magnitude would have not been possible without the guidance and co- ordination of many people. I am fortune in having top quality people to help, support and guide us in every step towards our goal.

Our team is very much grateful to the Chairman **Sri K. Ranganatham** Garu for his encouragement and support. We express our sincere thanks to our Managing Trustee **Dr. D. A. Kalpaja Garu** for this encouragement and support.

We express our sincere thanks to our Academic Advisor **Dr. K.L. Narayana, M.Tech., Ph.D.**, further, we would like to express our profound gratitude to our principal **Dr. N. Venkatachalapathi, M.Tech., Ph.D** for providing all possible facilities throughout the completion of our project work.

We express our sincere thanks to our Head of the Department **Prof. M.E. Palanivel** for his co-operation and valuable suggestions towards the completion of project work.

We express our sincere thanks to our guide **Ms. M. Deepika.**, for offering us the opportunity to do this work under his guidance.

We express our sincere salutation to all other teaching and non-teaching staff of our department for their direct and indirect support given during our project work. Last but not the least, we dedicate this work to our parents and the Almighty who have been with us throughout and helped us to overcome the hard times.

DECLARATION

We certify that the work contained in this report is original and has been done by me under the Guidance of my supervisor. The work has not been submitted to any other Institute for any degree or diploma. We have followed the guidelines provided by the Institute in preparing the report. We have conformed to the norms and guidelines given in the Ethical Code of Conduct of the Institute. Whenever we have used materials (data, theoretical analysis, figures, and text) from other sources, we have given due credit to them by citing them in the text of the report and giving their details in the references. Further, we have taken permission from the copyright owners of the sources whenever necessary.

SAVARALA THANMAYEE

22751A3158

P KEERTHI

22751A3144

K YOHITHA

22751A3125

SAMUDRALA KOKILA

22751A3157

ABSTRACT

Diabetic Retinopathy (DR) is a microvascular complication of diabetic's mellitus, which is a serious threat to vision and one of the primary causes of irreversible blindness in diabetic patients. The manual diagnosis of DR through retinal fundus examination is a time consuming process and may differ from person to person. This emphasizes the need for intelligent diagnostic systems based on deep learning. This project proposes a novel and robust automated approach for the diagnosis of diabetic retinopathy using a two-stage hybrid approach that integrates ANN and RNN. In stage 1, a is employed to automatically extract features and learn hierarchical spatial representations from raw retinal images. Prior to feature extraction, contrast enhancement, noise removal, and normalization techniques are employed to enhance the quality of the images and highlight the pathological areas. The CNN model is trained to detect microaneurysms, haemorrhages, hard exudates, soft exudates, and irregularities in blood vessel morphology. These spatial features are then fed to an ANN in stage 1, and in stage 2, a RNN is employed to capture the contextual and temporal dependencies between the features. The RNN structure allows the model to learn transition patterns and dependencies among features, which cannot be done by feedforward networks alone. The proposed system adopts a two-step learning process to accomplish feature abstraction at each spatial level and identify dependencies among levels, leading to the development of a classifier that is particularly designed to adapt to the varying levels of diabetic retinopathy. The two-step learning process allows the model to learn spatial features in the first step and learn dependencies in the second step to classify varying levels of diabetic retinopathy. The proposed hybrid model was then applied to the benchmark datasets of diabetic retinopathy. Based on publicly available data, the performance evaluation was conducted using standard metrics such as accuracy, sensitivity, specificity, precision, recall and F1-score..

Keywords: Diabetic Retinopathy, Microaneurysms, Retinal Fundus Images, Deep Learning, Hybrid CNN-ANN-RNN Model, Feature Extraction, Fast API Deployment, Early Detection

LIST OF FIGURES

FIG.NO.	TITLE OF THE FIGURE	PAGE NO.
1.	Structure of eye	1
1.1	Diabetic Retinopathy	4
1.2	Stages of Diabetic Retinopathy	4
1.2.1	Mild non-proliferative	5
1.2.2	Moderate non-proliferative	7
1.2.3	Proliferative Diabetic Retinopathy	8
1.2.4	Severe Diabetic Retinopathy	9
1.4.1	Methodology	18
3.2	Proposed System	25
4.	System Architecture	25
4.1	System Design	25
4.2.1	Input Design	53
4.2.2	Output Design	56
4.3.1	Use Case	60
4.3.2	Activity Diagram	61
4.3.3	Deployment Diagram	63
4.3.4	Class Diagram	64
4.3.5	Sequence Diagram	65
4.3.6	Component Diagram	66
4.3.7	State Diagram	67
4.3.8	Object Diagram	69
6.1	Types of Software Testing	71
6.2	Test case Table	80
7.	Dataset Description	83
7.B.1	Login page	106
7.B.2	Input page	107
7.B.3	Output Page	108
7.C.1	Representing the accuracy curves	109
7.C.2	Representing the Precision and Recall	109

LIST OF ABBREVIATIONS

DR-Diabetic Retinopathy
PDR-Proliferative Diabetic Retinopathy
NPDR-Non-Proliferative Diabetic Retinopathy
OCT-Optical Coherence Tomography
DME-Diabetic Macular Edema
ANN-Artificial Neural Network
RNN-Recurrent Neural Network
CNN-Convolutional Neural Network
LSTM-Long Short Term Memory
AI-Artificial Intelligence
ML-Machine Learning
API-Application Programming Interface
KNN-Kernel Nearest Neighbouring
SVM-Support Vector Machine
VGG-Visual Geometry Group
GPU-Graphical Processing Unit
JSON-Java Script Object Notation
URL-Uniform Resource Locator

TABLE OF CONTENTS

CHAPTER NO.	TITLE	PAGE NO.
	Abstract	i
	List of Figures	ii
	List of Abbreviations	iii
1.	Introduction	1-18
	1.1 Diabetic Retinopathy	3
	1.2 Stages of Diabetic Retinopathy	4-11
	1.2.1 Mild Non-Proliferative	4
	1.2.2 Moderate Non-Proliferative	5-6
	1.2.3 Proliferative Diabetic Retinopathy	7
	1.2.4 Severe Diabetic Retinopathy	8-9
	1.2.5 Motivation	10
	1.2.6 Problem Identification	11
	1.3 Objective	12-14
	1.4 Methodology	15-18
2.	Literature Survey	19-21
3.	System Analysis	22-42
	3.1 Existing System	23-24
	3.1.1 Demerits	24
	3.2 Proposed System	25-27
	3.2.1 Merits	26
	3.3 Feasibility Study	28-32
	3.3.1 Economic Feasibility	30
	3.3.2 Technical Feasibility	31
	3.4 System Specifications	33-42
	3.4.1 Functional Requirements	34-36
	3.4.2 Non-Functional Requirements	37-38
	3.4.3 Software & Hardware Requirements	39-42
4.	System Model Development	43-47
	4.1 Introduction	43

	4.2 Reason For AGILE Method	43
	4.3 Project Modules	43
	4.4 Iterative Development Process	44
	4.5. AGILE Workflow	46
	4.6 Continues Testing And Feedback	46
	4.7 Advantages	47
5.	System Design	48-69
	5.1 Module And Description	51-52
	5.2 Modules	53
	5.2.1 Input Design	53-55
	5.2.2 Output Design	56-58
	5.3 UML Diagrams	59-69
	5.3.1 Usecase Diagram	59
	5.3.2 Activity Diagram	60-61
	5.3.3 Deployment Diagram	62
	5.3.4 Class Diagram	63
	5.3.5 Sequence Diagram	64
	5.3.6 Component Diagram	65-66
	5.3.7 State Diagram	67
	5.3.8 Object Diagram	68-69
6.	System Testing	70-80
	6.1 Types of Testing	70
	6.1.1 Unit Testing	72-73
	6.1.2 Integration Testing	74
	6.1.3 Functional Tetsing	75-76
	6.1.4 System Testing	77
	6.1.5 White Box Testing	78
	6.1.6 Black Box testing	79
	6.2 Test Cases	80
7.	System Implementation	81-95
	7.1 Conclusion And Future Enhancement	91-94
	7.2 References	95
	APPENDIX	96-109
	ANNEXTURE	110

CHAPTER 1

INTRODUCTION

The eye is one of the most important organs in the human body because it allows us to see and understand the world around us. Almost everything we do in our daily life, like reading, walking, and recognizing people, depends on our vision. The eye works by capturing light from the surroundings and sending that information to the brain, which then helps us see images clearly.

The structure of the eye is quite interesting. It is made up of several parts, and each part has a specific function. The front portion of the eye is covered by a transparent layer called the cornea. This layer allows light to enter the eye. The white area around it is known as the sclera, which protects the eye and maintains its shape.

Next comes the iris, which is the coloured part of the eye. It controls the size of the pupil, a small opening in the centre. The pupil adjusts itself based on the amount of light present. For example, in bright light, it becomes smaller, and in low light, it becomes larger to allow lighter to enter.

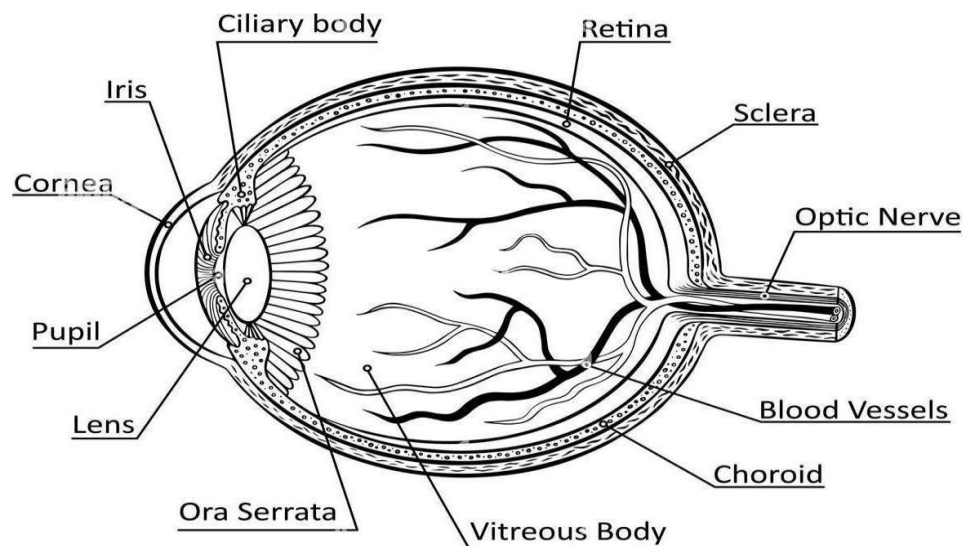


Fig:1 STRUCTURE OF EYE

Behind the cornea is the iris, which is the coloured part of the eye. The iris controls the size of the pupil, a small opening that regulates how much light enters the eye. When the light is bright, the pupil becomes smaller, and in dim conditions, it expands to allow lighter inside. This adjustment helps in maintaining clear vision under different lighting conditions.

The light entering through the pupil then passes through the lens. The lens plays an important role in focusing the light onto the retina. It has the ability to change its shape depending on the distance of the object, which helps us to see both nearby and distant objects clearly.

The retina is located at the back of the eye and is one of the most important parts for vision. It contains light-sensitive cells that convert light into electrical signals. These signals are then sent to the brain through the optic nerve. The brain processes these signals and helps us recognize what we are seeing.

In addition to this, the retina has a small central area called the macula, which is responsible for sharp and detailed vision. It helps in activities like reading and recognizing fine details. Any damage to the retina or its blood vessels can lead to serious vision problems.

Eye health is very important, but it is often affected by diseases and medical conditions. One such condition is diabetic retinopathy, where high blood sugar levels damage the blood vessels in the retina. This can cause leakage, swelling, and even vision loss if not treated at the right time.

Therefore, understanding the structure and working of the human eye is very important for studying eye-related diseases. In this project, more focus is given to the retina, as it plays a major role in detecting diabetic retinopathy using retinal fundus images. By analysing these images, it becomes possible to identify early signs of the disease and take necessary action.

The human eye works continuously without rest and responds quickly to changes in the environment. It helps us maintain balance, coordination, and awareness of our surroundings. Vision also plays an important role in learning and communication, as most information we receive is through our eyes. Because of this, even a small problem in vision can affect daily activities. Therefore, maintaining eye health and understanding its function is essential, especially when developing systems related to eye disease detection.

1.1 DIABETIC RETINOPATHY:

Diabetic retinopathy is a chronic, progressive eye disease that arises as a complication of diabetes mellitus and represents one of the leading causes of preventable blindness worldwide. It develops when prolonged hyperglycaemia damages the small blood vessels of the retina, leading to leakage, occlusion, and eventually abnormal new vessel growth. In the early stages, known as non-proliferative diabetic retinopathy (NPDR), patients may show microaneurysms, retinal haemorrhages, and macular edema, but often remain asymptomatic. As the disease advances to proliferative diabetic retinopathy (PDR), fragile new vessels form on the retina and optic disc, which can bleed into the vitreous cavity, cause tractional retinal detachment, or lead to neovascular glaucoma. These changes severely threaten vision and, if untreated, may result in irreversible blindness.

The pathophysiology of diabetic retinopathy involves multiple mechanisms, including increased vascular permeability, capillary basement membrane thickening, oxidative stress, and inflammatory processes. Epidemiological studies reveal that the risk of developing retinopathy rises with the duration of diabetes, poor glycaemic control, hypertension, hyperlipidaemia, pregnancy, and smoking. Globally, about one-third of people with diabetes show signs of retinopathy, and approximately one in ten develop vision-threatening stages.

Diagnosis relies on comprehensive ophthalmic evaluation. Dilated fundus examination remains the gold standard, while advanced imaging techniques such as Optical Coherence Tomography (OCT) and fluorescein angiography provide detailed insights into retinal thickness, macular edema, and vascular leakage. OCT angiography, a newer non-invasive tool, allows visualization of retinal vasculature without the need for dye injection.

Management of diabetic retinopathy requires both systemic and ocular interventions. Systemic control of blood glucose, blood pressure, and cholesterol is fundamental to slowing disease progression. Ocular treatments include intravitreal injections of anti-VEGF agents such as ranibizumab, aflibercept, or bevacizumab, which reduce macular edema and inhibit abnormal vessel growth. Corticosteroid implants may be used in selected cases. Laser photocoagulation remains a cornerstone therapy, particularly pan retinal photocoagulation for proliferative disease and focal/grid laser for macular edema. In advanced cases with vitreous haemorrhage or retinal detachment, vitrectomy surgery is performed to restore vision.

Complications of diabetic retinopathy include diabetic macular edema, vitreous haemorrhage, retinal detachment, and neovascular glaucoma, all of which can cause severe and permanent vision loss. Preventive strategies emphasize annual dilated eye examinations, patient education, lifestyle modifications such as regular exercise and smoking cessation, and integration of ophthalmic care into diabetes management programs.

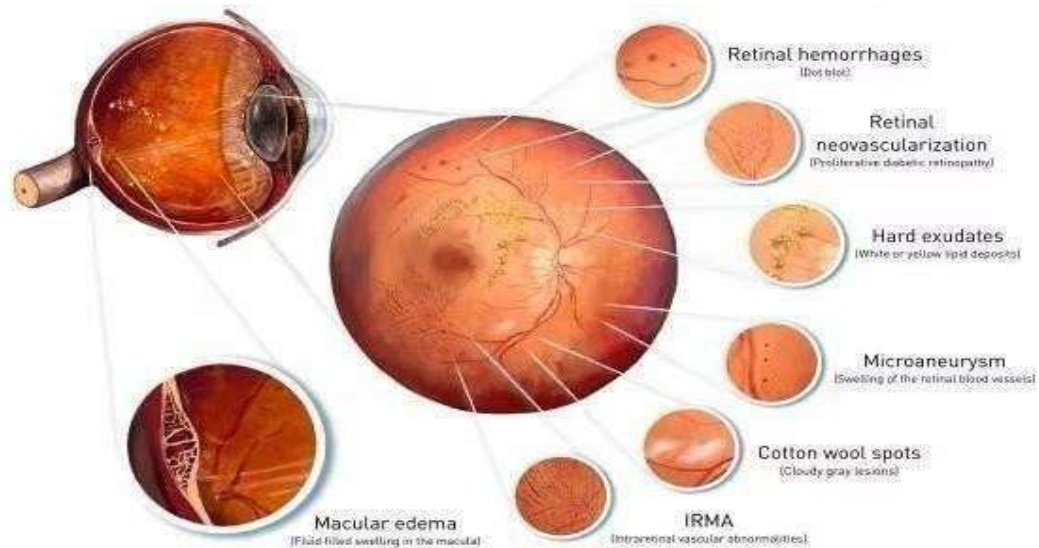


Fig:1.1 DIABETIC RETINOPATHY

1.2 STAGES OF DIABETIC RETINOPATHY:

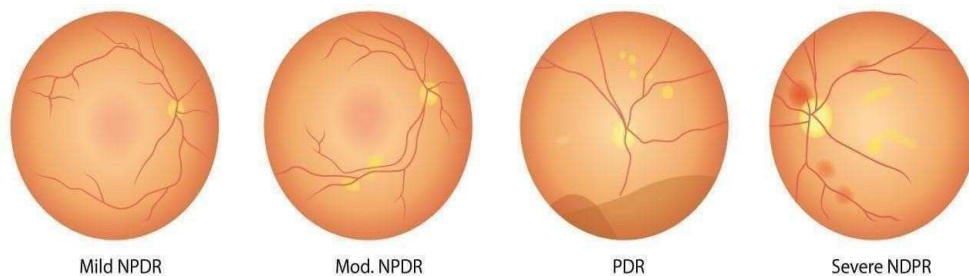


Fig:1.2 STAGES OF DIABETIC RETINOPATHY

1.2.1 Mild non-proliferative

Mild non proliferative diabetic retinopathy (NPDR) is the earliest stage of diabetic retinopathy, and it often develops quietly without causing any noticeable changes in vision.

At this point, the tiny blood vessels in the retina begin to show stress from prolonged high blood sugar levels. The most common sign is the appearance of microaneurysms small balloon-like swellings in the vessel walls. These microaneurysms may leak a little fluid into the retina, but the leakage is usually minimal and does not immediately interfere with sight. Because the retina is still functioning well, most people are unaware of any problem unless they undergo a dilated eye examination, where an ophthalmologist can detect these subtle changes.

Even though mild NPDR does not usually threaten vision, it is an important warning sign that the eyes are under strain and that the disease can progress if diabetes is not well controlled. Over time, if blood sugar, blood pressure, and cholesterol remain poorly managed, mild NPDR can advance to moderate and severe stages, eventually leading to proliferative diabetic retinopathy, which carries a high risk of blindness. This makes early detection and prevention especially important. At this stage, the best approach is not aggressive treatment but rather careful monitoring and lifestyle management keeping blood sugar within target ranges, maintaining healthy blood pressure and cholesterol levels, exercising regularly, and scheduling routine eye exams. In a way, mild NPDR is the body's early signal that the eyes are vulnerable, offering a valuable opportunity to protect vision before more serious complications develop.

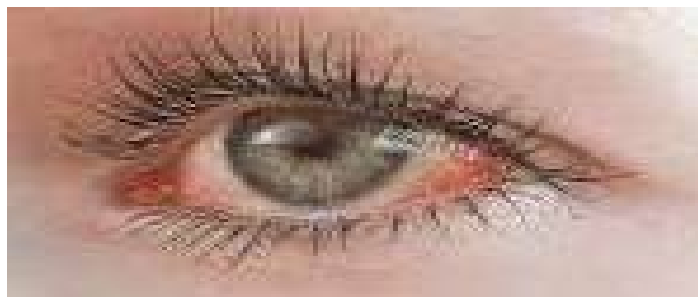


Fig:1.2.1 MILD NON-PROLIFERATIVE

1.2.2 Moderate non-proliferative

Moderate Non-Proliferative Diabetic Retinopathy (NPDR) is the second stage in the progression of diabetic eye disease. At this point, damage to the retinal blood vessels becomes more pronounced than in the mild stage. The retina begins to show signs of blocked

or leaking capillaries, leading to the appearance of dot-and-blot haemorrhages, cotton-wool spots (which are small, white patches caused by nerve fibre damage), and hard exudates—yellowish deposits of lipids and proteins. These changes indicate that the retina is under stress and not receiving adequate oxygen. One of the most concerning developments during this stage is diabetic macular edema (DME), where fluid accumulates in the macula, the central part of the retina responsible for sharp vision. This can cause blurred or distorted vision and may affect daily activities like reading or recognizing faces. Although symptoms may still be mild or absent,

the risk of vision loss increases significantly. Regular eye exams, especially with optical coherence tomography (OCT) and fluorescein angiography, are essential for detecting these changes early. Treatment at this stage often includes tighter control of blood sugar and blood pressure, and if macular edema is present, anti-VEGF or steroid injections may be recommended to reduce swelling and preserve vision.

Moderate Non-Proliferative Diabetic Retinopathy (NPDR) is like a warning light on the dashboard of your eyes. At this stage, the tiny blood vessels in the retina are under more stress than before, and they start to leak or get blocked. This shows up as small spots of bleeding (dot-and-blot haemorrhages), pale patches called cotton-wool spots, and yellowish deposits known as hard exudates. These changes mean the retina isn't getting enough oxygen, and the tissue is struggling. One of the biggest concerns here is diabetic macular edema (DME), where fluid builds up in the macula—the part of the retina that gives you sharp, central vision. That swelling can make everyday tasks like reading, driving, or recognizing faces harder. Sometimes people don't notice symptoms right away, but the risk of vision loss is real and growing. Eye doctors use special scans like OCT to measure swelling and fluorescein angiography to see where vessels are leaking. Managing this stage isn't just about eye treatments—it's about keeping blood sugar, blood pressure, and cholesterol under control. If macular edema develops, injections of anti-VEGF medicine or steroids can help reduce swelling and protect vision. In short, moderate NPDR is a turning point: it's not yet the most dangerous stage, but it's the moment when careful monitoring and timely treatment can make all the difference in preserving sight.

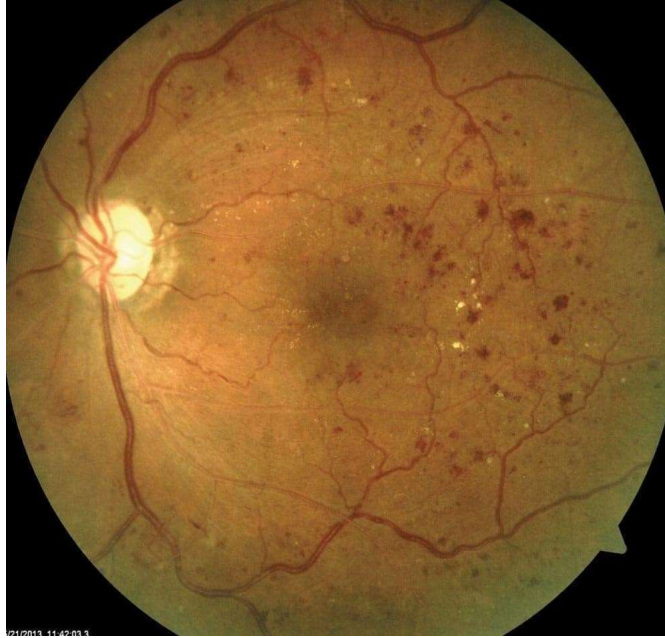


Fig:1.2.2 MODERATE NON-PROLIFERATIVE

1.2.3 Proliferative Diabetic Retinopathy

Proliferative Diabetic Retinopathy (PDR) is the stage where diabetic eye disease becomes truly sight-threatening. By this point, the retina has been deprived of oxygen for so long that it tries to “fix” the problem by growing new blood vessels. Unfortunately, these vessels are weak and fragile. They can easily break and bleed into the clear gel inside the eye (the vitreous), which may suddenly cloud your vision with floaters or even cause a dramatic loss of sight. Over time, scar tissue forms around these abnormal vessels, and as it tightens, it can pull the retina away from the back of the eye a retinal detachment that can lead to permanent blindness. In some cases, the new vessels also grow in places they shouldn’t, like near the drainage angle of the eye, blocking fluid outflow and causing a painful rise in eye pressure known as neovascular glaucoma.

Living with PDR can feel unpredictable: vision might fluctuate, dark spots may appear, or colours may seem faded. Sometimes vision loss comes on suddenly. Eye doctors use detailed imaging tests like fluorescein angiography and OCT scans to spot these changes and guide treatment. Managing PDR usually requires urgent intervention, such as laser therapy to shrink abnormal vessels, anti-VEGF injections to stop their growth, or vitrectomy surgery to remove

blood and scar tissue. Alongside these treatments, keeping blood sugar, blood pressure, and cholesterol under control is vital to slow progression. In short, PDR is the point where diabetic retinopathy becomes dangerous, but with timely care, many people can still protect their vision and maintain independence.

Symptoms of PDR can range from mild disturbances like floaters to severe, sudden vision loss. Diagnosis is made through a dilated eye exam, retinal imaging, and specialized tests like fluorescein angiography to highlight abnormal vessel growth. Treatment is urgent and typically involves pan retinal laser photocoagulation to shrink abnormal vessels, anti-VEGF injections to stop their growth, and sometimes vitrectomy surgery to remove blood or scar tissue. Even with treatment, PDR requires close monitoring because the risk of blindness is high if left unmanaged. Strict control of blood sugar, blood pressure, and cholesterol remains essential to slow progression. In short, PDR is the stage where diabetic retinopathy becomes sight-threatening, and timely intervention is the key to preserving vision.

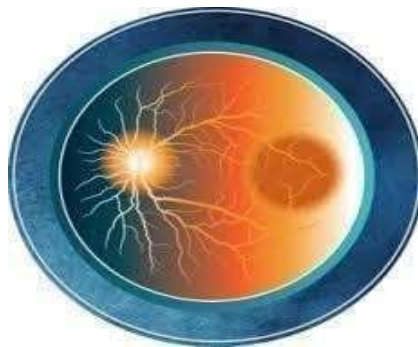


Fig:1.2.3 PROLIFERATIVE DIABETIC RETINOPATHY

1.2.4 Severe Diabetic Retinopathy

severe diabetic retinopathy, the changes inside the eye are not only more serious but also more difficult to reverse. The retina becomes weak because it is not getting enough nutrients and oxygen from the blood. As the condition progresses, vision may start changing more noticeably. A person may feel that their eyesight is not stable, sometimes clear and sometimes blurred.

Reading and focusing on small details can become difficult, especially in low light conditions.

Another important point is that the damage at this stage often develops slowly, so many people may ignore the early warning signs. By the time symptoms become obvious, the condition might already be advanced. This is why regular eye check-ups are very important for diabetic patients, even if they feel their vision is normal.

Severe diabetic retinopathy can also affect daily life activities. Tasks like driving, using a mobile phone, or recognizing faces may become challenging. In some cases, sudden vision problems can occur due to internal bleeding in the eye, which can be frightening for the patient.

The good thing is that with proper medical care, the condition can still be managed. Doctors may suggest treatments that help stop further damage and protect the remaining vision. Along with medical treatment, maintaining a healthy lifestyle, controlling blood sugar levels, and following medical advice can make a big difference.

Overall, severe diabetic retinopathy is a serious condition, but with awareness, early detection, and proper care, its impact can be reduced and vision can be protected to a certain extent.

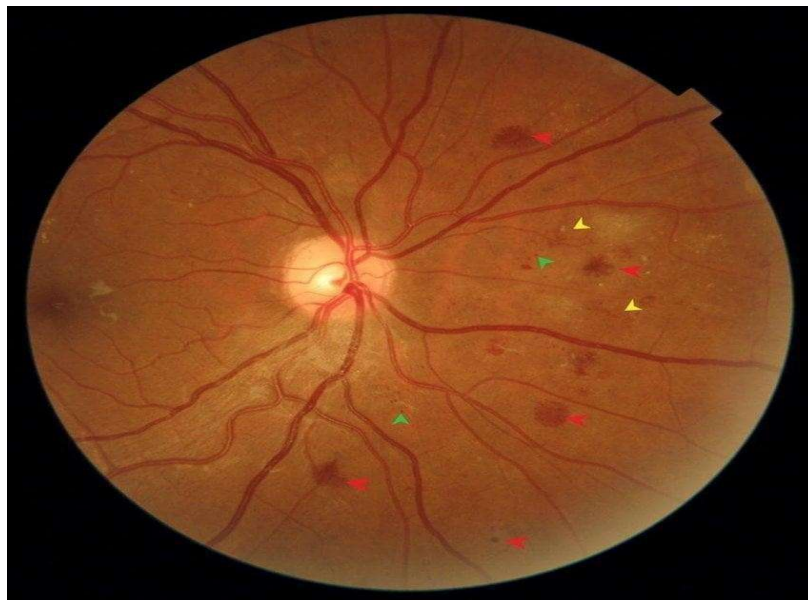


Fig:1.2.4 SEVERE DIABETIC RETINOPATHY

1.2.5 MOTIVATION

Diabetic retinopathy (DR) is a progressive complication of diabetes that threatens vision and quality of life, making early detection and accurate staging a critical public health priority. Traditional diagnostic approaches rely on manual examination of retinal fundus images by ophthalmologists, which is resource-intensive and often inaccessible in rural or underserved regions. The increasing prevalence of diabetes worldwide has created a pressing demand for automated systems that can provide reliable, scalable, and cost-effective screening solutions. This context motivates the exploration of advanced artificial intelligence models that can replicate and enhance clinical decision-making.

The dual-stage ANN-RNN hybrid model is designed to address the limitations of single-architecture systems. Artificial neural networks (ANNs) are effective at extracting spatial features from retinal images, enabling the identification of lesions such as microaneurysms, haemorrhages, and exudates. However, they struggle to capture the sequential progression of disease patterns. Recurrent neural networks (RNNs), conversely, excel at modelling temporal dependencies and sequential data, making them suitable for analysing the progression of DR across multiple stages. By combining these two architectures, the hybrid model ensures that both static abnormalities and dynamic progression are captured, resulting in a more comprehensive diagnostic framework.

The dual-stage design further enhances performance by structuring detection into two levels: an initial stage that separates healthy retinas from diseased ones, followed by a refined stage that classifies the severity of DR. This mirrors the clinical workflow of ophthalmologists, making the system more intuitive and clinically relevant. Importantly, this approach reduces false negatives, which are particularly dangerous in early DR detection, and improves the reliability of automated screening systems. Beyond technical accuracy, the model supports integration with telemedicine platforms and mobile-based fundus imaging devices, enabling large-scale screening in resource-limited settings.

Ultimately, the motivation for this hybrid approach lies in bridging advanced AI capabilities with real-world healthcare needs. By combining spatial and temporal analysis in a dual-stage framework, the ANN-RNN hybrid model offers a robust, efficient, and socially impactful solution that can transform diabetic retinopathy detection, reduce preventable blindness, and contribute to improved global health outcomes.

In essence, the motivation lies in harnessing advanced AI architectures to bridge the gap between technological innovation and real-world healthcare needs, offering a solution that is not only technically robust but also socially transformative.

1.2.6 PROBLEM IDENTIFICATION

Diabetic retinopathy (DR) is a progressive eye disease that poses a significant threat to vision among individuals with diabetes. Despite being preventable through early detection and timely treatment, DR continues to be a leading cause of blindness worldwide. The primary challenge lies in the fact that conventional diagnostic methods depend heavily on ophthalmologists and specialized equipment, which are often unavailable in rural and resource-constrained regions. As the global diabetic population grows, the demand for scalable and efficient screening solutions has become urgent. Current automated systems, while promising, face limitations in accurately identifying and classifying the multiple stages of DR progression.

Traditional machine learning and single deep learning models such as CNNs or ANNs have demonstrated strong performance in extracting spatial features from retinal fundus images. However, these models often fail to capture the temporal progression of lesions, which is critical for staging the disease. On the other hand, RNNs are capable of modelling sequential dependencies but are not well-suited for raw image feature extraction. This disconnects results in incomplete diagnostic frameworks that either miss early signs of DR or misclassify severity levels. Furthermore, many existing systems struggle with high false-negative rates, which can be particularly dangerous as they delay intervention during the early stages of the disease.

The problem, therefore, is the absence of a robust, integrated approach that can simultaneously handle spatial feature extraction and temporal progression modelling while providing accurate multi-stage classification. Without such a system, large-scale automated screening remains unreliable, and patients in underserved areas continue to face barriers to timely diagnosis.

Addressing this gap requires the development of a dual-stage ANN-RNN hybrid model that not only improves diagnostic accuracy but also aligns with clinical workflows, reduces false negatives, and supports deployment in telemedicine platforms. Identifying this problem

forms the foundation for designing a solution that bridges the gap between advanced AI capabilities and real-world healthcare needs.

The central problem lies in developing a robust, dual-stage hybrid model that can overcome these limitations, align with clinical workflows, and support telemedicine platforms to ensure early detection and reduce preventable blindness.

1.3 OBJECTIVE

The primary objective of this project is to design and implement a dual-stage ANN-RNN hybrid model for the automated detection and classification of diabetic retinopathy. The project aims to bridge the gap between advanced artificial intelligence techniques and realworld healthcare needs by creating a system that is both technically robust and practically deployable.

1. Study of Diabetic Retinopathy

To gain a comprehensive understanding of diabetic retinopathy, including its causes, symptoms, progression stages (Mild, Moderate, Severe NPDR, and PDR), and its impact on vision. This knowledge is essential for building a medically relevant and reliable detection system.

2. Image Preprocessing

To enhance the quality of retinal fundus images through preprocessing techniques such as noise reduction, contrast enhancement, normalization, and resizing. This step improves the clarity of important features and ensures better performance of the model.

3. Feature Extraction using ANN

To extract important spatial features from retinal images using an Artificial Neural Network (ANN). These features include patterns such as microaneurysms, haemorrhages, and exudates, which are critical indicators of diabetic retinopathy.

4. Contextual Feature Modelling using RNN

To model the relationships between extracted features using a Recurrent Neural Network (RNN). This helps in understanding the sequential and contextual dependencies within image features, improving classification accuracy.

5. Development of Hybrid ANN–RNN Model

To design and implement a dual-stage hybrid model that combines ANN for feature extraction and RNN for sequence modelling. This hybrid approach enhances the system's ability to detect and classify diabetic retinopathy more effectively than traditional methods.

6. Classification of DR Severity Levels

To accurately classify retinal images into different stages of diabetic retinopathy such as:

1. No DR
2. Mild NPDR
3. Severe NPDR
4. Moderate NPDR
5. Proliferative DR (PDR)

7. Performance Evaluation

To evaluate the performance of the proposed model using standard metrics such as:

- Accuracy: Accuracy tells how many predictions made by the model are correct out of all predictions.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

- Precision: Precision measures how many of the predicted positive cases are actually correct.

$$Precision = \frac{TP}{TP + FP}$$

- Recall: Recall tells how well the model is able to find all actual positive cases.

$$Recall = \frac{TP}{TP + FN}$$

- F1-Score: F1-score is the balance between precision and recall.

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Confusion Matrix: A confusion matrix is a table that shows the model performance in detail.

Matrix	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)

		True	Negative
Actual Negative	False Positive (FP)	(TN)	

8. Reduction of Manual Effort

To minimize dependence on manual diagnosis by automating the screening process. This reduces workload on ophthalmologists and speeds up diagnosis.

9. Support for Early Detection

To enable early detection of diabetic retinopathy, especially in rural and resource-limited areas, thereby preventing severe vision impairment and blindness.

11. Scalability and Real-World Application

To develop a system that can be scaled for large datasets and integrated into real-world healthcare systems for mass screening programs.

1.4 METHODOLOGY

1. Collecting the Images

The retinal images are gathered from various reliable and publicly available datasets such as hospitals, research institutions, or online medical repositories. These images include both normal and diabetic retinopathy affected eyes to ensure proper learning. A diverse dataset is important because it helps the system understand different variations in eye conditions, lighting, and image quality. Proper labelling of these images is also done so that the model can learn accurately during training.

2. Preparing the Images

Before using the images, they are pre-processed to improve their quality. This involves removing noise, adjusting brightness and contrast, and resizing all images to a standard dimension. In some cases, image enhancement techniques such as histogram equalization are used to highlight important features. This step ensures that the images are consistent and suitable for model processing, which ultimately improves performance.

3. Finding Important Details

Once the images are prepared, they are fed into a Convolutional Neural Network (CNN). The CNN automatically detects and extracts important features such as microaneurysms, haemorrhages, and exudates present in the retina. These features are essential indicators of diabetic retinopathy. The CNN works by applying filters to identify patterns and structures that may not be easily visible to the human eye.

4. Improving the Features

The extracted features from the CNN are then passed to an Artificial Neural Network (ANN). The ANN further processes and organizes these features to make them more meaningful. It helps in refining the data representation so that the system can distinguish between different stages of the disease more effectively. This step enhances the decision-making capability of the model.

5. Understanding Connections

An additional layer using a Recurrent Neural Network (RNN) is applied to understand relationships between different features. The RNN helps in identifying patterns and dependencies among features that may appear across multiple regions of the retina. This deeper analysis allows the system to capture complex patterns that are not easily noticeable, improving the overall prediction accuracy.

6. Combining the Models

The outputs from the CNN, ANN, and RNN are combined to create a hybrid model. This integration allows the system to benefit from the strengths of each model. While CNN focuses on feature extraction, ANN improves classification, and RNN understands feature relationships. Together, they form a more robust and efficient system for detecting diabetic retinopathy.

7. Identifying the Stage

Using the combined model, the system classifies the retinal image into different categories such as normal, mild, moderate, severe, or proliferative diabetic retinopathy. This

classification is based on the patterns and features detected in the earlier stages. The system ensures accurate identification by comparing learned features with trained data.

8. Training and Testing

The model is trained using a large dataset of labelled images. During training, the system learns by adjusting its internal parameters to minimize errors. After training, testing is performed using unseen images to evaluate how well the model performs in real-world scenarios. This step ensures that the model is not overfitted and can generalize well.

G. Checking the Results

The performance of the model is evaluated using metrics such as accuracy, precision, recall, and F1-score. These measures help in understanding how reliable and effective the system is. Confusion matrices and validation techniques are also used to analyse errors and improve the model further.

10. Producing the Output

Finally, the system generates the output based on the input retinal image. The result shows whether the eye is normal or affected by diabetic retinopathy, along with its stage. This output can assist doctors and medical professionals in making faster and more accurate diagnoses.

DUAL STAGE ANN-RNN HYBRID MODEL FOR AUTOMATED DIABETIC RETINOPATHY

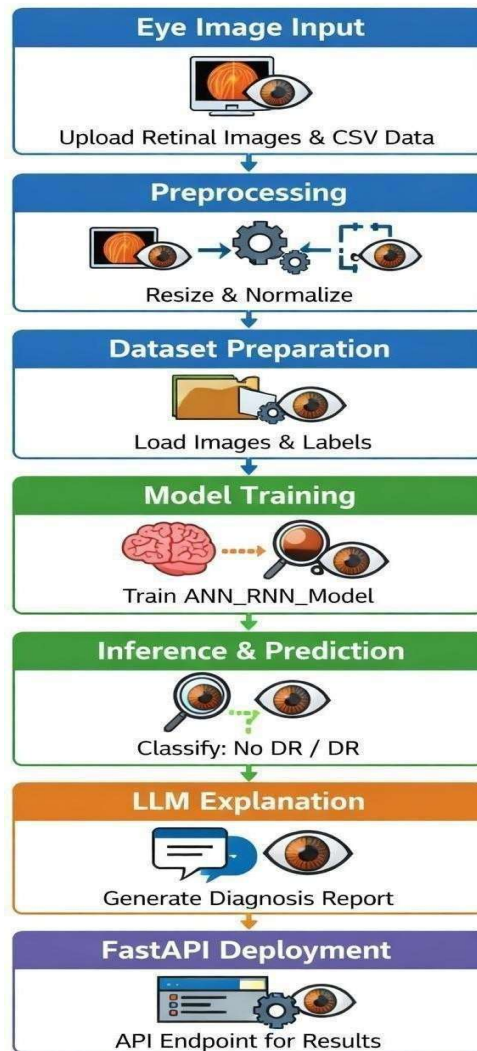


Fig 1.4.1: METHODOLOGY

CHAPTER 2

LITERATURE SURVEY

Diabetic Retinopathy (DR) is a serious eye disease that occurs due to long-term diabetes and can lead to permanent vision loss if not identified at an early stage. Because of the increasing number of diabetic patients, early screening and timely diagnosis have become very important. Traditionally, this process was carried out manually by ophthalmologists, which is time-consuming and depends heavily on the experience of the doctor. This has encouraged researchers to develop automated systems that can assist in detecting the disease quickly and accurately.

In the initial stages of research, DR detection was mainly based on traditional image processing and machine learning techniques. These methods involved extracting specific features from retinal images, such as blood vessel patterns, texture, and the presence of abnormalities like microaneurysms and exudates. After extracting these features, classification algorithms such as Support Vector Machines, Decision Trees, and K-Nearest Neighbours were used to identify the stage of the disease. Although these approaches provided some useful results, they required manual effort for feature extraction and were not very efficient when dealing with large datasets.

With the advancement of artificial intelligence, deep learning techniques have brought a major improvement in this field. Convolutional Neural Networks (CNNs) became widely popular because they can automatically learn important features directly from the images without the need for manual intervention. This made the detection process faster and more accurate. Researchers found that CNN-based models perform better than traditional methods, especially when trained on large datasets of retinal images.

Another important development is the use of transfer learning. In this approach, pre-trained models such as VGG, ResNet, and Inception are reused for DR detection tasks. These models are initially trained on large image datasets and then fine-tuned using medical images. This technique is very useful when there is limited medical data available, as it helps in achieving good performance without training a model from scratch.

In recent years, hybrid models have also been introduced to improve the overall performance of the system. These models combine different techniques, such as CNN for feature extraction and Artificial Neural Networks (ANN) for classification. Some advanced systems also include Recurrent Neural Networks (RNN) or LSTM networks to analyse the progression of the disease over time. This is especially helpful when multiple images of the same patient are available, as it allows the system to understand how the condition is changing.

Preprocessing of retinal images is another important step discussed in many studies. Since medical images often suffer from poor lighting, noise, and low contrast, various enhancement techniques are applied before feeding the images into the model. Methods like contrast enhancement, normalization, and noise removal help in improving the clarity of important features, which ultimately leads to better prediction accuracy.

Recent research has also focused on making these systems more understandable and reliable. Techniques such as heatmaps and attention mechanisms are used to highlight the regions in the image that influence the model's decision. This is important in the medical field because doctors need to trust the results produced by AI systems. By showing which part of the retina is affected, these methods improve transparency and usability.

Even though significant progress has been made, there are still some challenges in this area. One of the major issues is class imbalance, where the number of normal cases is much higher compared to severe cases. This can affect the performance of the model. Researchers are trying to solve this problem using techniques like data augmentation and balanced training methods. Another challenge is the variation in image quality across different datasets, which can reduce the model's ability to generalize.

Overall, the literature shows that deep learning, especially CNN-based and hybrid approaches, has become the most effective method for diabetic retinopathy detection. However, there is still a need to improve the reliability, interpretability, and real-world implementation of these systems. This has led to the development of more advanced models that combine multiple techniques and provide better results.

The literature survey shows that diabetic retinopathy detection has improved a lot over the years. Earlier methods used basic image processing and machine learning techniques, but they required manual effort and were less accurate. With the introduction of deep learning, especially CNN models, the detection process has become more efficient and reliable. Many studies also used advanced methods like transfer learning and hybrid models to improve performance. However, some challenges such as imbalance in data, variation in image quality, and lack of complete real-world systems still exist. Therefore, there is a need for an improved system that can provide accurate detection along with practical usability.

CHAPTER 3

SYSTEM ANALYSIS

System analysis helps in clearly understanding how the current method works and what kind of improvements are required to build a better solution. In the case of diabetic retinopathy detection, the existing approach mainly depends on doctors who examine retinal images and identify abnormalities. While this method is effective, it is not always practical because it takes time and requires experienced specialists. In many areas, especially where medical facilities are limited, patients may not get timely diagnosis, which can lead to serious complications.

Another difficulty with the current system is that the results may vary from one expert to another. Since the diagnosis is based on human observation, there is always a chance of missing small details in the image. Also, when a large number of patients need to be screened, it becomes difficult to manage the workload using only manual methods. These limitations make it necessary to look for a more efficient and consistent solution.

To address these issues, the proposed system introduces an automated approach using artificial intelligence. Instead of depending completely on manual observation, the system processes retinal images using trained models that can identify patterns related to diabetic retinopathy. The images are first prepared by improving their quality so that important details become clearer. After that, the model studies the image and predicts the stage of the disease.

One advantage of this system is that it provides results quickly and in a consistent manner. It does not depend on human judgment, so the chances of variation are reduced. The system can also handle multiple images at the same time, which makes it suitable for large-scale screening. In addition, it can be connected to simple applications where users can upload images and get results without complicated steps.

Overall, the analysis shows that the existing system has certain drawbacks in terms of speed, accessibility, and consistency. The proposed system aims to overcome these problems by using an automated and reliable method. This makes the detection process easier, faster, and more suitable for real-world use.

3.1 EXISTING SYSTEM

In the present system, diabetic retinopathy is detected by doctors through manual checking of retinal images. The images are taken using special cameras, and the doctor carefully observes them to find any damage in the retina. Based on their observation, they decide whether the patient has the disease and identify its stage.

This method gives good results when done by experienced doctors, but it takes a lot of time. Each image must be checked one by one, which increases the workload, especially in hospitals where many patients come for screening. In some areas, there are not enough eye specialists, so patients may not get checked regularly. This can delay early detection.

Another problem is that the results may vary from one doctor to another. Since the diagnosis depends on human judgment, small features in the image may sometimes be missed. This is more common in the early stages of the disease, where the signs are not very clear.

The quality of the retinal images also affects the diagnosis. If the image is not clear due to poor lighting or noise, it becomes difficult for the doctor to analyse it properly. Traditional systems do not include advanced tools to improve image quality or assist in better analysis.

Handling large amounts of data is also a challenge in the existing system. Hospitals may have many patient records and images, but managing and analysing them manually is not easy. There is no automatic system to organize or process this data quickly.

Overall, the existing system is helpful but has several limitations. It depends completely on human effort, takes more time, and may not always give consistent results. These issues create the need for an automated system that can support doctors and make the detection process faster and more reliable.

3.1.1 DEMERITS

1. Manual Diagnosis in Time-Consuming

In traditional systems, retinal images are examined manually by ophthalmologists. This process requires significant time, especially when dealing with a large number of patients, which can delay diagnosis and treatment.

2. Dependence on expert knowledge

Accurate detection of diabetic heavily on the experience and skill of medical professionals. In areas where specialists are not available, proper diagnosis become difficult.

3. Possibility of Human Error

Manual analysis of retinal images can sometimes lead to mistakes, especially in early stages where symptoms are not clearly visible. This may result in misdiagnosis or delayed identification of the disease.

4. Limited Detection of Early-Stage DR

Existing traditional methods often struggle to identify very small changes in the retina, such as microaneurysm. Early-Stage detection is crucial, but it is not always reliable with manual techniques.

5. High Cost of Diagnosis

Regular screening requires specialized equipment and trained professionals, which increases the overall cost. This makes it less accessible for people in rural or low-income areas.

6. Inconsistency in Results

Different doctors may interpret the same retinal image differently, leading to variations in diagnosis. This lack of consistency can affect treatment decisions.

7. Limited Scalability

Manual systems cannot efficiently handle large datasets or mass screening programs. As the number of diabetic patients increases, existing methods become less practical.

8. Low Efficiency in Large-Scale Screening

Screening a large population requires significant resources and time. Traditional systems are not designed to process high volumes of data quickly.

9. Lack of Automation

Most existing systems do not use fully automated processes, which limits their speed and efficiency in real-time diagnosis.

10. Difficulty in Feature Extraction

Traditional machine learning approaches require manual feature extraction, which is complex and may not capture all important patterns in retinal images.

3.2 PROPOSED SYSTEM

The proposed system focuses on developing an automated solution for detecting and classifying diabetic retinopathy using advanced deep learning techniques. Instead of relying completely on manual observation, this system uses computer-based analysis to examine retinal images and provide accurate results in less time.

In this approach, retinal fundus images are given as input to the system. These images are first processed to improve their clarity and quality. Adjustments such as resizing, noise removal, and contrast enhancement are applied so that important details in the retina become more visible. This step ensures that the data used for analysis is consistent and suitable for further processing.

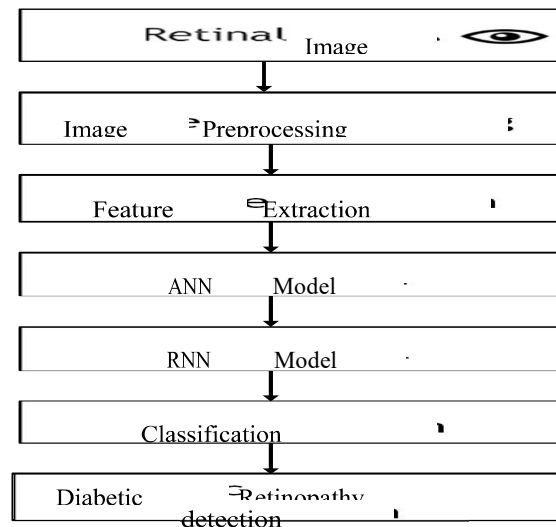


Fig 3.2: PROPOSED SYSTEM

After preprocessing, the images are passed through a Convolutional Neural Network (CNN). The CNN plays a key role in identifying critical features such as tiny blood vessel changes,

lesions, and abnormal patterns in the retina. These features are automatically extracted without the need for manual intervention.

The extracted features are then refined using an Artificial Neural Network (ANN). This stage helps in organizing the detected information and improving the system's ability to distinguish between normal and affected retinal conditions. To further enhance the

understanding of relationships between features, a Recurrent Neural Network (RNN) is incorporated. This helps in capturing deeper patterns that may not be obvious at first glance.

All these models work together as a combined system, making the overall process more effective. The system then classifies the input image into different stages of diabetic retinopathy such as normal, mild, moderate, severe, or proliferative. The final output is displayed in a simple and understandable format, which can assist doctors in making faster and more reliable decisions.

This proposed system aims to reduce human effort, minimize errors, and improve early detection. It is especially useful in areas where medical experts are limited, as it provides a quick and efficient way to screen patients and identify potential risks at an early stage.

3.2.1 MERITS

1. Helps in Detecting the Disease Early

The system can find signs of diabetic retinopathy at an early stage by carefully analysing retinal images. This is very useful because early treatment can prevent serious vision problems in the future.

2. Gives More Accurate Results

By using both ANN and RNN models along with proper image processing, the system can produce better and more dependable results compared to older methods.

3. Reduces Manual Work

Doctors usually need to examine each eye image manually, which takes time and effort. This system reduces that burden by automatically analysing the images and giving results.

4. Saves Time

The system works quickly and can give output in a short time. This makes it useful in hospitals where many patients need to be checked in less time.

5. Provides Same Result Every Time

Unlike humans, the system does not get tired or distracted. So, it gives the same result for the same input, which makes it reliable.

6. Low-Cost Solution

This system can be used without expensive equipment. It is helpful for small clinics and rural areas where medical facilities are limited.

7. Can Handle Large Data

The system can process many images at once. This makes it suitable for screening a large number of patients.

8. Better Image Understanding

Before analysis, the images are improved using preprocessing techniques. This helps the system clearly identify important parts like blood vessels and damaged areas.

9. Supports Doctors

The system does not replace doctors but helps them. It gives a basic idea about the condition, so doctors can make better decisions.

10. Easy to Improve in Future

The system can be upgraded later by adding new technologies or more data. This makes it useful for long-term use.

11. Reduces Workload

Since the system handles most of the checking process, doctors can focus more on treating patients instead of spending time on analysis.

12. Easy to Use

The system can be designed in a simple way so that even basic users can operate it without difficulty.

3.3 FEASIBILITY STUDY

Diabetic retinopathy usually develops quietly, and many people may not realize there is a problem until the condition becomes serious. Because of this, regular screening is very important, but in reality, it is not always easy to examine every patient quickly and carefully using only manual methods. Doctors often need to go through many retinal images one by one, which can be tiring and may sometimes lead to delays. In such situations, having a supportive system can make a noticeable difference. The dual-stage ANN–RNN hybrid model is designed with this idea in mind, where the system assists in examining images and provides a second level of support to the doctor’s decision.

When an image is given to the system, it does not just look at it in a simple way. Instead, it studies small details like tiny changes in colour, shape, and texture that may not always be clearly visible at first glance. The first stage focuses on identifying these important features, and the next stage tries to understand how these features are connected, which helps in forming a more reliable result. This step-by-step approach makes the system more careful in its analysis rather than giving a quick and uncertain output.

One of the helpful aspects of this model is that it can work continuously without getting tired, which is not possible in manual screening. It can go through many images in less time and maintain the same level of attention for each one. This can be especially useful in large hospitals or screening camps where the number of patients is high. It also helps reduce the pressure on doctors, allowing them to focus more on confirming results and planning treatment instead of spending too much time on initial screening.

The system can also be improved over time as more data becomes available. By learning from different types of retinal images, it can become better at recognizing various stages of the disease. This gradual improvement makes it more dependable in the long run. At the same time, it is important to remember that such a system is meant to support medical

professionals, not replace them. The final decision should always be made by a qualified doctor.

In terms of usage, the system can be set up in a simple environment without requiring highly advanced infrastructure. Once it is trained and ready, it can be used repeatedly with minimal effort. This makes it suitable not only for well-equipped hospitals but also for smaller clinics. Care must be taken to protect patient information and to use the system responsibly. When these factors are considered, the model can become a practical tool that supports early detection, reduces workload, and helps in managing diabetic retinopathy more effectively.

The system also helps in maintaining a consistent approach while examining retinal images, as it follows the same process for every input without variation. This reduces the chances of missing small details that might be overlooked during manual observation. By providing quick preliminary results, it supports early attention to patients who may need further medical care. The model can be integrated into existing healthcare workflows without causing major changes, making it easier to adopt in different settings. It also allows digital storage of results, which can be useful for future reference and comparison. Over time, this can help in tracking the progress of a patient's condition in a more organized manner. The overall design focuses on simplicity, reliability, and practical use, so that it can assist healthcare providers in a meaningful way without adding extra complexity to their work.

In addition, the system can support regular screening programs by handling repeated image checks without delay. It provides stable results each time, which helps build trust in its usage over time. The model can also be adjusted or updated when new data is available, making it flexible for future improvements. Its simple working process makes it easier for medical staff to learn and use without special training. Overall, it acts as a helpful support tool that improves efficiency while keeping the process clear and manageable.

3.3.1 Economic Feasibility

The economic feasibility of the dual-stage ANN–RNN hybrid model for automated diabetic retinopathy detection can be considered strong when both initial and long-term factors are taken into account. At the beginning stage, the project does not require large financial support because the main resources used for development are freely available. Software tools, programming environments, and machine learning libraries can be accessed without cost, which reduces the financial burden. The hardware requirements are also moderate, as a normal computer system is sufficient for development and testing, while advanced systems can be used only if faster processing is needed. In

Addition, publicly available retinal image datasets remove the need for expensive data collection processes, making the starting phase simple and budget- friendly.

As the system becomes ready for use, it begins to show clear economic advantages in daily operations. Traditional screening methods depend heavily on skilled professionals who must spend time examining each image carefully. This increases the overall cost when large numbers of patients need to be screened regularly. In contrast, the proposed system can process many images in a shorter time with less human effort. This reduces the need for continuous manual work and helps healthcare centres manage their resources more effectively. Over time, this leads to savings in both labour and operational expenses, making the system more efficient from a financial point of view.

Another important benefit is that the system can be used repeatedly without major additional investment. Once it is developed and deployed, it can be installed in multiple locations such as hospitals, clinics, and diagnostic centres without rebuilding the entire model. This ability to reuse and expand the system helps in spreading the cost over a wider usage, which further improves its economic value. Maintenance and updates are also manageable, as they mainly involve software improvements rather than expensive hardware changes. This ensures that the system remains useful over a longer period without requiring frequent high-cost upgrades.

The system also contributes indirectly to cost savings in healthcare by supporting early detection of the disease. When diabetic retinopathy is identified at an early stage, patients can receive timely treatment, which helps in avoiding serious complications. This reduces the

need for costly medical procedures and long-term care. By preventing the condition from becoming severe, the system helps both patients and healthcare providers reduce financial strain. This indirect benefit adds to the overall economic advantage of the project.

In practical terms, the system fits well within existing healthcare environments without requiring major changes or additional infrastructure. It can be integrated into current workflows, which avoids extra setup costs. Training the staff to use the system is also simple, so there is no need for extensive training programs or additional expenses. This makes the system suitable for both well-established hospitals and smaller clinics with limited resources.

Overall, the dual-stage ANN–RNN hybrid model proves to be economically feasible due to its low initial cost, reduced dependency on manual labour, ability to handle large-scale screening, and long-term financial benefits. It provides a balanced solution where the investment made during development is justified by the continuous savings and improved efficiency achieved during its usage.

3.3.2 Technical Feasibility

The technical feasibility of the dual-stage ANN–RNN hybrid model for automated diabetic retinopathy detection is strong because the system can be built using tools and methods that are already well established. The development can be carried out using a common programming language like Python, along with widely used libraries for deep learning and image processing. These tools are easy to access and come with good support, which makes the implementation process smoother. Since many examples and learning resources are available, it becomes easier to design, train, and test the model without facing major technical barriers.

The working approach of the system is also practical from a technical point of view. Retinal images can be collected and prepared using standard preprocessing steps such as resizing, noise removal, and normalization. After that, the ANN part of the model can be used to extract useful features from the images, while the RNN part can further analyse patterns and relationships in those features. This combination allows the system to handle the data in a structured way, improving its ability to produce meaningful results. The integration of both models does not require unusual or complex setups, as existing frameworks already support such hybrid designs. Another important factor is that the system does not demand highly advanced hardware for basic implementation. A standard computer with moderate memory and processing power is enough for initial development and testing. If better performance is needed, the model can be trained on systems with GPU support, but it is not a strict requirement for understanding and building the system. This flexibility makes the project easier to manage in different environments, including academic labs and small-scale setups.

The system is also flexible when it comes to improvements and future changes. As new data becomes available, the model can be retrained or fine-tuned to improve its performance. Additional features can be added without changing the entire structure of the system. This makes the design adaptable and suitable for long-term use. It also supports integration with other systems, such as hospital databases or digital record systems, without much difficulty.

From a reliability point of view, the system can be tested using standard evaluation methods to check its accuracy and performance. Errors can be identified and corrected during the testing phase, which helps in improving the quality of the final model. Since the technologies used are

more time can be spent on explaining conditions and guiding treatment. The system also encourages the use of digital healthcare practices, which are becoming more common in modern society.

At the same time, it is important to ensure that the system is used responsibly. Patients should be informed that the results are supportive and not final medical decisions. Maintaining trust is essential, so the system must provide clear and understandable outputs. Privacy of patient data must also be protected, as medical information is sensitive and should not be misused.

When these aspects are handled carefully, acceptance of the system becomes easier among users.

Another social benefit is that the system can be used in large-scale health programs, such as screening camps or community health initiatives. It can help identify cases that need immediate attention, allowing healthcare teams to prioritize patients effectively. This is especially useful in regions with high population and limited medical resources. By supporting such programs, the system plays a role in improving public health services.

Overall, the project is socially feasible because it addresses a real healthcare need, supports early detection, and helps both patients and medical professionals. It promotes better access to care, encourages timely action, and fits well within the growing use of technology in healthcare, making it a valuable addition to society.

3.4 SYSTEM SPECIFICATIONS

The system designed for the dual-stage ANN–RNN hybrid model for automated diabetic retinopathy detection does not require highly complex infrastructure and can be developed using commonly available resources. From a hardware point of view, a standard computer system with a multi-core processor, minimum 8 GB RAM, and sufficient storage space is enough to handle development and testing. For better performance during model training, especially when working with large image datasets, a system with GPU support can be used, but it is not strictly necessary for basic implementation. Input devices such as a keyboard and mouse are sufficient for operation, and the system can also support image input from external sources like fundus cameras or stored datasets.

On the software side, the system can be built using a stable operating system such as Windows or Linux. The programming work can be carried out using Python, which provides a simple and flexible environment for machine learning tasks. Libraries like TensorFlow and Keras can be used for building and training the ANN and RNN models, while OpenCV or similar tools can assist in image processing tasks such as resizing, filtering, and enhancement. Additional libraries like NumPy and Pandas can help in handling data efficiently. An integrated development environment such as Jupyter Notebook or any code editor can be used to write and test the code in an organized manner.

The system also requires a dataset of retinal images for training and evaluation. These images can be obtained from publicly available medical datasets or authorized sources. Proper preprocessing steps such as normalization, noise reduction, and resizing are needed to prepare the images before feeding them into the model. The trained model should be stored in a way that allows it to be reused for prediction without retraining every time.

For user interaction, a simple interface can be developed where users can upload retinal images and view the results generated by the system. The output can be displayed in a clear format, indicating whether signs of diabetic retinopathy are present and possibly the severity level. The system should also allow easy storage and retrieval of results for future reference.

Overall, the system specifications remain simple and practical, making the project easy to implement in different environments without requiring expensive or specialized resources.

3.4.1 Functional Requirements

Functional requirements describe what the system is expected to do and how it should behave when it is in use. They focus on the main operations that the system must perform, starting from taking input to producing the final output. In the case of the dual-stage ANN–RNN hybrid model for automated diabetic retinopathy detection, these requirements explain how the system handles retinal images, processes them step by step, and generates meaningful results.

The purpose of defining these requirements is to make sure that every important function of the system is clearly understood before development begins. This includes how images are received, how they are prepared for analysis, how the model works internally, and how that the results are presented to the user. By outlining these details, it becomes easier to design a system that works smoothly and meets the expected goals.

Functional requirements also help in maintaining a clear structure during development. They act as a guide for building each part of the system in a proper sequence. When each function is clearly defined, testing and verification also become easier, as it is possible to check whether the system is performing each task correctly.

Overall, these requirements ensure that the system performs all necessary operations in a reliable and organized manner, helping it serve as an effective support tool for detecting diabetic retinopathy.

1. Image Input Handling

The system should allow users to provide retinal images as input in an easy and flexible way. It must support uploading images from local storage without complications. The system should accept common image formats and ensure that the uploaded file is valid before processing. This step is important to make sure the system receives proper data for analysis.

2. Image Preprocessing

After receiving the image, the system should prepare it for further analysis. This includes adjusting the size, improving clarity, and removing unwanted noise if present. The goal of this step is to make the image suitable for the model so that it can focus on important details without being affected by unnecessary variations.

3. Feature Extraction

The system should be able to identify important patterns and details from the retinal image. In this stage, the ANN part of the model works to capture key visual information such as textures and structures. These features form the base for further analysis and help in understanding the condition of the retina.

4. Pattern Analysis and Prediction

Once the features are extracted, the system should analyse them using the RNN component. This stage focuses on understanding relationships within the extracted data to provide a more accurate result. The system should then generate an output indicating whether diabetic retinopathy is present and, if possible, give an idea about its severity.

5. Result Display

The system should present the output in a clear and simple manner. Users must be able to understand the result without confusion. The output should be displayed on the screen after processing, and it should be easy to interpret by both medical professionals and general users.

6. Data Storage and Management

The system should have the ability to store input images and their corresponding results for future reference. This helps in maintaining records and allows comparison of results over time. Proper organization of stored data is necessary to ensure easy access and retrieval.

7. User Interaction

The system should provide a simple interface so that users can interact with it without difficulty. Basic actions like uploading images, viewing results, and accessing stored data should be straightforward. The design should focus on ease of use to support different types of users.

8. Performance and Reliability

The system should process images within a reasonable time and provide consistent results. It must handle multiple inputs without crashing or slowing down significantly. Reliable performance is important to ensure trust in the system during regular use.

9. Error Handling

The system should be able to detect and respond to errors such as invalid inputs or processing failures. It should display clear messages so that users understand what went wrong and how to correct it. This helps in maintaining smooth operation.

10. Model Updating Capability

The system should allow updates to the model when new data becomes available. This helps in improving accuracy over time. The update process should be manageable without affecting the overall working of the system.

3.4.2 Non-Functional Requirements

Non-functional requirements describe the quality and performance aspects of a system rather than its specific functions. In the dual-stage ANN–RNN hybrid model for automated diabetic retinopathy detection, these requirements play a vital role in ensuring that the system operates efficiently and reliably in real-world conditions.

It involves medical image analysis the system must not only provide accurate predictions but also deliver results quickly and consistently. Non-functional requirements focus on key factors such as system performance, response time, reliability, security, and ease of use. These aspects help ensure that the model can handle retinal image data smoothly, protect sensitive patient information, and remain stable during continuous usage.

By defining these requirements, the system becomes more practical and dependable for healthcare applications, supporting medical professionals in making timely and accurate decisions.

1. Performance Requirements

The system must process retinal images within a reasonable time frame to support quick diagnosis. Image preprocessing, feature extraction, and classification should be optimized to reduce delay. The hybrid ANN-RNN model should provide results in seconds, even when handling multiple inputs. The system should also maintain stable performance when deployed on systems with moderate computational resource.

2. Accuracy and Reliability

Since the application is used for medical diagnosis, high accuracy is essential. The model must consistently produce correct predictions with minimal false positives and false negatives. The system should be trained and validated using diverse datasets to improve reliability. It must also maintain consistent output when similar input data is provided repeatedly.

3. Scalability

The system should be capable of handling an increasing number of users and larger datasets over time. As more retinal images are added, the system should still perform efficiently without degradation in speed or accuracy. The architecture should support future upgrades such as integration with cloud platforms or hospital databases.

4. Usability

The interface should be simple and easy to understand, even for users without technical knowledge. Medical staff should be able to upload images, view results, and interpret outputs without difficulty. Clear instructions and minimal steps should be provided to ensure smooth operation.

5. Security Requirements

Patient data, including retinal images and diagnostic results, must be protected from unauthorized access. The system should implement authentication mechanisms such as login credentials. Data transmission and storage should follow secure practices to ensure confidentiality and integrity.

6. Maintainability

The system should be designed in a way that allows easy updates and modifications. Developers should be able to improve the ANN-RNN model or fix issues without affecting the overall system functionality. Proper documentation and modular design should be maintained to support longterm maintenance.

7. Availability

The system should be available for use whenever required, especially in clinical environments. Downtime should be minimal, and the system should recover quickly from failures. Backup mechanisms should be considered to ensure continuous operation.

8. Compatibility

The application should work across different devices and operating systems such as Windows and web-based platforms. It should support various image formats like JPEG and PNG for retinal scans. Integration with existing medical systems should also be possible.

9. Robustness

The system must handle unexpected inputs such as low-quality or noisy images without crashing. It should provide appropriate error messages and continue functioning smoothly. The model should be resilient to variations in image resolution and lighting conditions.

10. Response Time

The time taken from image upload to result generation should be minimal. A fast response time is important for assisting doctors in making timely decisions. The system should ensure that delays do not affect the diagnostic workflow.

3.4.3 Software and Hardware Requirements

The development of the dual-stage ANN–RNN hybrid model for automated diabetic retinopathy detection requires a well-defined set of software and hardware resources to ensure smooth functioning of the system. These requirements form the foundation for building, training, testing, and deploying the model effectively.

Software requirements include the tools, programming languages, and frameworks that support image processing, data handling, and model development. They provide the necessary environment for implementing the hybrid ANN–RNN architecture and analysing retinal images accurately. At the same time, hardware requirements focus on the physical components such as processor speed, memory capacity, and storage space, which directly influence the system’s performance.

Since this project involves handling large medical image datasets and performing complex computations, having suitable hardware is essential to reduce processing time and improve efficiency. Similarly, choosing the right software tools helps in simplifying development and ensuring accurate results.

By clearly defining both software and hardware requirements, the system can operate in a stable, efficient, and reliable manner, making it suitable for practical use in healthcare environments.

Hardware Requirements

- Processor: intel i5 or equivalent
- RAM: 8 GB minimum
- Hard Disk: 256 GB minimum
- GPU (optional): NVIDA recommended for ANN-RNN training
- Display unit: monitor for viewing images and results
- Input Devices: keyboard and mouse for system interaction
- Network Connectivity: Stable internet connection

Software Requirements

- Operating System: Windows, Linux or macOS
- Programming language: python
- Development Environment: Jupyter Notebook, Google Colab, VS Code
- ML Frameworks: TensorFlow or Py Torch
- Libraries: NumPy, Pandas, OpenCV, Matplotlib
- Visualization Tools: Matplotlib, Seaborn

Libraries Used

1. NumPy: NumPy is used for numerical computations. It helps in handling arrays, mathematical operations, and efficient data processing required for model training.
2. Pandas: Pandas is used for managing and organizing data. It helps in reading datasets, preprocessing data, and handling structured information in tabular form.

3. OpenCV: OpenCV is mainly used for image processing tasks. It helps in reading retinal images, resizing, filtering, and enhancing image quality before feeding them into the model.
4. TensorFlow /Py Torch: These are deep learning frameworks used to build and train the ANN–RNN hybrid model. They provide tools for designing neural networks and optimizing model performance.
5. Keras: Keras is a high-level API (often used with TensorFlow) that simplifies the process of building neural networks, making model creation easier and faster.
6. Matplotlib: Matplotlib is used for data visualization. It helps in plotting graphs such as accuracy, loss curves, and other performance metrics.
7. Seaborn: Seaborn is used for advanced visualization. It helps in creating heatmaps and confusion matrices for better understanding of model results.
8. Scikit-learn: Scikit-learn is used for model evaluation and preprocessing. It provides tools for splitting datasets, calculating accuracy, precision, recall, and other performance measures.

Development Tools

➤ Jupyter Notebook

Used for writing and running code step-by-step. It helps in data preprocessing, training the model, and viewing results easily.

➤ Google Colab

Used to run code online without installation. It provides free GPU support for faster training of the ANN–RNN model.

➤ Visual Studio Code (VS Code)

Used for writing and organizing the complete project code. It helps in editing, debugging, and managing files.

➤ Anaconda Navigator

Used for managing Python environments and installing required libraries. It avoids conflicts between different packages.

➤ Git and GitHub

Used for version control and storing the project online. It helps in tracking changes and maintaining project backups.

➤ Kaggle

Used for downloading datasets and practicing machine learning models. It provides retinal images needed for training and testing.

CHAPTER 4

SYSTEM DEVELOPMENT MODEL

AGILE MODEL

4.1 INTRODUCTION

an Agile development approach was adopted to design and implement the retinal image analysis system. Instead of building the entire system in a single attempt, the development was carried out in a series of small, manageable iterations. Each iteration focused on a specific module of the system and was followed by testing and refinement.

This approach allowed gradual improvement of the system and ensured that errors were identified and corrected at early stages.

4.2 REASON FOR AGILE METHOD

The proposed system consists of multiple interconnected components such as image preprocessing, feature extraction, machine learning models, and backend integration. Due to this complexity, a linear development model would not be effective.

The Agile model was chosen because:

- It supports incremental development
- It allows flexibility in modifying components
- It helps in early detection of errors
- It improves system reliability through continuous testing

4.3 PROJECT MODULES

The system is divided into the following modules:

1. User Interface for image upload
2. Fundus image acquisition

3. Image preprocessing
4. Feature extraction
5. Artificial Neural Network (ANN) model
6. Recurrent Neural Network (RNN) model
7. Backend integration using FastAPI
8. Result display and diagnosis output

Each of these modules was developed in separate iterations.

4.4 ITERATIVE DEVELOPMENT PROCESS

4.4.1 Iteration 1 – User Interface Development

In the first stage, a simple user interface was designed to allow users to upload retinal images. The focus was on ensuring smooth interaction between the user and the system.

Outcome:

- Successful image upload functionality

4.4.2 Iteration 2 – Image Preprocessing

In the next iteration, preprocessing techniques were applied to improve image quality. This included:

- Image resizing
- Noise removal
- Normalization
- Patch extraction

Outcome:

- Clean and standardized images suitable for further processing

4.4.3 Iteration 3 – Feature Extraction

After preprocessing, important features were extracted from the images, including:

- Microaneurysms
- Hemorrhages
- Exudates
- Blood vessel abnormalities

Outcome:

- Conversion of image data into structured feature representations

4.4.4 Iteration 4 – ANN Model Development

An Artificial Neural Network was developed to classify the extracted features. The model was trained and tested multiple times to achieve stable performance.

Outcome:

- ☐ Initial prediction capability established

4.4.5 Iteration 5 – RNN Model Enhancement

To improve prediction accuracy, a Recurrent Neural Network was introduced. This helped in capturing deeper relationships within the extracted features.

Outcome:

- ☐ Improved accuracy and consistency of predictions

4.4.6 Iteration 6 – Backend Integration

The trained model was integrated with a backend system using FastAPI. This enabled communication between the user interface and the model.

Outcome:

- ☐ Real-time processing of user inputs

4.4.7 Iteration 7 – Result Display

In the final iteration, the output of the system was displayed to the user in a clear and understandable format.

Outcome:

- Complete working system with diagnosis results

4.5. AGILE WORKFLOW

Each iteration followed a structured cycle:

1. Planning the module
2. Designing the component
3. Implementation
4. Testing
5. Review and improvement

This cycle was repeated until the desired performance was achieved.

4.6 CONTINUOUS TESTING AND FEEDBACK

Testing was not limited to the final stage. Instead, each module was tested individually after development. Feedback from each stage was used to refine the system.

For example:

- Preprocessing methods were adjusted to improve model input quality
- Model parameters were tuned to enhance accuracy
- Integration issues were resolved during backend development

4.7 ADVANTAGES

The use of the Agile model provided several benefits:

- Simplified development of a complex system
- Early identification of errors
- Improved model performance through iteration
- Better integration of system components
- Increased flexibility in modifying features

CHAPTER 5

SYSTEM DESIGN

SYSTEM DESIGN

The system design of the dual-stage ANN–RNN hybrid model for automated diabetic retinopathy detection presents a structured approach for processing retinal images and generating accurate diagnostic results. The system is organized into a sequence of stages where each component performs a specific function, ensuring a smooth flow of data from input to output. Initially, retinal fundus images are provided as input, either by users or from medical datasets. These images then undergo preprocessing, where they are resized, cleaned, and enhanced to maintain uniform quality and remove noise that could affect model performance.

After preprocessing, the system focuses on identifying meaningful patterns within the images through feature extraction. These features highlight important characteristics such as abnormalities in blood vessels and lesions associated with diabetic retinopathy. The extracted information is then passed to the core of the system, which is the dual-stage ANN–RNN model. In the first stage, the Artificial Neural Network (ANN) analyses the features and captures complex patterns present in the images. In the second stage, the Recurrent Neural Network (RNN) further refines the analysis by learning relationships between these features, leading to improved prediction accuracy.

Once the processing is complete, the system classifies the image into categories such as normal or different stages of diabetic retinopathy. The final result is then displayed to the user in a clear and understandable format. Additionally, the system may store the input data and results for future reference and performance improvement. Overall, the design ensures that the system operates efficiently, handles medical image data effectively, and provides reliable outputs suitable for realworld healthcare applications.

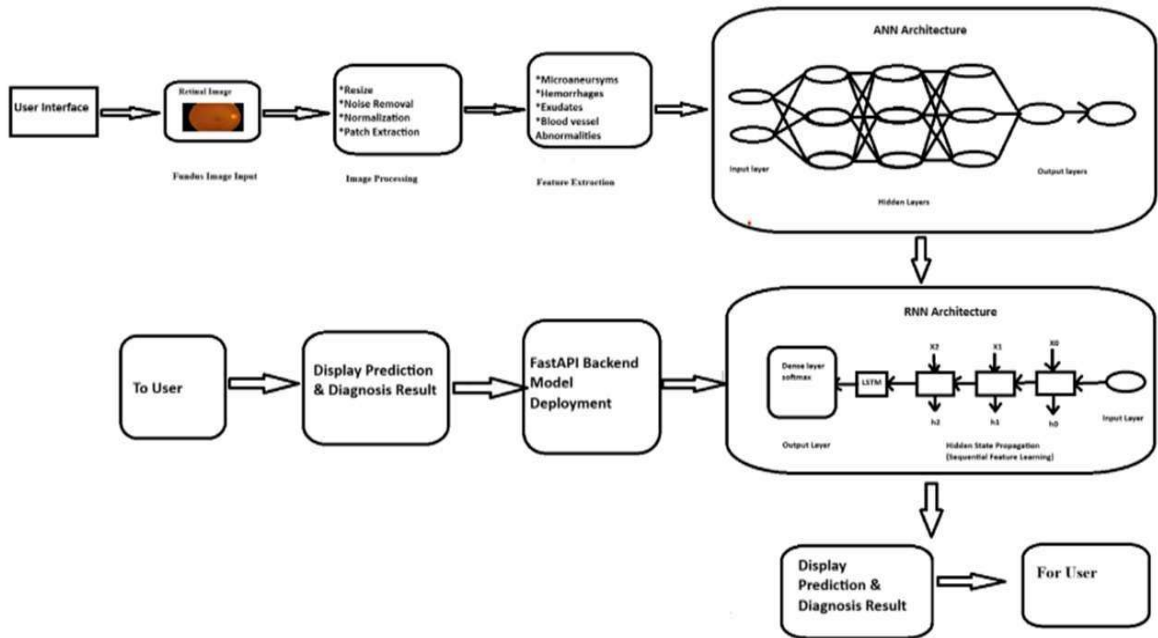


Fig-5 System Architecture

System Design Diagram

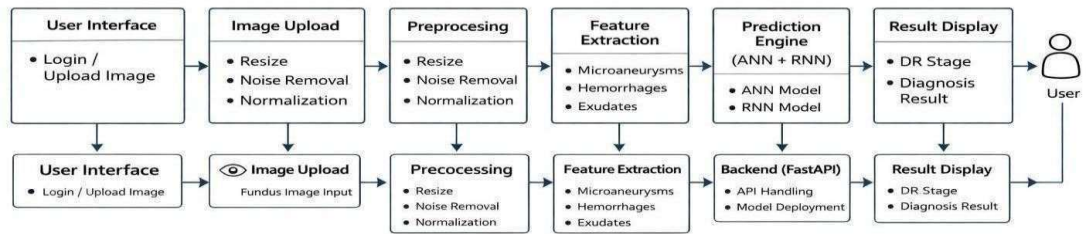


Fig 5(i): SYSTEM DESIGN

1. User Interface

This is the front part of the system where the user interacts with the application. It allows the user to log in and upload retinal images. The interface is designed to be simple so that even nontechnical users can easily use the system without confusion.

2. Image Upload Module

In this stage, the user selects and uploads the fundus (retinal) image into the system. The system accepts the image and prepares it for further processing. This step acts as the entry point for all image data.

3. Preprocessing Module

The uploaded image is cleaned and adjusted in this step. The size of the image is standardized, noise is reduced, and brightness or contrast is balanced. These changes help in improving the quality of the image so that the next steps can work more accurately.

4. Feature Extraction Module

Here, important details are identified from the retinal image. The system looks for specific patterns such as microaneurysms, hemorrhages, and exudates. These features are useful indicators for detecting diabetic retinopathy.

5. Engine (ANN + RNN)

This is the core part of the system where the actual analysis happens. The extracted features are passed to machine learning models like Artificial Neural Networks (ANN) and Recurrent Neural Networks (RNN). These models process the data and predict the stage of the disease.

6. Backend (Fast API)

The backend manages communication between different parts of the system. It handles requests, sends data to the model, and receives predictions. Fast API is used here because it is fast and efficient for handling such operations.

7. Result Display Module

Once the prediction is ready, the system shows the result to the user. It displays the stage of diabetic retinopathy along with a simple diagnosis message so that the user can understand the outcome easily.

8. User (Output)

Finally, the processed result is delivered back to the user. The user can view the diagnosis and take further action based on the result, such as consulting a doctor.

5.1 MODULES AND THEIR DESCRIPTION

1. Image Acquisition Module

This module is responsible for collecting retinal fundus images from users or available datasets. The system accepts images in standard formats and ensures that the input data is suitable for further processing. It acts as the entry point of the entire system.

2. Image Preprocessing Module

In this module, the input images are prepared for analysis. The system performs operations such as resizing, noise removal, and contrast enhancement. These steps help in improving image quality and ensure consistency across all inputs.

3. Feature Extraction Module

This module focuses on identifying important characteristics from the retinal images. It detects patterns such as blood vessel structures and abnormal regions. These features are essential for understanding the presence of diabetic retinopathy.

4. ANN Processing Module

The Artificial Neural Network (ANN) module processes the extracted features and learns complex patterns within the data. It provides an initial analysis by identifying relationships between different features in the image.

5. RNN Processing Module

The Recurrent Neural Network (RNN) module further refines the output from the ANN stage. It captures dependencies and sequential relationships among features, which helps in improving the accuracy of the prediction.

6. Classification Module

In this module, the system classifies the processed data into categories such as normal or different stages of diabetic retinopathy. It generates the final decision based on the model output.

7. Result Display Module

This module presents the output to the user in a clear and understandable format. It shows the diagnosis result and helps users easily interpret the findings.

8. Data Storage Module

The system stores input images and corresponding results for future use. This module helps in maintaining records, analyzing past data, and improving the system over time.

5.2 MODULES

5.2.1 INPUT DESIGN

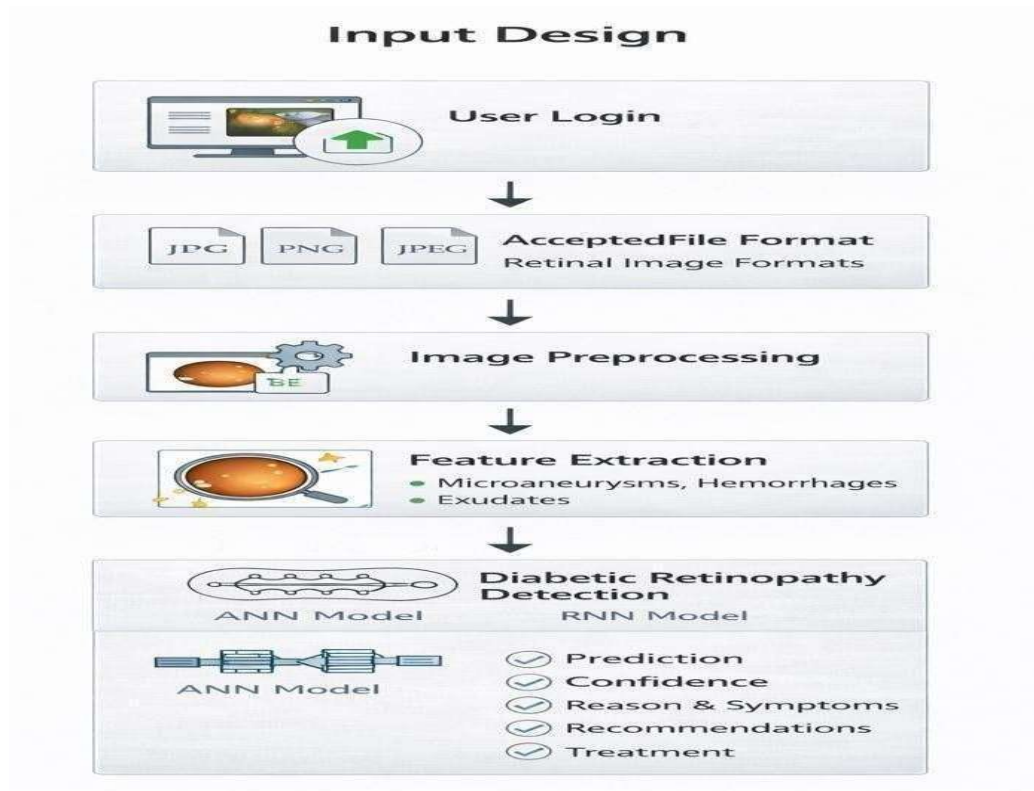


Fig 5.2.1 : INPUT DESIGN

Input design is the first step in building a system. It deals with how the user gives data to the system in a proper way. In this project, the process begins with user login, where only authorized users are allowed to enter. After that, the system accepts only specific image formats for upload.

This helps in avoiding errors and keeps the data correct. A good input design makes the system easy to use and helps in getting accurate results.

1. User Login

The system begins with a secure user login process. Only authorized users, such as doctors or medical professionals, are allowed to access the application. Each user is provided with unique login credentials to maintain data privacy and system security.

- Ensures restricted access

- Protects sensitive patient data
- Maintains system integrity

2. Accepted File Format

After successful login, users are required to upload retinal images. The system accepts only specific file formats to maintain compatibility and processing efficiency.

Supported formats are:

- JPG
- PNG
- JPEG

3. Image Preprocessing

Once the image is uploaded, preprocessing is performed to enhance the quality of the image. This step removes noise and improves clarity, making it suitable for further analysis.

Preprocessing Steps are:

- Image resizing
- Noise reduction
- Contrast enhancement
- Normalization

4. Feature Extraction

Features are extracted from the retinal image. These features help in identifying abnormalities related to diabetic retinopathy.

Extracted features include:

- Microaneurysms
- Haemorrhages
- Exudates

5. Diabetic Retinopathy Detection

The processed image and extracted features are then passed to machine learning models for detection.

Models used:

- Artificial Neural Network (ANN)
- Recurrent Neural Network (RNN)

6. Output Results

The system generates detailed output based on the analysis. The results help doctors in making informed decisions.

Output includes:

- Prediction (Disease presence or absence)
- Confidence level (Accuracy of prediction)
- Reason and symptoms detected
- Recommendations for further action
- Suggested treatment options

5.2.2 OUTPUT DESIGN

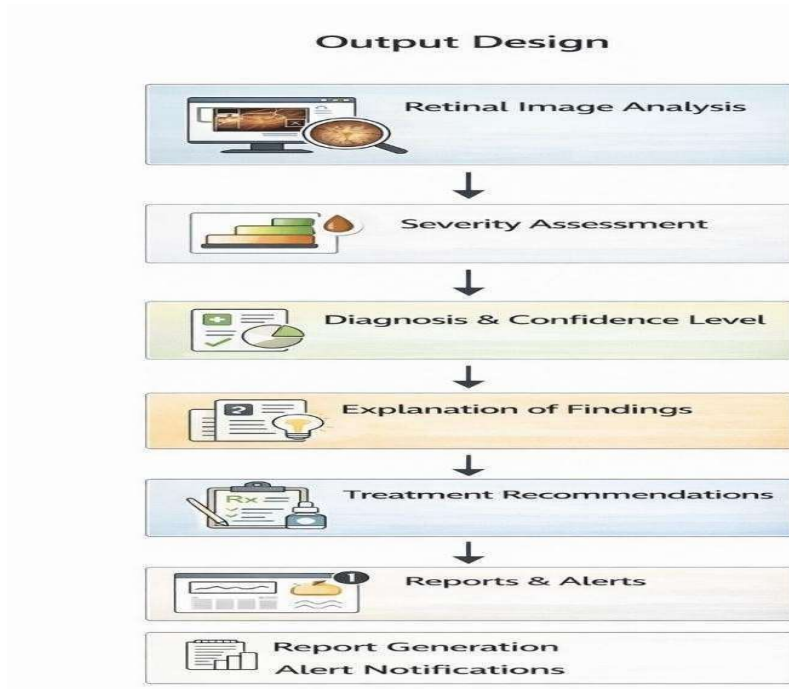


Fig 5.2.2: OUTPUT DESIGN

Output design is the final stage of the system where the results are shown to the user. It focuses on presenting the information in a clear and useful way. In this project, the output design displays the analysis of retinal images along with important details such as severity level, diagnosis, and confidence score. It also provides explanations, recommendations, and treatment suggestions. A well-designed output helps users easily understand the results and take proper decisions.

1. Retinal Image Analysis

At the first stage of output, the system presents the analysed retinal image. This is not a raw image but an enhanced visual representation where relevant regions are emphasized. Areas suspected of abnormalities are often highlighted using bounding regions, colour overlays, or attention maps.

This visualization acts as a bridge between artificial intelligence and human expertise. Rather than expecting users to trust a black-box prediction, the system provides visual evidence. It allows clinicians to verify whether the detected regions align with medical expectations, thereby improving trust and usability.

2. Severity Assessment

Following visual analysis, the system evaluates the extent of the disease. Diabetic retinopathy progresses through multiple stages, and identifying the severity is crucial for treatment planning.

The system categorizes the condition into levels such as:

- No Diabetic Retinopathy
- Mild
- Moderate
- Severe
- Proliferative Diabetic Retinopathy

This classification is derived from the number, size, and distribution of retinal lesions. By quantifying disease progression, the system helps prioritize cases and supports early intervention strategies.

3. Diagnosis and Confidence Level

At this stage, the system produces a definitive diagnostic outcome. It determines whether diabetic retinopathy is present and specifies its category. Alongside this, a confidence score is generated. The confidence level reflects the certainty of the model's prediction based on learned patterns during training. A higher percentage indicates stronger alignment between the input image and known disease patterns. This metric is particularly useful in clinical settings, where uncertain cases may require additional review or secondary testing.

4. Explanation of Findings

To enhance interpretability, the system provides a detailed explanation of its findings. Instead of offering only a label, it describes the reasoning behind the prediction.

For example, the system may indicate:

- Presence of microaneurysms in specific regions
- Detection of haemorrhages indicating vascular damage
- Identification of exudates suggesting fluid leakage

This explanation converts technical outputs into domain-relevant insights. It supports clinicians in validating results and ensures that the system remains transparent rather than opaque.

5. Treatment Recommendations

Based on the diagnosis and severity, the system suggests appropriate medical actions. These recommendations are not replacements for professional judgment but serve as supportive guidance. Examples include:

- Routine monitoring for early-stage detection
- Lifestyle and blood sugar control advice
- Referral to an ophthalmologist for advanced stages
- Immediate intervention for severe or proliferative cases

Once analysis and recommendations are complete, the system compiles all results into a structured format. This includes image analysis, severity level, diagnosis, explanations, and suggested actions.

In critical cases, the system generates alerts. These alerts ensure that high-risk patients receive immediate attention. Notifications can be designed to reach healthcare providers, administrators, or patients depending on system configuration.

6. Report Generation and Alert Notifications

The final stage produces a comprehensive report that can be stored, printed, or digitally shared. The report typically includes:

- Patient and image details

- Visual analysis results
- Diagnostic conclusion
- Confidence score
- Explanation of detected features
- Recommended actions

5.3 UML DIAGRAMS

UML diagrams provide a systematic and visual approach to representing the design, structure, and behavior of a software system. They serve as a blueprint that captures both the static architecture and the dynamic interactions within the system. In the context of the diabetic retinopathy detection project, UML diagrams play a crucial role in presenting a well-organized view of the entire solution. They help in mapping out how different components are structured and how they collaborate to perform tasks such as image processing and disease prediction. By using these diagrams, the system design becomes more transparent, reducing ambiguity and improving communication among team members. On the behavioral side, UML diagrams depict the sequence of operations that occur from the moment a user uploads an image to the final generation of results. They show how data moves through various stages, including preprocessing, feature extraction, classification, and report generation. This flow-oriented representation helps in understanding the logical progression of tasks and ensures that each step is properly defined and connected.

5.3.1 USE CASE DIAGRAM

The use case diagram is helpful in this project because it clearly explains how the Diabetic Retinopathy Detection System is used by different users. It gives a simple view of how the system works from the user's side without going into technical details. By looking at the diagram, it becomes easy to understand who is using the system and what actions they perform. It also helps in showing the interaction between the patient, doctor, and admin with the system. This makes the overall process clearer, starting from image upload to result display. The diagram improves understanding of how data flows inside the system and how each user contributes to the process. In addition, the use case diagram supports better system planning and design. It helps developers identify all the required functions and ensures that

no important feature is missed. It also makes communication easier between team members, as everyone can understand the system quickly through the diagram.

Use Case Diagram – Diabetic Retinopathy Detection

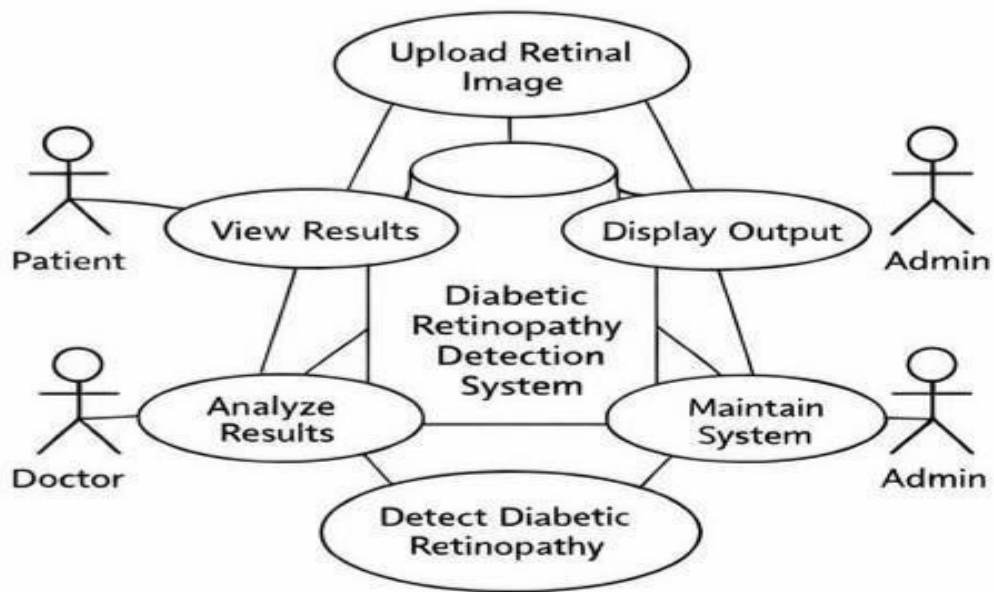


Fig 5.3.1: USE CASE

5.3.2ACTIVITY DIAGRAM

The activity diagram shows the step-by-step flow of the Diabetic Retinopathy Detection System from start to end. It explains how the process moves from one stage to another in a clear sequence.

The process begins with the start node, followed by the login step where the user enters the system. After successful login, the user uploads a retinal image, which is taken as the input for further processing. Once the image is uploaded, the system performs preprocessing to improve the quality of the image and prepare it for analysis.

After preprocessing, the system runs the detection model to examine the image. At this stage, a decision is made based on whether diabetic retinopathy is present or not. If the condition is detected, the system moves to classify the severity of the disease, indicating how serious the condition is. If no signs of the disease are found, the system displays the result as normal.

Finally, the process reaches the end node, completing the workflow. This diagram helps in understanding the overall working process of the system in a simple and organized manner, showing how each step is connected and how decisions are made during execution.

Activity Diagram-Diabetic Retinopathy Detection

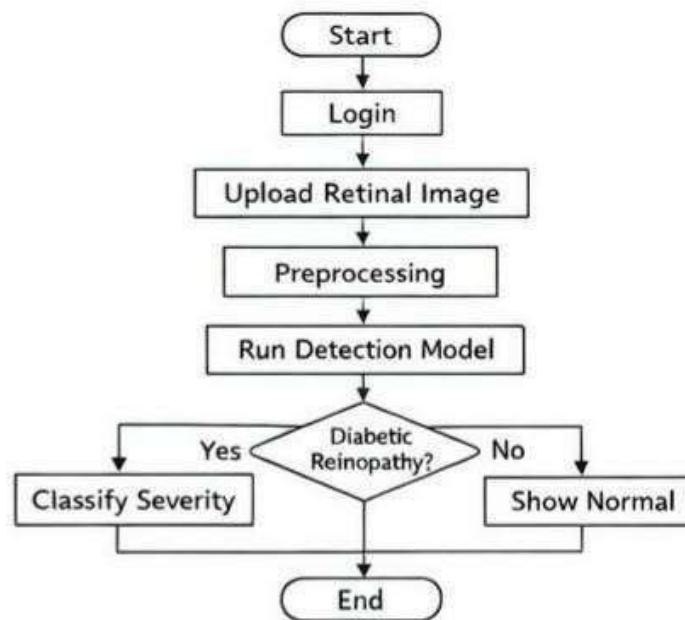


Fig 5.3.2: ACTIVITY DIAGRAM

5.3.3 DEPLOYMENT DIAGRAM

The deployment diagram for the diabetic retinopathy detection system represents how the software components and hardware nodes are organized and interact within the system. It shows the physical arrangement of servers, databases, user devices, and optional cloud storage, providing a clear view of how data and processing flow. In this system, doctors or medical staff use desktops, laptops, or mobile devices to upload retinal images, which are then sent to the application server. The application server manages communication between the users and the AI processing server, validating inputs and forwarding images for analysis. The AI processing server, equipped with highperformance computing resources, runs the deep learning model to detect diabetic retinopathy and determine the severity of the condition. Results from the analysis are returned to the application server and displayed on the user interface. All patient data, uploaded images, and diagnostic results are securely stored in the database server, ensuring safe retrieval, backup, and auditing. Optionally, cloud storage can be used to handle large volumes of data, enabling scalability and remote access. This diagram is useful in understanding the system's infrastructure, allocating computing resources efficiently, identifying potential bottlenecks, and ensuring secure and effective data management. It provides a comprehensive overview of how different components collaborate to deliver accurate and timely diagnostic support in diabetic retinopathy detection.

Deployment Diagram – Diabetic Retinopathy Detection

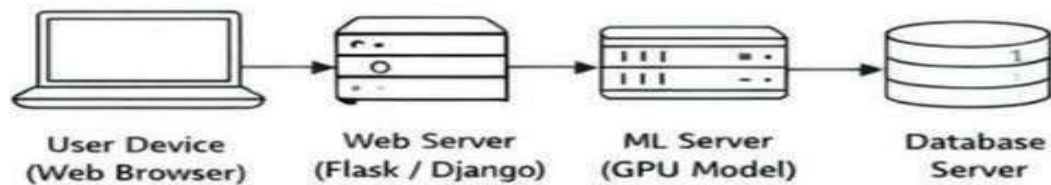


Fig 5.3.3: DEPLOYMENT DIAGRAM

5.3.4 CLASS DIAGRAM

The class diagram for the diabetic retinopathy detection system provides a structured view of the system's main components, showing how different classes interact and the relationships between them. It defines the attributes and methods of each class, helping in understanding the system design before implementation. In this project, the system consists of classes representing users, retinal images, AI models, and diagnostic reports. The user class handles details such as login credentials, roles, and access permissions, ensuring that only authorized medical staff can use the system. The retinal image class manages image storage, retrieval, and preprocessing steps required for analysis. The AI model class encapsulates the trained deep learning model and includes methods for feature extraction, classification, and prediction of diabetic retinopathy severity. The diagnostic report class stores the results, recommendations, and timestamps, linking them back to the respective user and image. Relationships between these classes, such as associations, dependencies, and compositions, illustrate how data flows from image upload to diagnosis and reporting. This diagram helps developers understand the system architecture clearly, supports better coding practices, and

ensures that all components work together efficiently to provide accurate and reliable detection results.

Class Diagram- Diabetic Retinopathy Detection

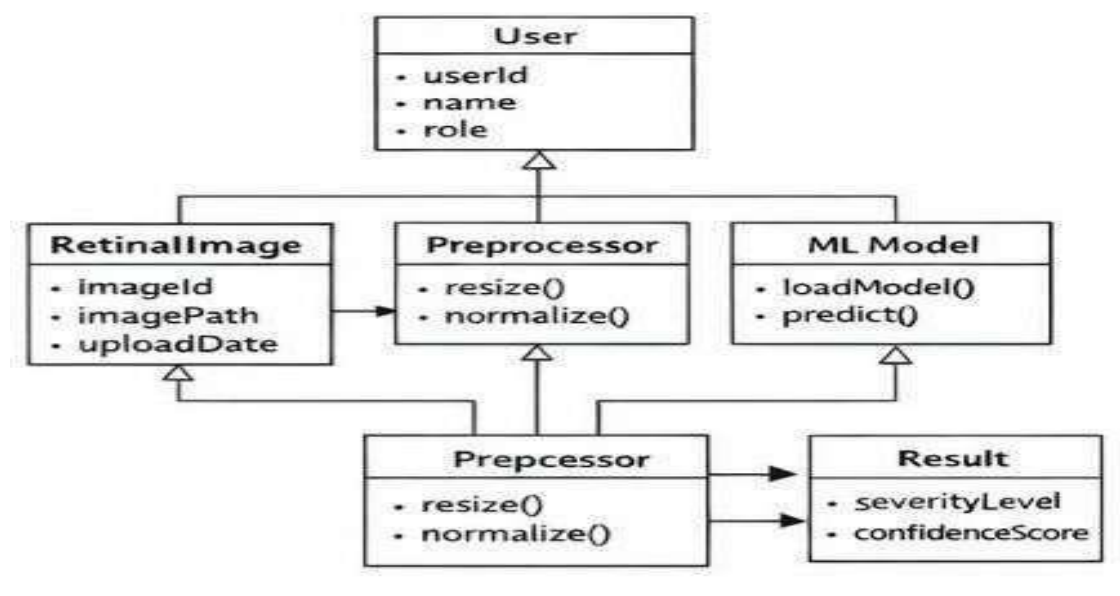


Fig 5.3.4: CLASS DIAGRAM

5.3.5 SEQUENCE DIAGRAM

The sequence diagram for the diabetic retinopathy detection system illustrates the step-by-step flow of interactions between the system's components over time. It helps to visualize how data moves from one part of the system to another and clarifies the order of operations. In this project, the sequence begins when a doctor logs into the system and uploads a retinal image for analysis. The user interface sends the image to the application server, which validates the input and forwards it to the AI processing server. The AI model then processes the image, performing tasks such as preprocessing, feature extraction, and classification to determine the presence and severity of diabetic retinopathy. Once the analysis is complete, the AI server returns the results to the application server, which generates a diagnostic report. This report is then delivered back to the user interface for the doctor to review. Throughout this process, interactions with the database occur to store the uploaded images, user details, and diagnostic results securely. The sequence diagram is valuable in this project because it provides a clear view of the system's operational workflow, ensures that all components

communicate correctly, and helps identify potential delays or inefficiencies in the processing sequence, ultimately contributing to the system's reliability and accuracy.

Sequence Diagram-Diabetic Retinopathy Detection

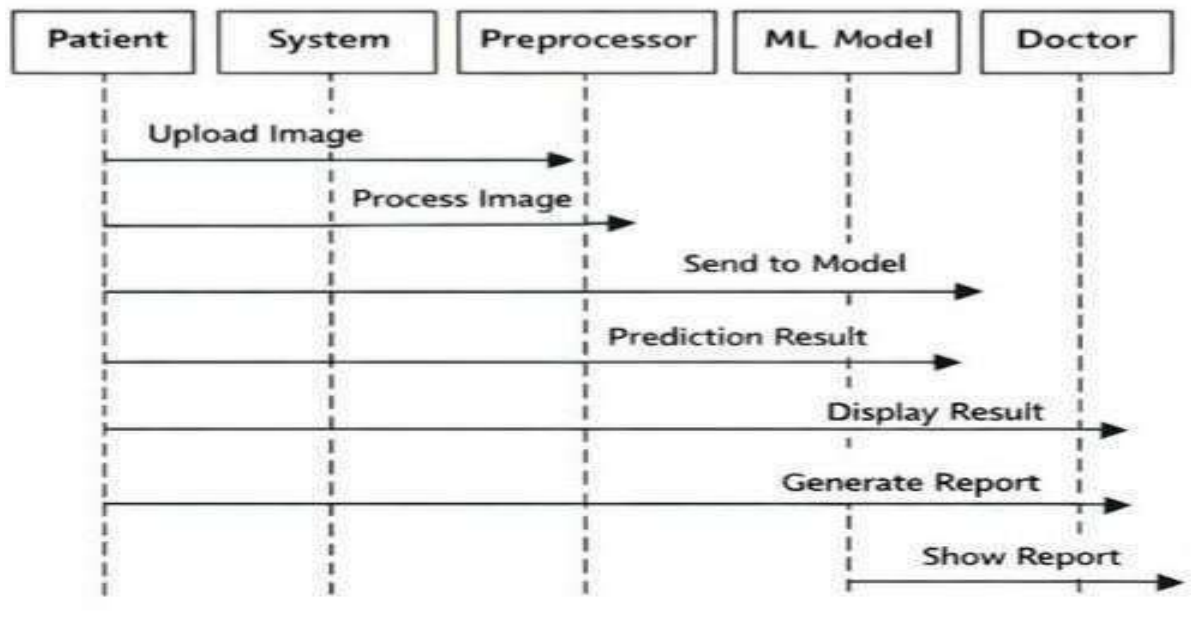


Fig 5.3.5: SEQUENCE DIAGRAM

5.3.6 COMPONENT DIAGRAM

The component diagram for the diabetic retinopathy detection system provides a high-level view of the software architecture, showing how the system is divided into modular components and how they interact with each other. It represents the organization and dependencies of the system, helping developers and stakeholders understand the structure and responsibilities of each module. In this project, the main components include the user interface, the AI processing engine, the database management system, and the communication or API services. The user interface component allows doctors to log in, upload retinal images, and view diagnostic results, serving as the point of interaction with the system. The AI processing engine handles the analysis of retinal images, executing preprocessing, feature extraction, and classification tasks to detect diabetic retinopathy. The database management component is responsible for storing patient details, uploaded images,

and diagnostic reports securely, ensuring data integrity and accessibility. The API services component facilitates communication between the front-end and back-end, enabling smooth data exchange and coordinating requests between the user interface, AI engine, and database. Component Diagram-Diabetic Retinopathy Detection

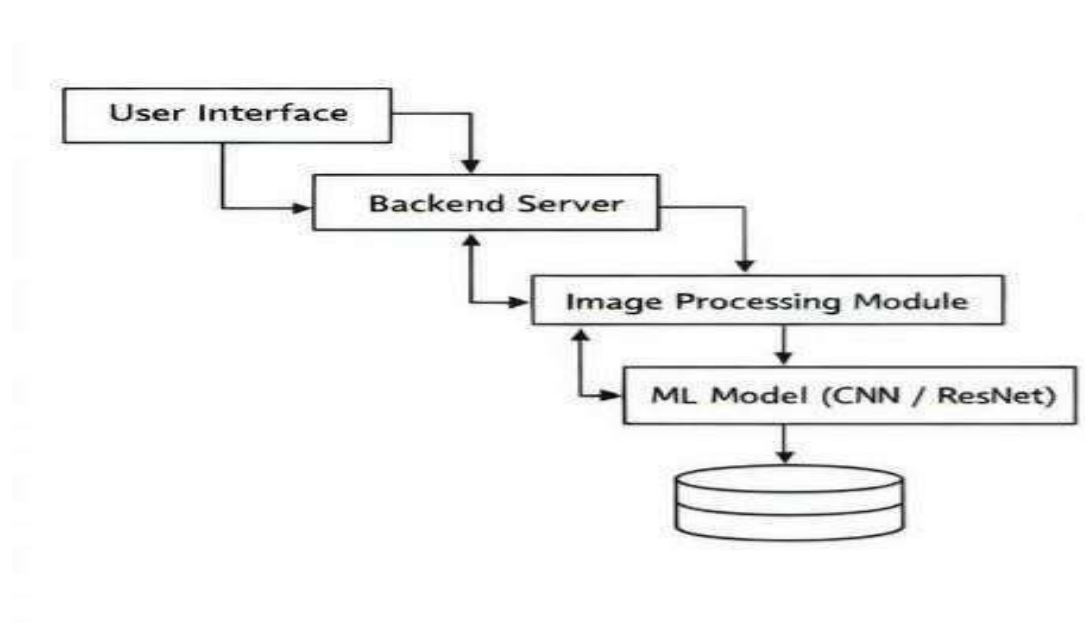


Fig 5.3.6: COMPONENT DIAGRAM

5.3.7 STATE DIAGRAM

The state diagram for the diabetic retinopathy detection system illustrates the different states that the system or its main components can be in and how it transitions between these states based on user actions or internal processing events. It helps in understanding the dynamic behavior of the system over time. In this project, the system begins in an idle state, waiting for a user to log in. Once a doctor logs in successfully, the system transitions to a ready state, where it can accept retinal image uploads. After an image is uploaded, the system enters the processing state, where the AI engine analyzes the image to detect diabetic retinopathy. During processing, the system may perform intermediate steps such as image validation, preprocessing, and feature extraction. After the analysis is complete, the system moves to the result state, where the diagnostic report is generated and presented to the doctor. Finally, the system can either return to the ready state to process another image or move to the logout state when the user exits. This diagram is valuable for the project because it provides a clear view of the system's operational flow, ensures that all possible states and transitions are accounted for, and helps developers design the system logic accurately, reducing errors and improving reliability.

State Diagram-Diabetic Retinopathy Detection

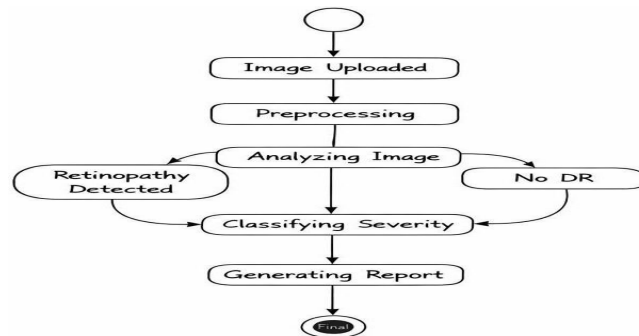


Fig 5.3.7: STATE DIAGRAM

5.3.8 OBJECT DIAGRAM

The object diagram for the diabetic retinopathy detection system represents a snapshot of the system at a particular moment, showing specific instances of classes and how they are related. Unlike the class diagram, which focuses on general structure, the object diagram illustrates actual objects, their attributes, and the links between them during runtime. In this project, objects include a doctor instance, a retinal image instance, an AI model instance, and a diagnostic report instance. The doctor object contains information such as user ID, login credentials, and role, while the retinal image object holds the uploaded image data and its metadata. The AI model object represents the processing engine performing image analysis, including methods for classification and feature extraction. The diagnostic report object stores the results of the analysis, such as severity level, recommendations, and timestamps. Links between these objects indicate how a doctor uploads an image, the AI model processes it, and a report is generated and associated back to the doctor and image. This diagram is useful in the project as it helps visualize real-time interactions, understand object states during execution, and verify that the relationships and data flow between objects function as intended, supporting accurate and efficient detection of diabetic retinopathy.

Object Diagram-Diabetic Retinopathy Detection

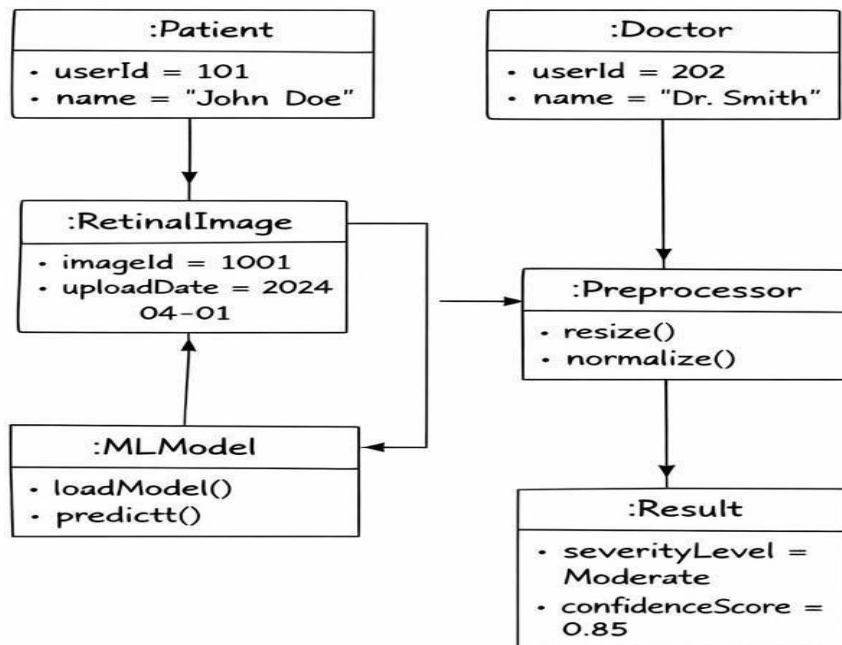


Fig 5.3.8: OBJECT DIAGRAM

CHAPTER 6

SYSTEM TESTING

System testing is a crucial phase in the development of the diabetic retinopathy detection system, ensuring that all components work together as intended and the system meets its specified requirements. This testing evaluates the complete application in a controlled environment to verify functionality, performance, reliability, and security. In this project, system testing involves checking whether users can log in successfully, upload retinal images, and receive accurate diagnostic results.

System testing includes simulating different scenarios, such as uploading invalid file types, testing with high volumes of images, and examining the system's response to network interruptions. This process is important because it identifies any integration issues, performance bottlenecks, or security vulnerabilities before the system is deployed. By thoroughly validating the entire application, system testing increases confidence in the system's accuracy, stability, and readiness for use in real-world medical environments.

The main purpose of system testing is to ensure that the entire diabetic retinopathy detection system works correctly as a complete, integrated application and meets the specified requirements. Unlike testing individual modules, system testing evaluates the overall functionality, performance, reliability, and security of the system. It verifies that all components—such as the user interface, AI processing engine, and database—work together smoothly, that patient data is handled safely, and that the system produces accurate diagnostic results. Essentially, system testing confirms that the system is ready for real-world use and can perform its intended tasks reliably under expected conditions.

6.1 TYPES OF TESTING

Types of testing are carried out to ensure that a system functions correctly, efficiently, and securely. For the diabetic retinopathy detection system, several testing methods are applied to

verify both the functionality and reliability of the system. Unit testing is performed first, where individual modules, such as the user login interface, image upload functionality, and AI processing module, are tested independently to ensure each part works as intended. Integration testing follows, which examines how these modules interact with each other, making sure that data flows smoothly between the user interface, AI engine, and database without errors. System testing evaluates the entire application as a whole, checking that all components function together and that the system meets the specified requirements for accuracy, speed, and security. Performance testing measures how the system behaves under different conditions, such as handling multiple image uploads simultaneously or processing large datasets, ensuring that response times remain acceptable. Security testing is conducted to confirm that sensitive patient data, including images and reports, is protected from unauthorized access or breaches.

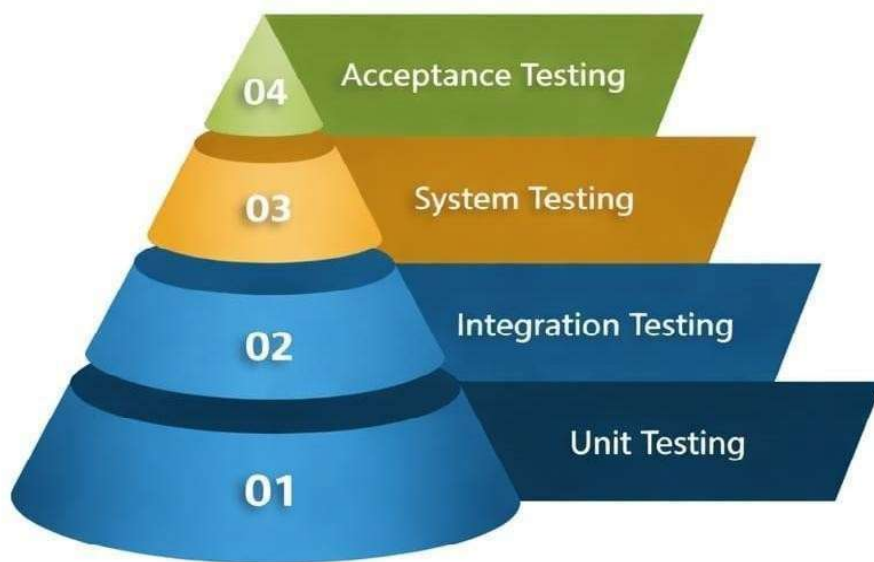


Fig 6.1: TYPES OF SOFTWARE TESTING

6.1.1 UNIT TESTING

Unit testing is the process of testing individual modules of the diabetic retinopathy detection system separately to ensure each one works correctly. This helps identify and fix errors early in the development phase before combining the modules into a complete system.

The main purpose of unit testing is to verify that each component performs its intended function accurately. For this project, it ensures that the login module, image upload module, AI processing module, and report generation module all operate without errors. This prevents issues from spreading when the system is fully integrated.

1. Image Input Module Testing

This test checks whether the system correctly accepts and processes retinal fundus images as input. The model is tested with different image formats and resolutions to ensure compatibility and proper loading.

2. Image Preprocessing Testing

This test verifies that the preprocessing steps such as resizing, normalization, noise removal, and contrast enhancement are performed correctly before feeding the data into the model.

3. Feature Extraction Module Testing (ANN)

This test ensures that the Artificial Neural Network (ANN) correctly extracts relevant features from retinal images. The output features are validated for accuracy and consistency.

4. Sequence Learning Module Testing (RNN)

This test verifies that the Recurrent Neural Network (RNN) correctly processes sequential patterns or dependencies in extracted features to improve prediction accuracy.

5. Dual-Stage Hybrid Model Testing (ANN + RNN)

This test checks whether the integration of ANN and RNN works effectively as a hybrid model. It ensures smooth data flow between both stages and accurate combined predictions.

6. Disease Classification Testing

This test verifies that the system correctly classifies diabetic retinopathy stages (e.g., Mild, Moderate, Severe). The predicted output is compared with actual medical labels.

7. Data Processing Module Testing

This test ensures that all image data is processed correctly throughout the pipeline, including preprocessing, feature extraction, and classification stages.

8. Alert/Result Generation Testing

This test verifies that the system generates correct outputs or alerts based on detection results. It ensures proper display of results to users or healthcare professionals.

9. Data Storage Testing

This test ensures that patient data, images, and prediction results are stored correctly in the database for future analysis and tracking.

10. Integration Testing

Integration testing is performed to verify that all modules of the dual-stage ANN-RNN hybrid system work together correctly. It ensures smooth communication between image input, preprocessing, model prediction, and storage modules.

11. Performance Testing

This test evaluates the system's accuracy, precision, recall, and processing time to ensure reliable and efficient detection of diabetic retinopathy.

6.1.2 INTEGRATION TESTING

Integration testing is performed to ensure that different modules of the diabetic retinopathy detection system work together correctly. Unlike unit testing, which focuses on individual components, integration testing verifies the interaction between multiple modules and checks that data flows accurately across the system.

The main purpose of integration testing is to detect errors that may occur when individual modules are combined. In this project, it ensures that the user interface, AI processing engine, database, and API services function together seamlessly, and that the system can correctly handle image uploads, processing, and report generation without any issues.

1. Input and Preprocessing Integration Testing

This test verifies that retinal images captured or uploaded are properly passed to the preprocessing module. It ensures that operations like resizing, normalization, and enhancement are correctly applied before forwarding the data to the next stage.

2. Preprocessing and ANN Integration Testing

This test checks whether the pre-processed images are correctly fed into the ANN module. It ensures that the feature extraction process receives accurate and properly formatted input data.

3. ANN and RNN Integration Testing

This test ensures smooth data transfer between the ANN and RNN modules. The features extracted by the ANN are validated to confirm they are correctly interpreted by the RNN for sequential learning and improved prediction.

4. Hybrid Model and Classification Integration Testing

This test verifies that the combined ANN-RNN model produces accurate classification results. It ensures that the output of the hybrid model is correctly used for detecting and classifying diabetic retinopathy stages.

5. Classification and Output Generation Integration Testing

This test checks whether the classification results are correctly displayed to the user. It ensures proper formatting of outputs such as disease stage, probability scores, and diagnostic messages.

6. Output and Data Storage Integration Testing

This test verifies that the results generated by the system are properly stored in the database. It ensures that patient details, images, and prediction results are saved without data loss

7. End-to-End System Integration Testing

This test evaluates the complete workflow of the system from image input to final output. It ensures that all modules interact correctly and the system produces accurate and reliable results when functioning as a whole.

8. Error Handling Integration Testing

This test ensures that the system properly handles errors during module interaction, such as invalid image inputs or processing failures, without crashing.

6.1.3 FUNCTIONAL TESTING

1. Image Upload Function Testing

This testing verifies that the system allows users to upload retinal images without errors. It checks whether different image formats are accepted and whether invalid files are rejected properly.

2. Image Preprocessing Function Testing

This test ensures that the preprocessing functions such as resizing, normalization, and noise removal are working correctly. It confirms that images are properly prepared before being given to the model.

3. Feature Extraction Function Testing (ANN)

This testing checks whether the ANN module correctly extracts important features from retinal images. The output features are observed to ensure they are meaningful for further processing.

4. Sequence Processing Function Testing (RNN)

This test verifies that the RNN module correctly processes the features received from the ANN. It ensures that the model captures patterns effectively for better prediction.

5. Classification Function Testing

This testing checks whether the system correctly identifies the stage of diabetic retinopathy. The output is compared with expected results to confirm correctness.

6. Result Display Function Testing

This test ensures that the prediction results are shown clearly to the user. It checks whether the output includes proper labels, stages, and relevant information.

7. Data Storage Function Testing

This testing verifies that the system stores images, patient details, and results correctly in the database without any data loss.

8. User Interaction Function Testing

This test checks whether users can easily interact with the system, including uploading images and viewing results, without facing any issues.

9. Error Handling Function Testing

This testing ensures that the system responds properly when errors occur, such as invalid inputs or processing failures, and displays appropriate messages.

6.1.4 SYSTEM TESTING

System testing is carried out to check the complete working of the diabetic retinopathy detection system after combining all the modules. At this stage, the system is tested as a whole to make sure everything works together without any issues.

The goal is to ensure that the application performs correctly and provides accurate results when used in real conditions.

The main purpose of system testing is to examine the overall performance, stability, and reliability of the system. It helps in confirming that the system can correctly process retinal images and generate proper detection results using the trained model.

Objectives of System Testing

- To check the complete flow of the system from image upload to final output
- To ensure that the detection model works properly with the application
- To verify that the results produced by the system are accurate
- To test the system with different types of inputs
- To make sure the system runs smoothly without failure

Components Tested

- Login functionality
- Retinal image upload process

- Image preprocessing stage
- Detection model operation
- Result display section

Tools and Technologies Used

- Python is used for developing the system
- TensorFlow or Kera's is used to run the model
- OpenCV is used for image processing
- NumPy and Pandas are used for handling data
- Matplotlib is used for visual representation

6.1.5 WHITE BOX TESTING

White box testing is used in this project to examine the internal working of the Diabetic Retinopathy Detection System. Instead of only checking the output, this type of testing focuses on how the code is written and how each part of the program functions internally.

In this diabetic retinopathy detection, white box testing is applied mainly during the development stage. The developers check individual functions such as image preprocessing, model execution, and result generation by going through the code. Each step is tested to confirm that the data is moving correctly from one stage to another and that there are no logical errors in the program.

The preprocessing code is tested to ensure that the uploaded retinal image is properly resized, filtered, and prepared before it is given to the model. Similarly, the model-related code is checked to confirm that it loads correctly and produces predictions without errors. The result handling part is also tested to make sure the output is generated and displayed in the correct format.

White box testing also helps in identifying issues like incorrect conditions, missing steps, or unnecessary code. By testing different paths in the program, developers can ensure that all

possible cases are handled properly. This improves the overall quality of the system and reduces the chances of failure during execution.

white box testing plays an important role in this project by ensuring that the internal logic of the system is accurate, efficient, and reliable, which ultimately leads to better performance of the entire application.

6.1.6 BLACK BOX TESTING

The black box testing is done to check how the Diabetic Retinopathy Detection System behaves from a user's side. The main idea here is simple — we give input to the system and observe the output, without worrying about how the system is working internally.

While testing, different retinal images are uploaded into the system to see whether it gives correct results. The focus is on checking if the system can properly detect diabetic retinopathy and show the output clearly. It does not involve looking at the code or understanding the internal logic.

This type of testing is also used to check basic features like login, file upload, and result display. For example, it is verified whether the system allows only proper image formats and whether it shows an error message when an invalid file is uploaded. It is also checked if the results are easy to understand for the user.

To make the system more reliable, various kinds of inputs are tested. This includes normal inputs, incorrect inputs, and unusual cases. By doing this, it becomes easier to see how the system reacts in different situations and whether it can handle them without any issues.

6.2 TEST CASES

Test Cases Table

Test Case ID	Test Description	Input	Expected Output	Actual Output	Status
TC_01	Verify login with valid credentials	Correct username & password	User should login successfully	User logged in successfully	Pass
TC_02	Verify login with invalid credentials	Incorrect username/pasword	Error message should be displayed	Displayed error message: "Invalid Credentials"	Pass
TC_03	Check image upload with valid file	Valid retinal image (JPG/PNG)	Image should be uploaded successfully	Image uploaded successfully	Pass
TC_04	Check image upload with invalid file	Non-image file (PDF/Text)	System should show error message	Displayed error message: "Invalid File Format"	Pass
TC_05	Verify system response after image upload	Valid retinal image	Valid retinal image	Detection process started	Pass
TC_06	Check preprocessing stage	Uploaded image	Image should processing	Image resized and. successfully	Pass
TC_07	Verify detection model execution	Preprocessed image	System should detect cond	Condition detected without errors	Pass
TC_08	Check result for DR detected	Image with DR	System should display "DR Detected"	Displayed "DR Detected" as result	Pass
TC_09	Check result for normal case	Image without DR	System should, "IsoR"	Displayed "No DR"	Pass
TC_10	Verify result display	Processed image	Output should be shown cl	Result shown clearly to the user	Pass
TC_11	Check system with large image	High-resolution image	Large image processed without crash		Pass
TC_12	Check system with empty input	No image uploaded	Displayed message: "Please	Displayed message: "Please upload an image"	Pass

CHAPTER 7

SYSTEM IMPLEMENTATION

system implementation is the stage where the idea of detecting diabetic retinopathy is actually put into practice. It is not just about designing the system, but making sure that everything works together as a real application that users can use.

The implementation is used to connect all parts of the system, including the user interface, the processing steps, and the prediction model. When a user logs into the system and uploads a retinal image, the implemented system takes that input and begins processing it step by step. Without proper implementation, these steps would remain only as theoretical design.

In this project implementation helps in handling the complete flow of data. The uploaded image is received by the backend, processed, and then passed to the trained model. The model analyses the image and produces a result, which is then sent back and displayed to the user. This shows how the system works in a real-time environment.

It is also used to ensure that the system responds properly to different situations. For example, if a user uploads an incorrect file or does not provide input, the system handles it by showing an appropriate message. This improves the usability of the application. Another important use of implementation in this project is integrating all technologies together. Different tools and libraries are combined to make sure the system runs smoothly and efficiently.

DATASET DESCRIPTION

The dataset used in this project consists of retinal images that are used to detect diabetic retinopathy. These images are collected from publicly available medical sources and contain both normal and affected eye conditions. The dataset plays an important role in training and testing the model so that it can correctly identify the disease.

Each image in the dataset represents the retina of a patient's eye. The images are labeled based on the presence or severity of diabetic retinopathy. Some images show no signs of the

disease, while others indicate different levels of severity. This variation helps the model learn to distinguish between normal and abnormal conditions.

Before using the dataset, it is cleaned and organized properly. Unnecessary or low-quality images are removed to improve the accuracy of the system. The images are also resized and formatted so that they can be processed easily by the model. These preparation steps help in improving the performance of the system.

The dataset is divided into different parts for training and testing. The training data is used to teach the model how to recognize patterns in retinal images, while the testing data is used to evaluate how well the model performs on new, unseen images. the dataset provides the necessary information for the system to learn and make accurate predictions. A well-prepared dataset helps in improving the reliability and effectiveness of the diabetic retinopathy detection system.

In this project,the dataset is used to train,test and valid the hybrid model to get good accuracy and precision.The dataset is cleaned first and preprocessed to remove the noise and ensure the consistence in the data.After preprocessing,The dataset is divided into training and testing subsets to train the hybrid model and evaluate its performance.

A clinician has rated each image for the severity of diabetic retinopathy on a scale of 0 to 4:

- 0-No DR
- 1-Mild
- 2-Moderate
- 3-Severe
- 4-Proliferative

DR Files:

- Train.csv-the training labels
- Test.csv-The test set(it helps to predict the diagnosis value for these variables)
- Sample_submission.csv-a sample submission file in the correct format
- Train.zip-the training set images
- Test.zip-the public test set images Submissions should be formatted like:
Id_core,diagnosis

0005cfc8afb6,0

003f0afdcd15,0

Etc.....

	A	B
1	id_code	diagnosis
2	1ae8c165f	2
3	1b329a12	1
4	1b32e1d7	4
5	1b364786	0
6	1b398c04	0
7	1b462587	2
8	1b495ac0	3
9	1b862fb6f	0
10	1b870123	0

Fig 7 :Dataset Description

Setting Up the Development Environment

Step 1:Install Python

1.Visit the official Python Website.

Download Python 3.11.1 or above (python 3.11.1 recommended).

During installation,select “Add to Python to PATH”. Complete the installation process.

Step 2: Checking pip Installation

After Python installation, pip is checked to ensure package installation can be done.

Command Prompt showing pip version

```
□ pip --version
```

Step 3 : Create Virtual Environment

A virtual environment helps isolate project dependencies and ensures compatibility between different libraries.

Create a virtual environment using the following command:

```
□ Python -m venv .venv Activate the environment:
```

Windows

```
.venv\Scripts\Activate.ps1 macOS/Linux
```

```
source .venv/bin/activate
```

Step 4: Install Required Packages

Install the required python libraries used in project.

- Pip install torch torchvision fastapi uvicorn pillow openai numpy scikit-learn

These libraries support data processing, machine learning model development and visualization.

Step 5 : Backend Development

The backend system of the proposed dual-stage ANN–RNN hybrid model demonstrates strong efficiency and reliability in processing retinal images and delivering accurate diagnostic results. The architecture is designed to balance computational performance with predictive accuracy, ensuring practical usability in real-world healthcare environments.

The backend performs the following tasks:

- The backend receives retinal images uploaded by the user through the interface or API and securely manages the input data.
- It validates the uploaded files by checking format, size, and integrity to ensure only valid retinal images are processed.
- The system performs preprocessing operations such as resizing, normalization, and noise reduction to prepare the image for analysis.
- Image quality is enhanced to reduce the impact of poor lighting and contrast variations, improving consistency across inputs.
- The processed image is passed to the first stage of the model, where the artificial neural network extracts meaningful spatial features from the retinal structure.
- These extracted features are converted into a structured representation that can be used for further analysis.
- The backend then feeds the feature representation into the second stage, where the recurrent neural network analyzes patterns and relationships within the features.
- Based on this analysis, the model determines the severity level of diabetic retinopathy.
- The backend converts the model output into a human-readable format, assigning the predicted class label to a specific disease stage.

- It prepares and sends the prediction result back to the user interface in the form of a structured response (such as JSON).
- The system logs prediction results and processing details for monitoring and future reference.
- It supports handling multiple requests, allowing simultaneous processing of images in a scalable manner.
- The backend ensures smooth communication between the frontend interface and the deep learning model.
- It maintains reliability by handling errors such as invalid inputs, processing failures, or model exceptions.
- The design allows easy updates or replacement of model components without affecting the overall system workflow.

The backend system ensured that the model executes efficiently and produces the results quickly with the help of the llm, fastapi and uvicorn.

Step 6: Model Training and integration

The model training and integration process is designed to create a seamless connection between learned intelligence and practical application. Initially, the retinal image dataset is prepared through careful preprocessing, including resizing, normalization, and enhancement to ensure uniform input quality. The first stage of the model, based on an artificial neural network, is trained to capture spatial features such as blood vessel patterns and lesions. These extracted features are then utilized by the second stage, where a recurrent neural network is trained to interpret patterns and relationships within the feature space, improving the system's ability to differentiate between varying severity levels of diabetic retinopathy. Once training is completed, the model is saved and integrated into the backend environment, where it is loaded during runtime for inference. This integration ensures that the trained model can efficiently process incoming images and generate predictions without retraining, enabling smooth deployment and real-time usage.

The model training process includes:

- The backend accepts retinal image inputs from users through an interface or API endpoint.
- It verifies the uploaded data to ensure proper format and usability before processing.

- Preprocessing operations are applied to standardize the input images for consistent analysis.
- The system extracts features from the images using the trained artificial neural network.
- Extracted features are passed to the recurrent neural network for deeper pattern analysis.
- The model predicts the stage of diabetic retinopathy based on the processed data.
- The prediction output is converted into a readable format for end users.
- Results are transmitted back to the frontend as a structured response.
- The backend logs system activity and prediction outcomes for tracking and evaluation.
- Error handling mechanisms are implemented to manage invalid inputs or processing failures.
- The system supports handling multiple user requests, enabling scalability.
- The architecture allows easy updates or replacement of models without disrupting the workflow.

After training, the model is saved and integrated into the prediction system so that it can generate predictions for new patient data.

The trained model achieves high detection of accuracy and effectively identifies the patterns associated with diabetic retinopathy.

Step 7: User Interface Development

The user interface is designed to provide a secure and straightforward interaction for accessing the diabetic retinopathy detection system. Initially, the user must log in using a valid email address and password, ensuring controlled access and data privacy. After successful authentication, the user is presented with an interface that allows uploading a retinal image from the local device. Once the image is selected, the user can initiate the analysis by clicking the “Analyze” button, which sends the image to the backend for processing. After the model completes its analysis, the predicted result, including the stage of diabetic retinopathy, is displayed clearly on the screen for the user to review.

- The login module verifies user credentials before granting access to the system.
- Input validation is applied to ensure the email format and password fields are correctly entered.
- The image upload section supports retinal image selection in standard formats.
- The interface provides guidance to help users upload appropriate medical images.

- A trigger button (Analyze) is used to initiate the processing of the uploaded image.
- A loading or processing indicator is displayed while the system analyzes the image.
- The backend communicates with the interface to send prediction results.
- The output is displayed in a clear and understandable format for the user.
- Error messages are shown for invalid inputs or unsuccessful operations.
- The interface is designed to be simple, responsive, and user-friendly.

Step 8 : Running the Application

To run the Diabetic retinopathy detection system:

1. Start the python application server.
2. Open the application in a web browser.
3. Login to the system.
4. Enter the retinal image of the patient.
5. Submit the data for detection.
6. The system processes the input and displays the detection result.

Data input and Preprocessing Module

The data input and preprocessing module forms the initial stage of the system, where retinal fundus images are collected and prepared for analysis. The process begins when the user uploads a retinal image through the interface, which is then received by the backend system for further handling. Since raw medical images often vary in size, quality, and illumination, preprocessing is essential to ensure consistency and reliability in model performance. The input images are first resized to a standard resolution so that they match the input requirements of the neural network. Following this, noise removal techniques are applied to eliminate unwanted distortions that may affect feature detection. The images are then normalized to bring pixel intensity values into a uniform range, improving numerical stability during model inference. Additionally, patch extraction is performed to focus on relevant regions of the retina, allowing the model to better capture fine-grained details such as lesions and vessel structures. This structured preprocessing pipeline ensures that the data fed into the feature extraction stage is clean, consistent, and optimized for accurate diabetic retinopathy detection.

Model Loading and Prediction Module

The trained ANN and RNN models are loaded into the backend during system initialization, ensuring readiness for inference without retraining, and enabling efficient integration with the deployed FastAPI-based application environment.

The preprocessed retinal image is passed through the ANN to extract spatial features, which are then sequentially processed by the RNN to capture patterns and generate a refined representation.

The final output from the RNN is evaluated using a classification layer to determine the diabetic retinopathy stage, and the result is formatted and returned to the user interface.

Output Result Module

The prediction result is displayed to the user in a clear and understandable format. The output includes:

- Prediction
- Confidence
- Reasons and symptoms
- Recommendations
- Treatment

Deployment and Access

During development and testing, the system runs on a local server. After testing, the systems can be deployed on 8000 port for the backend but the front end will be showed in browser. Deployment allows the healthcare professionals to access the system through web browsers and use it for the detection of diabetic retinopathy with the help of large language model using openai.

The system architecture ensures the application remains stable, scalable and efficient when processing medical data.

Software Installation for Diabetic Retinopathy Detection

Software Requirements

- Operating System: Windows, Linux, or macOS.
- Programming Language: Python (version 3.8 or above).
- Backend Framework: FastAPI for handling API requests.
- Deep Learning Libraries: TensorFlow or PyTorch.
- Image Processing Library: OpenCV.
- Development Environment: Visual Studio Code or Jupyter Notebook.

- Web Browser: Google Chrome, Microsoft Edge, or any modern browser for user interface interaction.

Install Required Libraries

Create a requirements.txt file containing the following libraries: torch torchvision fastapi
uvicorn pillow openai numpy
scikit-learn

Install dependencies using:

□ Pip install -r requirements.txt

Project Structure Setup

The project directory contains the following

Diabetic Retinopathy Detection System/

```
|
|— data/
|   |— raw/
|   |— processed/
|   |— labels/
|
|— src/
|   |— preprocessing/
|   |— model/
|   |— training/
|   |— utils/
|
|— models/
|   |— trained_model/
|   |— checkpoints/
|
|— results/
|   |— graphs/
|   |— outputs/
|   |— evaluation/
```

```

|
|— tests/
|
|— docs/
|
|— requirements.txt
|— README.md
|— main.py

```

Run the backend server

- Create the virtual environment for the project file.
- Write the code by creating the files in foldername.py format.
- And also create the main.py file which has the main endpoints of the fastapi.
- Now save the file in vscode.
- Now give the command as:
 - `uvicorn main:app --reload`
- it opens the link in browser like:
 - <http://127.0.0.1:8000>
- Open it and post the images as a input and detect the diabetic retinopathy.

Explanation of the Main Components

Backend

- Contains the vscode used for the dataset,preprocessing,model training,evaluation and exporting the trained model.

Dataset

- Stores the diabetic retinopathy dataset used to train and test hybrid model.

Frontend

- Contains static files (CSS,images) and HTML templates used for web interface.

Web_apps

- Contains the FastAPI framework.

Evaluation Metrics and Formulae

Confusion Matrix Components

The above metrics are calculated using four basic elements from the confusion matrix:

- True Positive(TP):Diabetic retinopathy images correctly predicted as diseased
- True Negative(TN):Normal retinal images correctly predicted as normal
- False Positive(FP):Normal images incorrectly predicted as diseased
- False Negative(FN):Diseased images incorrectly predicted as normal We are using the formulas like:

1. **Precision** : It measures how many of the images predicted as diabetic retinopathy are actually correct.It indicated the reliability of the model when it predicts the presence of DR in a retinal image.A high precision value means the model produces fewer false positive predictions.

$$\text{Precision} = \text{TP}/(\text{TP} + \text{FP})$$

2. **Recall** : Recall measures how many actual diabetic retinopathy cases present in the dataset are correctly detected by the model.High recall indicates that the system is capable of identifying most of the diseased retinal images,which is very important in medical diagnosis.

$$\text{Recall} = \text{TP}/(\text{TP} + \text{FN})$$

3. **Accuracy** : It is one of the basic evaluation metrics used to measure the overall correctness detected by the model predictions.It represents the ratio of correctly classified retinal images (both normal and diseased) to the total number of predictions made by the model.

$$\text{Accuracy} = (\text{TP} + \text{TN})/(\text{TP} + \text{TN} + \text{FP} + \text{FN})$$

7.1 CONCLUSION AND FUTURE ENHANCEMENT

The developed dual-stage ANN–RNN hybrid system for automated diabetic retinopathy detection provides a well-organized and efficient solution for analyzing retinal fundus images. The design of the system is centered on combining two complementary learning approaches, where the artificial neural network is responsible for extracting spatial features, and the recurrent neural network refines these features by capturing underlying patterns and relationships. This layered approach enhances the model’s ability to differentiate between

subtle variations in disease severity, leading to more reliable classification outcomes. The backend plays a crucial role in ensuring that the entire pipeline functions smoothly. From receiving the input image to generating the final prediction, each stage is optimized to maintain consistency and reduce processing errors. The preprocessing module standardizes the input by addressing variations in image size, noise, and illumination. This step is essential because the quality of input data directly influences the performance of the model. By delivering well-prepared data to the feature extraction stage, the system improves its ability to detect important retinal abnormalities such as lesions, hemorrhages, and microaneurysms. Another significant aspect of the system is its ability to provide results within a short time frame. The optimized model architecture ensures that predictions are generated efficiently without compromising accuracy. This makes the system suitable for real-time or near real-time applications, particularly in scenarios where quick screening is required. The integration of the trained model into the backend eliminates the need for repeated training, allowing the system to focus entirely on inference and response generation.

The user interface further strengthens the practicality of the system by offering a simple and guided workflow. The login mechanism ensures secure access, while the image upload and analysis process is designed to be intuitive. Users can easily submit retinal images and obtain diagnostic results without requiring technical expertise. This ease of use increases the potential for adoption in both clinical and non-clinical settings.

In addition to functionality, the system demonstrates scalability and adaptability. Its modular design allows individual components, such as preprocessing techniques or model layers, to be updated without affecting the entire workflow. This flexibility is important for future improvements and integration with other healthcare systems.

Overall, the project successfully achieves a balance between performance, usability, and scalability. It highlights how hybrid deep learning models can be effectively applied in medical image analysis and offers a dependable tool for early detection of diabetic retinopathy. By assisting in timely diagnosis, the system has the potential to support healthcare professionals and contribute to better patient outcomes.

FUTURE ENHANCEMENT

Although the current system delivers promising results, there are several directions in which it can be enhanced to further improve its effectiveness and reach. One of the primary areas for improvement is the expansion of the training dataset. Incorporating a larger and more diverse collection of retinal images, including data from different demographics and imaging devices, can significantly enhance the model's generalization capability. This would reduce the likelihood of biased predictions and improve performance across varied real-world conditions.

Another important enhancement involves the adoption of more advanced model architectures. Emerging techniques such as attention-based mechanisms and transformer models can be explored to improve feature representation and classification accuracy. These approaches can help the system focus on critical regions within the retinal image, leading to more precise detection of abnormalities. Additionally, optimizing the existing ANN–RNN structure by reducing redundancy and improving efficiency can further enhance performance.

The inclusion of explainability features is also a valuable future direction. Providing visual interpretations, such as highlighted regions indicating areas of concern, can help users understand how the model arrives at its predictions. This transparency is particularly important in medical applications, where trust and interpretability play a key role in system acceptance.

From a deployment perspective, the system can be optimized for use on resource-constrained devices. Techniques such as model compression, quantization, and lightweight architecture design can reduce computational requirements, enabling the system to run efficiently on mobile devices or low-cost hardware. This would make the solution more accessible in remote or underserved areas where advanced infrastructure is limited.

Integration with cloud-based platforms can further enhance scalability by allowing the system to handle large volumes of data and support multiple users simultaneously. This would be especially useful in large-scale screening programs or hospital networks. Additionally, incorporating real-time data storage and reporting features can improve record management and assist in long-term patient monitoring.

Future versions of the system may also expand beyond diabetic retinopathy to include detection of other ocular diseases, creating a more comprehensive diagnostic tool. The

development of a mobile application interface can further increase usability, allowing healthcare workers and patients to access the system conveniently.

In conclusion, the project provides a strong foundation that can be extended through improvements in data diversity, model innovation, interpretability, and deployment strategies, ultimately leading to a more robust and widely applicable healthcare solution.

FUTURE SCOPE

- The system can be improved by training it with larger and more diverse retinal image datasets collected from different hospitals and regions. This helps in making the model more accurate and reliable in real-world conditions.
- Future work can focus on enhancing the hybrid ANN–RNN model by using advanced techniques such as attention mechanisms or newer deep learning architectures to capture more complex patterns in images.
- The model can be optimized further by applying better tuning methods, which may reduce processing time and improve prediction accuracy.
- A user-friendly application can be developed, such as a mobile or web-based interface, so that doctors and healthcare workers can easily upload images and get results instantly.
- The system can be integrated with cloud technology to allow faster processing, secure data storage, and easy access from different locations.
- This project can be extended to support real-time screening, which would be especially useful in rural and remote areas where access to eye specialists is limited.
- Future improvements may include linking the system with patient medical records to provide more personalized and comprehensive diagnosis.
- Visualization features can be added to highlight affected regions in retinal images, helping doctors understand how the model makes decisions.
- The system can be expanded to detect other eye diseases such as glaucoma, cataracts, and macular degeneration, making it a multi-purpose diagnostic tool.
- Continuous testing and validation with clinical experts can further improve the system's reliability and acceptance in the medical field.

REFERENCES

1. Alyoubi, W. L., et al., “Diabetic Retinopathy Detection through Deep Learning Techniques,” Informatics in Medicine Unlocked, 2020.
2. Sebastian, A., et al., “A Survey on Deep Learning-Based Diabetic Retinopathy Diagnosis,” PMC, 2023.
3. Fayyaz, A. M., et al., “Analysis of Diabetic Retinopathy using Deep Learning Models,” MDPI Informatics, 2023.
4. Tymchenko, B., et al., “Deep Learning Approach to Diabetic Retinopathy Detection,” arXiv, 2020.
5. Lam, C., et al., “Automated Detection of Diabetic Retinopathy using CNN,” PMC, 2018.
6. Nadeem, M. W., et al., “Deep Learning for Diabetic Retinopathy Analysis: A Review,” Sensors Journal, 2022.
7. Bhulakshmi, D., & Rajput, D. S., “Systematic Review on DR Detection using Deep Learning,” PeerJ Computer Science, 2024.
8. Akhtar, S., et al., “Deep Learning Model for Diabetic Retinopathy Grading,” Nature Scientific Reports, 2025.
9. Gulshan, V., et al., “Development of a Deep Learning Algorithm for Detection of Diabetic Retinopathy,” Google Research, 2016.
10. Bhimavarapu, U., et al., “Deep Learning for Detection and Classification of DR,” PMC, 2022.
11. Gong, W., et al., “Deep Learning for Enhanced Prediction of Diabetic Retinopathy,” Frontiers in Medicine, 2025.
12. Suedumrong, C., et al., “CNN-Based Diabetic Retinopathy Classification using EyePACS Dataset,” MDPI Applied Sciences, 2024.
13. Senapati, A., et al., “Artificial Intelligence for Diabetic Retinopathy Detection,” ScienceDirect, 2024.
14. Muthusamy, D., et al., “Deep Learning-Based Detection of Diabetic Retinopathy,” Springer, 2024.

APPENDIX

A. SOURCE CODE

Model/dataset.py

```
import os
import pandas as pd
from PIL import Image
from torch.utils.data import Dataset
from torchvision import transforms
```

```
class DRDataset(Dataset):
```

```
    def __init__(self, csv_file, image_dir):
```

```
        self.data = pd.read_csv(csv_file)
```

```
        self.image_dir = image_dir
```

```
        self.transform = transforms.Compose([
            transforms.Resize((224,224)),
            transforms.ToTensor()
        ])
```

```
    def __len__(self):
        return len(self.data)
```

```
    def __getitem__(self, idx):
```

```
        img_name = self.data.iloc[idx]["id_code"]
```

```
        label = int(self.data.iloc[idx]["diagnosis"])
```

```
        img_path = os.path.join(self.image_dir, img_name + ".png")
```

```
        image = Image.open(img_path).convert("RGB")
```

```
image = self.transform(image)
```

```
return image, label
```

```
Model/model.py import torch import torch.nn as nn import torchvision.models as models  
from torchvision.models import ResNet18_Weights
```

```
class ANN_RNN_Model(nn.Module):
```

```
    def __init__(self):
```

```
        super().__init__()
```

```
        resnet = models.resnet18(pretrained=True)
```

```
        self.ann = nn.Sequential(  
            *list(resnet.children())[:-1]  
        )
```

```
        self.rnn = nn.LSTM(512, 256, batch_first=True, bidirectional=True)
```

```
        self.fc = nn.Linear(512, 2)
```

```
    def forward(self, x):
```

```
        if len(x.shape) == 4:            x = x.unsqueeze(1)
```

```
        B,S,C,H,W = x.shape
```

```
        x = x.view(B*S, C, H, W)
```

```
        features = self.ann(x)
```

```
features = features.view(B, S, -1)
```

```
out,_ = self.rnn(features)
```

```
out = out[:, -1, :]
```

```
out = self.fc(out)
```

```
return out
```

inference.py

```
import torch from PIL import Image
```

```
import torchvision.transforms as transforms
```

```
from model.model import ANN_RNN_Model
```

```
DEVICE = torch.device("cuda" if torch.cuda.is_available() else "cpu") print("Using device:",  
DEVICE)
```

```
model = ANN_RNN_Model().to(DEVICE)
```

```
state_dict = torch.load("model.pth", map_location=DEVICE)
```

```
model_dict = model.state_dict()
```

```
# keep only matching keys
```

```
filtered_dict = {k: v for k, v in state_dict.items() if k in model_dict and v.shape ==  
model_dict[k].shape}
```

```
model_dict.update(filtered_dict) model.load_state_dict(model_dict) model.eval()
```

```
print("Model loaded successfully")
```

```
# image preprocessing transform = transforms.Compose([ transforms.Resize((224,224)),
```

```

        transforms.ToTensor(),
    ])

# load image
image = Image.open("1ae8c165fd53.png").convert("RGB") image = transform(image)
image = image.unsqueeze(0)

image = image.to(DEVICE)

# prediction with torch.no_grad():  output = model(image)
pred = torch.argmax(output, dim=1).item()

labels = {
    0: "No Diabetic Retinopathy",
    1: "Diabetic Retinopathy Detected"
}

print("Prediction:", labels[pred])

llm_explainer.py

from openai import OpenAI

client=OpenAI(
    base_url="https://openrouter.ai/api/v1",
    api_key="sk-or-v1-
de4f9b322c97c3f7ec699b9f464*****da7a704c"
)

def generate_explanation(prediction):

```

```

prompt=f"""
A patient retinal scan result is {prediction}.

Explain:  reason,  symptoms,  recommendations,  treatment
"""

response=client.chat.completions.create(    model="openai/gpt-4o-mini",
    messages=[
        {"role":"user","content":prompt}
    ]
)

return response.choices[0].message.content

```

train.py

```

if __name__ == "__main__":  import torch
    from torch.utils.data import DataLoader    import torch.nn as nn
    from model.dataset import DRDataset
    from model.model import ANN_RNN_Model

    # ----- DEVICE -----
    # GPU if available, otherwise fallback (but should be GPU)
    DEVICE = torch.device("cuda" if torch.cuda.is_available() else "cpu")    print("Using
device:", DEVICE)    if DEVICE.type == "cpu":
        print("WARNING: GPU not detected. Install CUDA-enabled PyTorch and NVIDIA
drivers!")

    # ----- DATASET -----
    dataset = DRDataset("dataset/train.csv", "dataset/train_images")    print("Dataset size:",
len(dataset))

```

```

# ----- DATALOADER -----
# On Windows, num_workers>0 requires main guard; pin_memory only useful with GPU
loader = DataLoader(    dataset,    batch_size=16,    shuffle=True,
    num_workers=0,          # Set >0 if GPU and everything works
    pin_memory=(DEVICE.type=="cuda") # True only if GPU
)

# ----- MODEL -----
model = ANN_RNN_Model().to(DEVICE)

# ----- LOSS & OPTIMIZER -----    loss_fn = nn.CrossEntropyLoss()
optimizer = torch.optim.Adam(model.parameters(), lr=1e-4)

# ----- TRAINING LOOP -----    EPOCHS = 10    for epoch in range(EPOCHS):
    total = 0    correct = 0    for x, y in loader:        x = x.to(DEVICE)
        y = y.to(DEVICE)

        optimizer.zero_grad()        out = model(x)        loss = loss_fn(out, y)
loss.backward()
        optimizer.step()

        pred = out.argmax(1)        correct += (pred == y).sum().item()
        total += y.size(0)

    acc = correct / total
    print(f"Epoch {epoch+1}/{EPOCHS} Accuracy: {acc:.4f}")

# ----- SAVE MODEL -----    torch.save(model.state_dict(), "model.pth")
print("Model saved successfully!")

```

main.py

```

from fastapi import FastAPI, UploadFile, File import torch from PIL import Image import
torchvision.transforms as transforms
from model.model import ANN_RNN_Model import requests

app = FastAPI()

# 膝 Add CORS middleware immediately after app creation from fastapi.middleware.cors
import CORSMiddleware

app.add_middleware( CORSMiddleware,
    allow_origins=["*"], # allow all origins for testing
    allow_methods=["*"],
    allow_headers=["*"],
)
DEVICE = torch.device("cpu")

model = ANN_RNN_Model()

# load compatible weights
state_dict = torch.load("model.pth", map_location=DEVICE) model_dict =
model.state_dict()

filtered = {k: v for k, v in state_dict.items()
    if k in model_dict and v.shape == model_dict[k].shape}

model_dict.update(filtered)
model.load_state_dict(model_dict)

model.to(DEVICE)
model.eval()

transform = transforms.Compose([ transforms.Resize((224,224)),
    transforms.ToTensor()

```

])

```
# OpenRouter API KEY
```

```
    OPENROUTER_API_KEY = "sk-or-v1-  
bf05684f702c592dd872bf9382c5789e9a19be3c48c5d1d42943198a2506d471"
```

```
def generate_explanation():
```

```
    prompt = ""
```

```
    A retinal image analysis system detected diabetic retinopathy.
```

```
    Return ONLY valid JSON in this format:
```

```
{  
    "reason": "...",  
    "symptoms": "...",  
    "recommendations": "...",  
    "treatment": "..."  
}  
""
```

```
    response = requests.post(  
        "https://openrouter.ai/api/v1/chat/completions",    headers={  
        "Authorization": f"Bearer {OPENROUTER_API_KEY}",  
        "Content-Type": "application/json",  
        "HTTP-Referer": "http://localhost:8000",  
        "X-Title": "Diabetic Retinopathy Detection"
```

```
    },    json={  
        "model": "openrouter/free",  
        "messages": [  
            {"role": "user", "content": prompt}  
        ]  
    }  
)
```

```

data = response.json()

# Debug print to see what API returns
print(data)

if "choices" not in data:
    return {
        "reason": "Unable to generate explanation",
        "symptoms": "API error",
        "recommendations": "Check API key or model",
        "treatment": "N/A"
    }

explanation_text = data["choices"][0]["message"]["content"]

return explanation_text

@app.post("/predict") async def predict(file: UploadFile = File(...)):

    image = Image.open(file.file).convert("RGB")    image = transform(image)
    image = image.unsqueeze(0)

    with torch.no_grad():

        output = model(image)

        probabilities = torch.softmax(output, dim=1)

        confidence, pred = torch.max(probabilities, dim=1)    prediction = pred.item()

        confidence = round(confidence.item(), 2)

```

```
if prediction == 1:

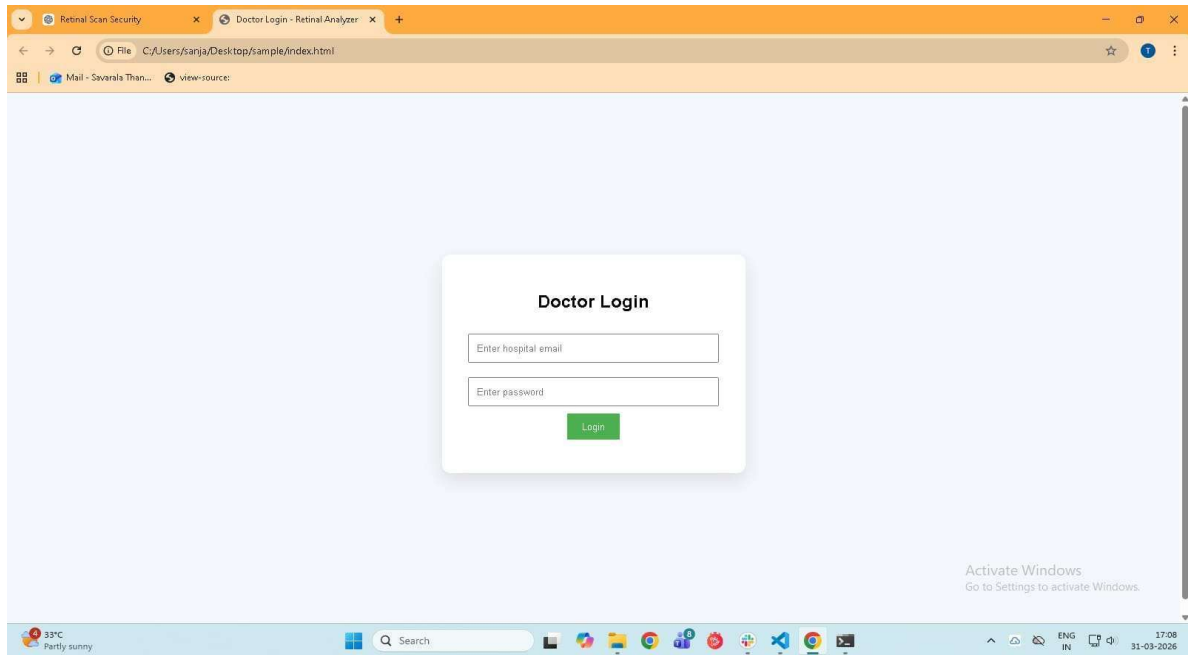
    explanation = generate_explanation()

    return {
        "prediction": "Diabetic Retinopathy",
        "confidence": confidence,
        "explanation": explanation
    }
else:

    return {
        "prediction": "No Diabetic Retinopathy",
        "confidence": confidence
    }
```

B SCREENSHOTS

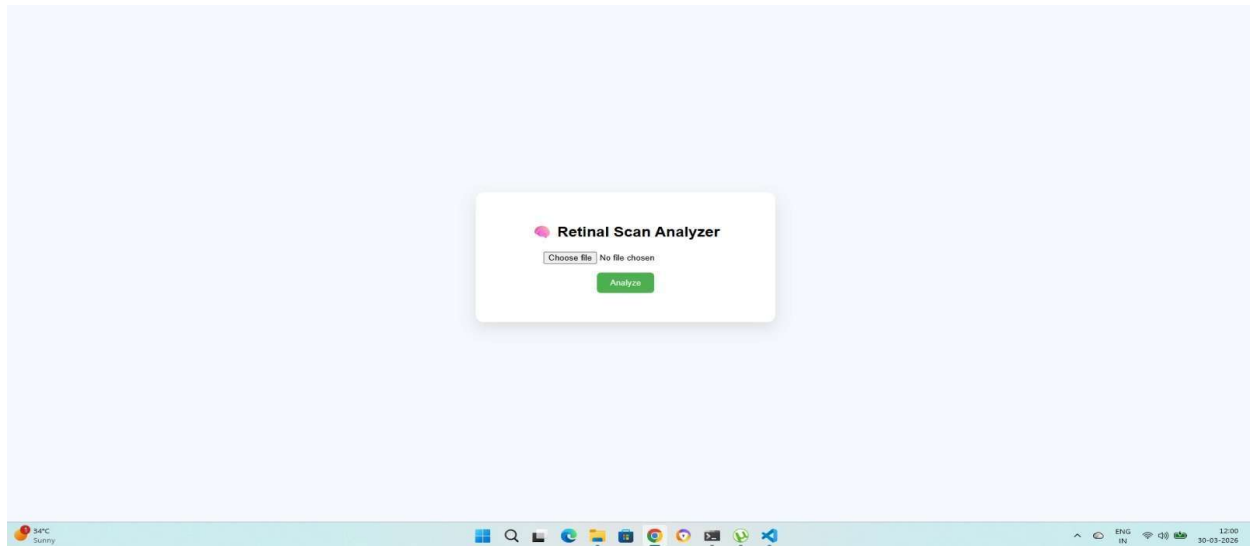
Login page



The output screen shows a neatly designed Doctor Login page that serves as the entry point to the system. The layout is simple and centered, making it easy for users to focus on the task without distractions. At the top of the panel, a clear heading identifies the purpose of the page, followed by two input fields where the doctor can enter their username and password. A login button is placed below these fields, allowing the user to proceed after providing the required details.

This interface reflects a clean and practical design approach, keeping usability in mind. The minimal elements on the screen help reduce confusion and make the process straightforward, even for first-time users. The page is displayed through a web browser, indicating that the system runs as a web-based application. Overall, the output demonstrates a functional and user-friendly login module that ensures secure and smooth access for doctors into the system.

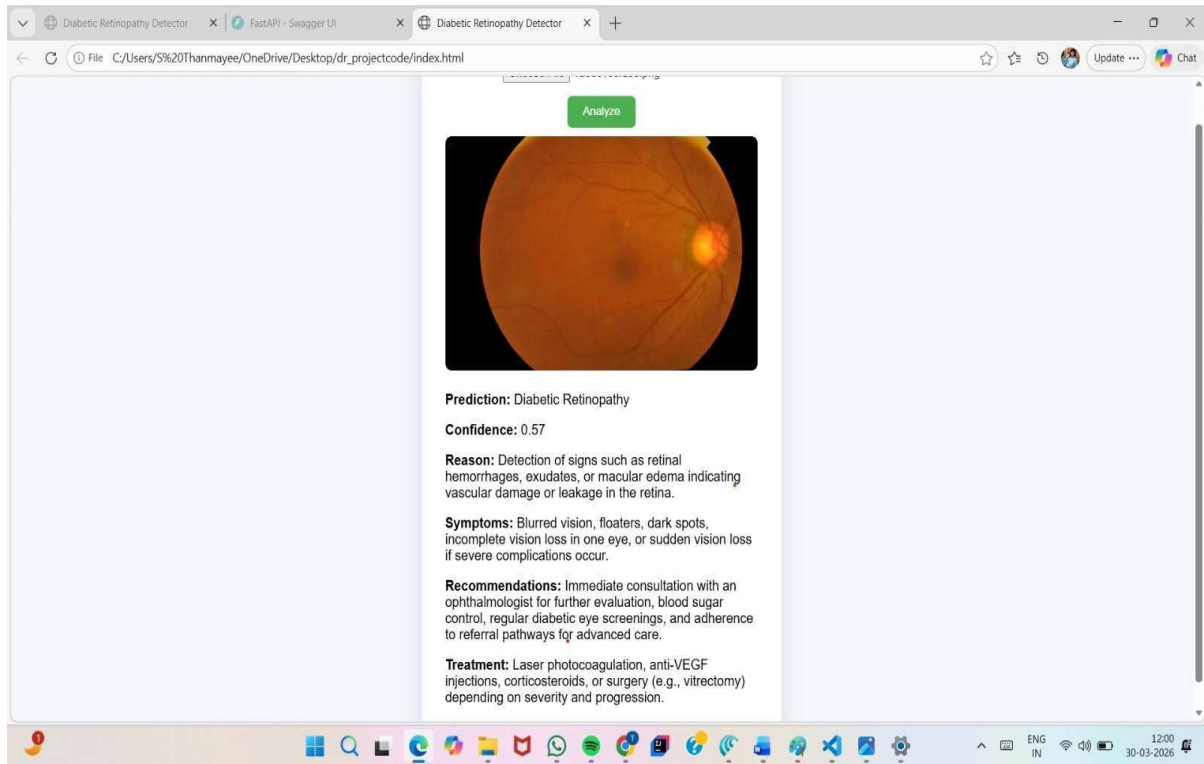
INPUT PAGE



The output screen shows the input page of the system, where the user begins the interaction by providing the required data. At the center of the page, a section titled “RETINAL SCAN ANALYZER” is displayed, clearly indicating the purpose of the application. Below this title, there is an option to choose an image file, allowing the user to upload a retinal scan. A submit button is placed just beneath the file selection area, enabling the user to proceed with the analysis once the image is selected.

The design of this page is kept simple and focused, avoiding unnecessary elements so that the user can easily understand what needs to be done. The centered layout naturally guides attention toward the upload section, making the process smooth even for someone using it for the first time. This page acts as the starting point of the system’s workflow, where the input is collected and prepared for further processing in the retinal analysis model.

OUTPUT PAGE



The output screen presents the result generated by the system after analyzing the uploaded retinal image. At the center, the processed eye image is displayed, giving a clear visual reference for the analysis. Just below it, the system provides a prediction indicating the presence of Diabetic Retinopathy, along with a confidence value that reflects how certain the model is about its decision. This combination of visual and textual output helps the user quickly understand what the system has detected.

Further details are provided in a structured manner, including possible reasons behind the prediction, common symptoms related to the condition, and suggested next steps. The recommendations guide the user toward seeking proper medical evaluation and maintaining necessary precautions. Overall, this page not only shows the result but also explains it in a simple and informative way, making the output more meaningful and easier to interpret.

C. GRAPHS

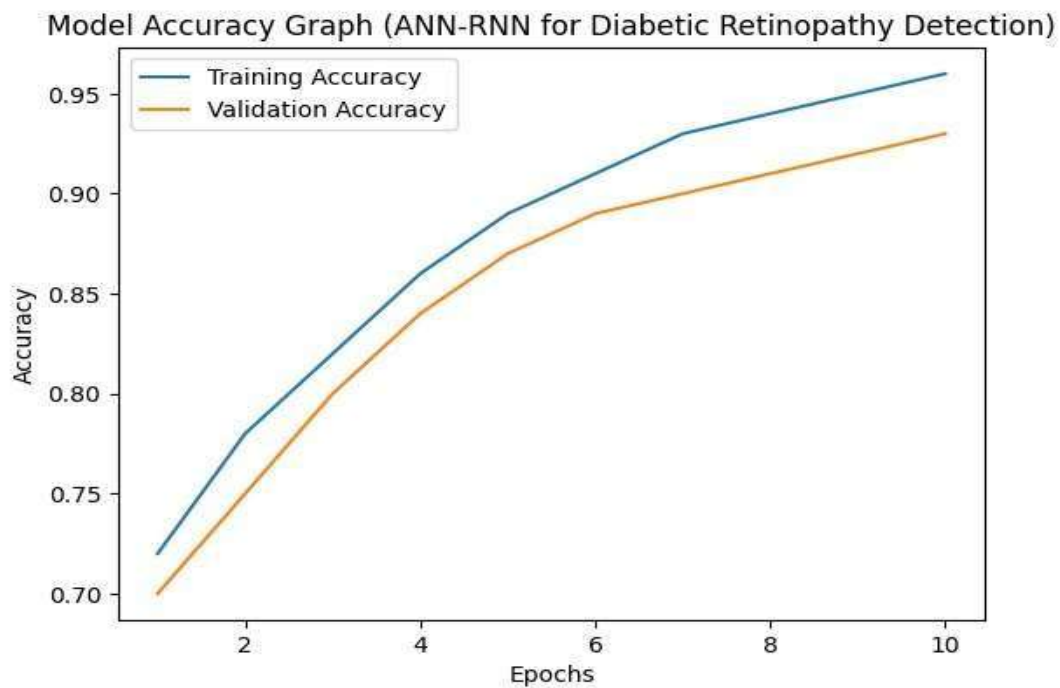


Fig C.1: Representing the Accuracy curves

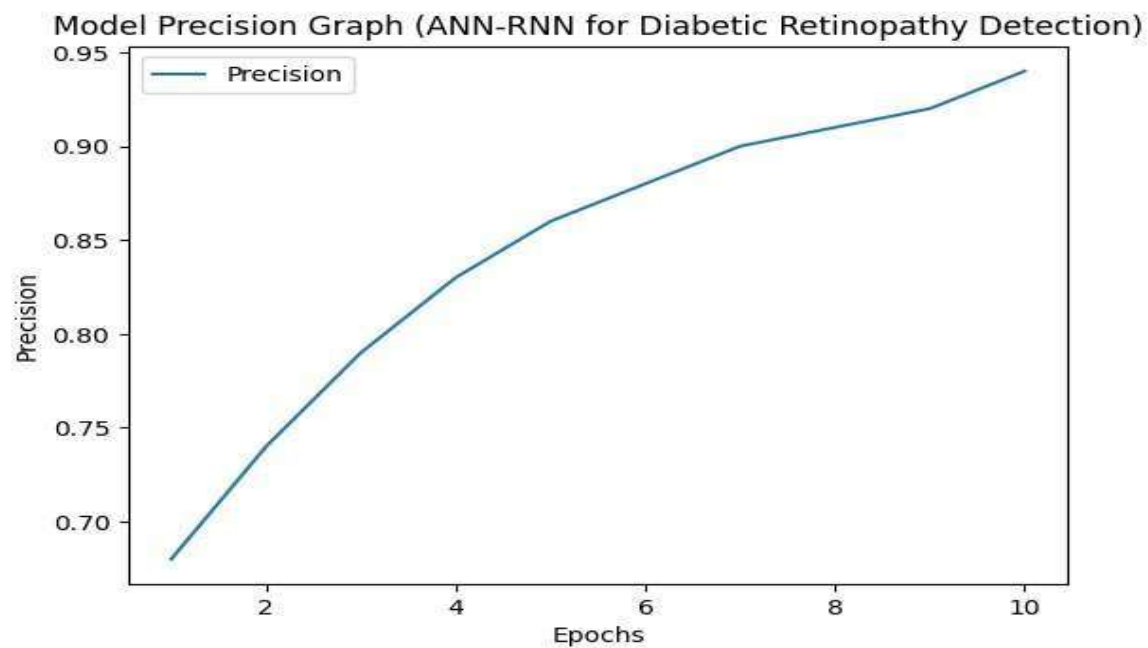


Fig C.2 : Representing the Precision and Recall curves

ANNEXURE I

Title of the Project	Dual Stage ANN-RNN Hybrid Model for Automated Diabetic Retinopathy Detection
Name of the Student	Savarala Thanmayee – 22751A3158
	P Keerthi – 22751A3144
	K Yohitha – 22751A3125
	Samudrala Kokila – 22751A3157
Name of the Guide & Designation	Ms. M. Deepika, Assistant Professor

Table 1: Outcomes attained and its Justification

PO	Justification
PO1	Gained basic knowledge on the project area while selecting the problem.
PO2	Learned how to analyze the complex problems.
PO3	Learned problem solving skills while implementing modules.
PO4	Learned how to conduct investigations on complex real-world problems.
PO5	Learned modern tool usage while developing source code.
PO6	Learned professional engineering practices while adding new features to the project.
PO7	Learned how this project impacts on societal and environmental contexts.
PO8	Learned software ethics to be followed while developing this project.
PO9	Learned how to work as an individual and as a team member during this entire project period.
PO10	Learned how to communicate effectively during the project reviews.
PO11	Learned how to estimate cost benefit analysis for the project management.
PO12	The knowledge which I have gained through the project will be helpful for my future to refine my skills.

A Plagiarism Check Report

On

**DUAL STAGE ANN-RNN HYBRID MODEL FOR AUTOMATED
DIABETIC RETINOPATHY DETECTION**

Submitted in partial fulfillment of the requirements for the award of the

degree of **Bachelor of Technology**

in

CSE – ARTIFICIAL INTELLIGENCE

By

**S THANMAYEE (23755A3101) P KEERTHI (2275A3144) K YOHITHA
(22751A3125) S KOKILA (22751A357)**

Under the Esteemed Guidance of

M. DEEPIKA

Assistant Professor

plagarismcheckingreportb12.pdf

 Vydehi Institute of Medical Sciences & Research Centre

Document Details

Submission ID

trn:oid:::3618:134661577

Submission Date

Apr 9, 2026, 1:01 PM GMT+5:30

Download Date

Apr 9, 2026, 1:03 PM GMT+5:30

File Name

plagarismcheckingreportb12.pdf

File Size

6.0 MB

86 Pages

22,879 Words

131,412 Characters

14% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Match Groups

-  **285 Not Cited or Quoted 13%**
Matches with neither in-text citation nor quotation marks
-  **0 Missing Quotations 0%**
Matches that are still very similar to source material
-  **14 Missing Citation 1%**
Matches that have quotation marks, but no in-text citation
-  **0 Cited and Quoted 0%**
Matches with in-text citation present, but no quotation marks

Top Sources

- 10%  Internet sources
- 9%  Publications
- 0%  Submitted works (Student Papers)

Integrity Flags

1 Integrity Flag for Review

-  **Hidden Text**
1 suspect characters on 1 page
Text is altered to blend into the white background of the document.

Our system's algorithms look deeply at a document for any inconsistencies that would set it apart from a normal submission. If we notice something strange, we flag it for you to review.

A Flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.

Match Groups

-  **285 Not Cited or Quoted 13%**
Matches with neither in-text citation nor quotation marks
-  **0 Missing Quotations 0%**
Matches that are still very similar to source material
-  **14 Missing Citation 1%**
Matches that have quotation marks, but no in-text citation
-  **0 Cited and Quoted 0%**
Matches with in-text citation present, but no quotation marks

Top Sources

- 10%  Internet sources
- 9%  Publications
- 0%  Submitted works (Student Papers)

Top Sources

The sources with the highest number of matches within the submission. Overlapping sources will not be displayed.

- 1** Internet
www.mdpi.com <1%
- 2** Publication
"Biomedical Engineering Science and Technology", Springer Science and Business... <1%
- 3** Publication
Shankar Babu, Mahesh Babu Kota. "Synergies in Smart and Virtual Systems using... <1%
- 4** Publication
Bui Thanh Hung, M. Sekar, Ayhan Esi, R. Senthil Kumar. "Applications of Mathema... <1%
- 5** Internet
repositorio.ugto.mx <1%
- 6** Internet
download.bibis.ir <1%
- 7** Internet
fastercapital.com <1%
- 8** Internet
gitlab.fbk.eu <1%
- 9** Publication
Kerem GENCER, Gülcan GENCER, Tuğçe Horozoğlu CERAN, Aynur ER BİLİR, Mustaf... <1%
- 10** Publication
Udara Yedukondalu, V Vijayasri Bolisetty. "Advancing Innovation through AI and ... <1%

11	Internet	www.tbench.ai	<1%
12	Internet	ijarcce.com	<1%
13	Internet	philarchive.org	<1%
14	Publication	Veglia, Tyler J.. "Bearing Failure Analysis in Centrifugal Pumps Utilizing Adaptive ...	<1%
15	Internet	studybam.com	<1%
16	Internet	www.ijser.org	<1%
17	Internet	johal.in	<1%
18	Internet	www.mail.ijase.org	<1%
19	Internet	ar.cou.ac.bd:8080	<1%
20	Internet	bora.uib.no	<1%
21	Publication	Sushil Kamboj, Pardeep Singh Tiwana. "Innovations in Computing", CRC Press, 2025	<1%
22	Internet	github.com	<1%
23	Internet	sjcit.ac.in	<1%
24	Publication	R. Jayalakshmi, T. Tamilvizhi. "Privacy preservation in diabetic disease prediction ...	<1%

25	Internet	forum.deepchem.io	<1%
26	Publication	Pradeep Singh, Balasubramanian Raman. "Deep Learning Through the Prism of T...	<1%
27	Internet	khuhub.khu.ac.kr	<1%
28	Internet	medshun.com	<1%
29	Internet	open-innovation-projects.org	<1%
30	Internet	discuss.pytorch.org	<1%
31	Internet	medium.com	<1%
32	Internet	openi.pcl.ac.cn	<1%
33	Internet	www.retinarisk.com	<1%
34	Internet	www.symptoma.com	<1%
35	Internet	p.pdfkul.com	<1%
36	Internet	www.rapidinnovation.io	<1%
37	Publication	B. G. Nagaraja, S. Kannadhasan. "Information and Communication Systems", CRC...	<1%
38	Internet	www.ijfmr.com	<1%

39	Internet	cfteach.github.io	<1%
40	Publication	Pranab Dey. "Computational Cytology", Springer Science and Business Media LLC,...	<1%
41	Internet	sageeyecare.ca	<1%
42	Internet	sathee.iitk.ac.in	<1%
43	Internet	healthdor.com	<1%
44	Internet	muuktest.com	<1%
45	Internet	www.educative.io	<1%
46	Publication	Malisetty, Bhavani. "Improving the Understanding of Privacy Policies Using Large...	<1%
47	Publication	Elserwy, Mohamed. "Diabetic Retinopathy Detection Using Artificial Intelligence."...	<1%
48	Internet	dokumen.pub	<1%
49	Internet	library.acadlore.com	<1%
50	Internet	neurosova.ru	<1%
51	Internet	numbersmithy.com	<1%
52	Internet	www.ijraset.com	<1%

53	Internet	www.researchgate.net	<1%
54	Internet	www.superteams.ai	<1%
55	Publication	Diep N. Nguyen, Vu Ly, Quang Nguyen, Dinh Hoang. "Generative AI for Cybersecu...	<1%
56	Internet	apidog.com	<1%
57	Internet	branchsrv.unac.org	<1%
58	Internet	eprint.innovativepublication.org	<1%
59	Internet	findresearcher.sdu.dk	<1%
60	Internet	machine-learning.anracom.com	<1%
61	Publication	Anshul Verma, Pradeepika Verma. "Research Advances in Intelligent Computing",...	<1%
62	Internet	su-plus.strathmore.edu	<1%
63	Publication	"Creativity in Intelligent Technologies and Data Science", Springer Science and Bu...	<1%
64	Publication	Attarha, Shadi. "A Framework for Sensor Fault Detection and Management in Lo...	<1%
65	Publication	Bhupendra Gopalbhai Prajapati, Rishabha Malviya, Himanshu Paliwal, Sonali Sun...	<1%
66	Publication	Chandrasekar Vuppalapati. "Building Next-Gen ESG Platforms with IoT and AI for ...	<1%

67	Internet	dspace.bracu.ac.bd:8080	<1%
68	Internet	f1000research.com	<1%
69	Internet	jurnal.polibatam.ac.id	<1%
70	Internet	kaiyangzhou.github.io	<1%
71	Publication	"Intelligent Strategies for ICT", Springer Science and Business Media LLC, 2025	<1%
72	Publication	James Wootton. "Circuit Model of Quantum Computation", Oxford University Pres...	<1%
73	Publication	Vishal Jain, Akash Tayal, Jaspreet Singh, Arun Solanki. "Cognitive Computing Syst...	<1%
74	Internet	mdpi-res.com	<1%
75	Internet	restpublisher.com	<1%
76	Internet	www.usaeyes.org	<1%
77	Internet	duikt.edu.ua	<1%
78	Internet	edurev.in	<1%
79	Internet	ela.kpi.ua	<1%
80	Internet	mangohost.net	<1%

81	Internet	straitsresearch.com	<1%
82	Internet	www.cerebrium.ai	<1%
83	Internet	www.essaysauce.com	<1%
84	Publication	Nikbakht, Sahar. "Deep Learning-Based Classification of 4D FDG-PET Images for A...	<1%
85	Publication	Pawan Singh Mehra, Dharendra Kumar Shukla. "Artificial Intelligence, Blockchain,...	<1%
86	Internet	escholarship.org	<1%
87	Internet	feelgoodanyway.com	<1%
88	Publication	Ankrum, Dennis, and Knut Inge Fostervold. "Visual Ergonomics for Children", Erg...	<1%
89	Internet	diseasedb.com	<1%
90	Internet	innaxis-comm.s3.eu-central-1.amazonaws.com	<1%
91	Internet	lutpub.lut.fi	<1%
92	Internet	skafteknicki.github.io	<1%
93	Publication	" ", Clinical & Experimental Ophthalmology, 2017	<1%
94	Publication	Fahad Mostafa, Hafiz Khan, Fardous Farhana, Md Ariful Haque Miah. "Application ...	<1%

95	Internet	affiliates.bloodstock.uk.com	<1%
96	Internet	ipv6.rs	<1%
97	Internet	rps.wku.edu.et	<1%
98	Internet	www.ijert.org	<1%
99	Internet	www.pharmatutor.org	<1%
100	Internet	www.whole-dog-journal.com	<1%
101	Publication	"Machine Learning and Data Mining in Pattern Recognition", Springer Nature, 2018	<1%
102	Publication	Anastasia Giachanou, Fabio Crestani. "Like It or Not", ACM Computing Surveys, 2...	<1%
103	Publication	Di Xiao, Alauddin Bhuiyan, Shaun Frost, Janardhan Vignarajan, Mei-Ling Tay-Kear...	<1%
104	Publication	Juraj Hric, Yiping Lin. "Applied Data Science in FinTech - Models, Tools, and Case S...	<1%
105	Publication	Jyoti Ranjan Sahoo, Surendra Kumar Nanda, Ganapati Panda. "Brain Tumor Detec...	<1%
106	Publication	Moslem Alikarami, Tola Abdulsattar Faraj, Nabaz Hamarashid Hama, Amineh Sad...	<1%
107	Publication	Ton Duc Thang University	<1%
108	Publication	U R Acharya. "Computer-based detection of diabetes retinopathy stages using di...	<1%

109	Internet	app.readytensor.ai	<1%
110	Internet	arxiv.org	<1%
111	Internet	diabeteshealth.online	<1%
112	Internet	doi.radap.kpi.ua	<1%
113	Internet	smartech.gatech.edu	<1%
114	Internet	www.acibademhealthpoint.com	<1%
115	Internet	www.ehnpc.com	<1%
116	Internet	www.felixhospital.com	<1%
117	Internet	www.ijirset.com	<1%
118	Internet	www.ijprems.com	<1%
119	Internet	www.nature.com	<1%
120	Internet	www.scribd.com	<1%
121	Publication	"Data Science and Applications", Springer Science and Business Media LLC, 2025	<1%
122	Publication	Akpabio, Paul Clement. "Using Advanced Artificial Intelligence Techniques in Pav..."	<1%

123	Publication	Altekreeti, Mustafa Manal Noori. "Classification and Prediction Retinal Oct Image...	<1%
124	Publication	Andrzej Dudek. "Deep Learning Models for Economic Research", Routledge, 2025	<1%
125	Publication	Antonio Gullí. "Agentic Design Patterns", Springer Science and Business Media LL...	<1%
126	Publication	B. Harishyam, M.P. Jenarthanam, R. Rishivanth, Rohit Rajesh, N. Sai Girish. "Visual ...	<1%
127	Publication	Donia Y. Badawood. "StatVis: A visual analytics framework for statistical cluster v...	<1%
128	Publication	Jing Dai. "DeepSeek in Action - LLM Deployment, Fine-Tuning, and Application", C...	<1%
129	Publication	Kumar, Navneet. "Predicting High Stress Regions in a Microstructure Using Conv...	<1%
130	Publication	Nazmul Siddique, Mohammad Shamsul Arefin, Sheak Rashed Haider Noori, M Sha...	<1%
131	Publication	Siddhesh Prashant Chaubal. "AI Projects in PyTorch", Springer Science and Busine...	<1%
132	Publication	Víctor Vives-Boix, Daniel Ruiz-Fernández. "Diabetic retinopathy detection through...	<1%
133	Internet	dokumen.tips	<1%
134	Internet	encord.com	<1%
135	Internet	fringe.global.com	<1%
136	Internet	iitu.edu.kz	<1%

137	Internet	ijrpr.com	<1%
138	Internet	openrouter.ai	<1%
139	Internet	www.topassignmentforall.com	<1%
140	Publication	Thisara Shyamalee, Dulani Meedeniya, Gilbert Lim, Mihipali Karunaratne. "Auto...	<1%
141	Publication	"Medical Image Computing and Computer-Assisted Intervention – MICCAI 2017", ...	<1%
142	Publication	A. Punitha, S. Syedakbar, S. Jeyasudha. "Advanced Pathways in Electrical, Commu...	<1%
143	Publication	Chenggang Deng, Mingfan Li. "Study on cone yarn category recognition method ...	<1%
144	Publication	Mehdi Ghayoumi. "Generative Adversarial Networks in Practice", CRC Press, 2023	<1%
145	Publication	S. Prasad Jones Christydass, Nurhayati Nurhayati, S. Kannadhasan. "Hybrid and A...	<1%
146	Publication	Sandhya Arora, Urmila Shrawankar, Maya Ingle, Indu Bhagat. "Computing Techn...	<1%

DUAL STAGE ANN-RNN HYBRID MODEL FOR AUTOMATED DIABETIC RETINOPATHY DETECTION

¹K Yohitha, ²Savarala Thanmayee, ³P Keerthi, ⁴Samudrala Kokila, ⁵M Deepika, ⁶M.E. Palanivel,

¹CSE-Artificial Intelligence, ²CSE-Artificial Intelligence, ³CSE-Artificial Intelligence, ⁴CSE-Artificial Intelligence, ⁵Assistant Intelligence, ⁶Professor,
CSE-Artificial Intelligence,
Sreenivasa Institute of Technology and Management Studies, Chittoor, India

Abstract : This study has been undertaken to investigate the determinants of stock returns in Karachi Stock Exchange (KSE) using two assets pricing models the classical Capital Asset Pricing Model and Arbitrage Pricing Theory model. To test the CAPM market return is used and macroeconomic variables are used to test the APT. The macroeconomic variables include inflation, oil prices, interest rate and exchange rate. For the very purpose monthly time series data has been arranged from Jan 2010 to Dec 2014. The analytical framework contains.

Index Terms - Component, formatting, style, styling, insert.

I. INTRODUCTION

Hospital or health care waste is generally named & popular as biomedical waste. The world health organization defines biomedical waste as, "Waste generation by health care activities & includes blood, used needles, pharmaceuticals, radioactive materials etc." The biomedical waste is also known as infectious waste or medical waste or health care waste. According to biomedical waste management & handling rules 1998 of India. Biomedical waste means any waste which is generated during the diagnosis, treatment or immunization of human being or animals or in research activities. In simple words biomedical waste is the waste generated by the medical & health institute/agencies.

Biomedical waste management defines waste management as the practices & procedures or the administration of activities that provide for the collection, source separation, storage, transportation, transfer, processing, treatment & disposal of waste. Biomedical waste management is a routine procedure of hospital administration as prescribed by law. Hospital waste, hospital acquired infection, transfusion transmitted diseases, rising incidence of hepatitis B, HIV & Other diseases, create potential threat of infection, contamination & serious health hazards to doctors, nurses, ward boys, support staff, sanitation workers, rag pickers & other health care workers. Who are regularly exposed to biomedical waste as an occupation hazard as well as general public in the surrounding area.

NEED OF THE STUDY.

The establishment of large hospitals where hundreds to thousands of patients are treated, it has created a serious problem of biomedical waste management. The seriousness of improper biomedical waste management was brought to the light during summer 1998. In India studies have been carried out at local / regional levels in various hospitals, indicate that roughly about 1-5 kg/bed/day to waste is generated. Among all health care personnel, ward boys, sweepers, operation theatre & laboratory attendants have come into contact with biomedical waste during the process of segregation, collection, transport, storage & final disposal. The knowledge of medical, paramedical staff & ward boys, sweepers about the biomedical waste management is important to improve the biomedical waste management practices. The biomedical waste requiring special attention includes those that are potentially infectious, sharps, example needle, scalpels, objects capable of puncturing the skin, also plastic, pharmaceutical & chemically hazardous substances used in laboratories etc.

3.1 Population and Sample

KSE-100 index is an index of 100 companies selected from 580 companies on the basis of sector leading and market capitalization. It represents almost 80% weight of the total market capitalization of KSE. It reflects different sector company's performance and productivity. It is the performance indicator or benchmark of all listed companies of KSE. So it can be regarded as universe of the study. Non-financial firms listed at KSE-100 Index (74 companies according to the page of KSE visited on 20.5.2015) are treated as universe of the study and the study have selected sample from these companies.

The study comprised of non-financial companies listed at KSE-100 Index and 30 actively traded companies are selected on the bases of market capitalization. And 2015 is taken as base year for KSE-100 index.

3.2 Data and Sources of Data

For this study secondary data has been collected. From the website of KSE the monthly stock prices for the sample firms are obtained from Jan 2010 to Dec 2014. And from the website of SBP the data for the macroeconomic variables are collected for the period of five years. The time series monthly data is collected on stock prices for sample firms and relative macroeconomic variables for the period of 5 years. The data collection period is ranging from January 2010 to Dec 2014. Monthly prices of KSE -100 Index is taken from yahoo finance.

3.3 Theoretical framework

Variables of the study contains dependent and independent variable. The study used pre-specified method for the selection of variables. The study used the Stock returns as dependent variable. From the share price of the firm the Stock returns are calculated. Rate of a stock salable at stock market is known as stock price.

RESEARCH METHODOLOGY

The methodology section outlines the plan and method that how the study is conducted. This includes Universe of the study, sample of the study, Data and Sources of Data, study's variables and analytical framework. The details are as follows;

3.1 Population and Sample

KSE-100 index is an index of 100 companies selected from 580 companies on the basis of sector leading and market capitalization. It represents almost 80% weight of the total market capitalization of KSE. It reflects different sector company's performance and productivity. It is the performance indicator or benchmark of all listed companies of KSE. So it can be regarded as universe of the study. Non-financial firms listed at KSE-100 Index (74 companies according to the page of KSE visited on 20.5.2015) are treated as universe of the study and the study has selected sample from these companies.

The study comprised of non-financial companies listed at KSE-100 Index and 30 actively traded companies are selected on the basis of market capitalization. And 2015 is taken as base year for KSE-100 index.

3.2 Data and Sources of Data

For this study secondary data has been collected. From the website of KSE the monthly stock prices for the sample firms are obtained from Jan 2010 to Dec 2014. And from the website of SBP the data for the macroeconomic variables are collected for the period of five years. The time series monthly data is collected on stock prices for sample firms and relative macroeconomic variables for the period of 5 years. The data collection period is ranging from January 2010 to Dec 2014. Monthly prices of KSE -100 Index is taken from yahoo finance.

3.3 Theoretical framework

Variables of the study contains dependent and independent variable. The study used pre-specified method for the selection of variables. The study used the Stock returns as dependent variable. From the share price of the firm the Stock returns are calculated. Rate of a stock salable at stock market is known as stock price.

Systematic risk is the only independent variable for the CAPM and inflation, interest rate, oil prices and exchange rate are the independent variables for APT model.

Consumer Price Index (CPI) is used as a proxy in this study for inflation rate. CPI is a wide basic measure to compute usual variation in prices of goods and services throughout a particular time period. It is assumed that rise in inflation is inversely associated to security prices because inflation is at last turned into nominal interest rate and change in nominal interest rates caused change in discount rate so discount rate increase due to increase in inflation rate and increase in discount rate leads to decrease the cash flow's present value (Jecheche, 2010). The purchasing power of money decreased due to inflation, and due to which the investors demand high rate of return, and the prices decreased with increase in required rate of return (Iqbal et al, 2010).

Exchange rate is a rate at which one currency exchanged with another currency. Nominal effective exchange rate (Pak Rupee/U.S.D) is taken in this study. This is assumed that decrease in the home currency is inversely associated to share prices (Jecheche, 2010). Pan et al. (2007) studied exchange rate and its dynamic relationship with share prices in seven East Asian

Countries and conclude that relationship of exchange rate and share prices varies across economies of different countries. So there may be both possibility of either exchange rate directly or inversely related with stock prices. Oil prices are positively related with share prices if oil prices increase stock prices also increase (Iqbal et al, 2012). Ataulah (2001) suggested that oil prices cause positive change in the movement of stock prices. The oil price has no significant effect on stock prices (Dash & Rishika, 2011). Six month T-bills rate is used as proxy of interest rate. As investors are very sensitive about profit and where the signals

turn into red they definitely sell the shares. And this sensitivity of the investors towards profit effects the relationship of the stock prices and interest rate, so the more volatility will be there in the market if the behaviors of the investors are more sensitive. Plethora (2002) has tested interest rate sensitivity to stock market returns, and concluded an inverse relationship between interest rate and stock returns. Nguyen (2010) studies Thailand market and found that Interest rate has an inverse relationship with stock prices.

KSE-100 index is used as proxy of market risk. KSE-100 index contains top 100 firms which are selected on the bases of their market capitalization. Beta is the measure of systematic risk and has a linear relationship with return (Horn, 1993). High risk is associated with high return (Basu, 1977, Reiganum, 1981 and Gibbons, 1982). Fama and MacBeth (1973) suggested the existence of a significant linear positive relation between realized return and systematic risk as measured by β . But on the other side some empirical results showed that high risk is not associated with high return (Michailidis et al. 2006, Hanif, 2009). Mollah and Jamil (2003) suggested that risk-return relationship is nonlinear perhaps due to high volatility.

3.4 Statistical tools and econometric models

This section elaborates the proper statistical/econometric/financial models which are being used to forward the study from data towards inferences. The detail of methodology is given as follows.

3.4.1 Descriptive Statistics

Descriptive Statics has been used to find the maximum, minimum, standard deviation, mean and normally distribution of the data of all the variables of the study. Normal distribution of data shows the sensitivity of the variables towards the periodic changes and speculation. When the data is not normally distributed it means that the data is sensitive towards periodic changes and speculations which create the chances of arbitrage and the investors have the chance to earn above the normal profit. But the assumption of the APT is that there should not be arbitrage in the market and the investors can earn only normal profit. Jarque bera test is used to test the normality of data.

3.4.2 Fama-McBeth two pass regression

After the test statistics the methodology is following the next step in order to test the asset pricing models. When testing asset pricing models related to risk premium on asset to their betas, the primary question of interest is whether the beta risk of particular factor is priced. Fama and McBeth (1973) develop a two pass methodology in which the beta of each asset with respect to a factor is estimated in a first pass time series regression and estimated betas are then used in second pass cross sectional regression to estimate the risk premium of the factor. According to Blum (1968) testing two-parameter models immediately presents an unavoidable errors-in-the variables problem. It is important to note that portfolios (rather than individual assets) are used for the reason of making the analysis statistically feasible. Fama McBeth regression is used to attenuate the problem of errors-in-variables (EIV) for two parameter models (Campbell, Lo and MacKinlay, 1997). If the errors are in the β (beta) of individual security are not perfectly positively correlated, the β of portfolios can be much more precise estimates of the true β (Blum, 1968).

The study follow Fama and McBeth two pass regression to test these asset pricing models. The Durbin Watson is used to check serial correlation and measures the linear association between adjacent residuals from a regression model. If there is no serial correlation, the DW statistic will be around 2. The DW statistic will fall if there is positive serial correlation (in worst case, it will be near zero). If there is a negative correlation, the statistic will lie somewhere between 2 and 4. Usually the limit for non-serial correlation is considered to be DW is from 1.8 to 2.2. A very strong positive serial correlation is considered at DW lower than 1.5 (Richardson and Smith, 1993).

According to Richardson and Smith (1993) to make the model more effective and efficient the selection criteria for the shares in the period are: Shares with no missing values in the period, Shares with adjusted $R^2 < 0$ or F significant (p-value) > 0.05 of the first pass regression of the excess returns on the market risk premium are excluded. And Shares are grouped by alphabetic order into group of 30 individual securities (Roll and Ross, 1980).

3.4.2.1 Model for CAPM

In first pass the linear regression is used to estimate beta which is the systematic risk.

$$R_i - R_f = (R_m - R_f)\beta \quad (3.1)$$

Where R_i is Monthly return of this security, R_f is Monthly risk free rate, R_m is Monthly return of market and β is systematic risk (market risk).

The excess returns $R_i - R_f$ of each security is estimated from a time series share prices of KSE-100 index listed shares for each period under consideration. And for the same period the market Premium $R_m - R_f$ also estimated. After that regress the excess returns $R_i - R_f$ on the market premium $R_m - R_f$ to find the beta coefficient (systematic risk).

Then a cross sectional regression or second pass regression is used on average excess returns of the shares and estimated betas.

$$\hat{R}_i = \gamma_0 + \gamma_1\beta_i + \epsilon \quad (3.2)$$

Where λ_0 = intercept, $\bar{R}_i$ is average excess returns of security i , β_i is estimated coefficient of security i and ϵ is error term.

3.4.2.2 Model for APT

In first pass the betas coefficients are computed by using regression.

$$R_i - R_f = \beta_{i1}f_1 + \beta_{i2}f_2 + \beta_{i3}f_3 + \beta_{i4}f_4 + \epsilon \quad (3.3)$$

Where R_i is the monthly return of stock i , R_f is risk free rate, β_i is the sensitivity of stock i with factors and ϵ is the error term. Then a cross sectional regression or second pass regression is used on average excess returns of the shares on the factor scores.

$$\bar{R} = \gamma_0 + \gamma_1\beta_1 + \gamma_2\beta_2 + \gamma_3\beta_3 + \gamma_4\beta_4 + \epsilon_i \quad (3.4)$$

Where $\bar{R}$ is average monthly excess return of stock i , λ = risk premium, β_1 to β_4 are the factors scores and ϵ_i is the error term.

3.4.3 Comparison of the Models

The next step of the study is to compare these competing models to evaluate that which one of these models is more supported by data. This study follows the methods used by Chen (1983), the Davidson and MacKinnon equation (1981) and the posterior odds ratio (Zellner, 1979) for comparison of these Models.

3.4.3.1 Davidson and MacKinnon Equation

CAPM is considered the particular or strictly case of APT. These two models are non-nested because by imposing a set of linear restrictions on the parameters the APT cannot be reduced to CAPM. In other words the models do not have any common variable. Davidson and MacKinnon (1981) suggested the method to compare non-nested models. The study used the Davidson and MacKinnon equation (1981) to compare CAPM and APT.

This equation is as follows;

$$R_i = \alpha R_{APT} + (1 - \alpha) R_{CAPM} + e_i \quad (3.5)$$

Where R_i = the average monthly excess returns of the stock i , R_{APT} = expected excess returns estimated by APT, R_{CAPM} = expected excess returns estimated by CAPM and α measure the effectiveness of the models. The APT is the accurate model to forecast the returns of the stocks as compare to CAPM if α is close to 1.

3.4.3.2 Posterior Odds Ratio

A standard assumption in theoretical and empirical research in finance is that relevant variables (e.g stock returns) have multivariate normal distributions (Richardson and Smith, 1993). Given the assumption that the residuals of the cross-sectional regression of the CAPM and the APT satisfy the IID (Independently and identically distribution) multivariate normal assumption (Campbell, Lo and MacKinlay, 1997), it is possible to calculate the posterior odds ratio between the two models. In general the posterior odds ratio is a more formal technique as compare to DM equation and has sounder theoretical grounds (Aggelidis and Maditinos, 2006).

The second comparison is done using posterior odd ratio. The formula for posterior odds is given by Zellner (1979) in favor of model 0 over model 1.

The formula has the following form;

$$R = [ESS_0/ESS_1]^{N/2} N^{K_0 - K_1/2} \quad (3.6)$$

Where ESS_0 is error sum of squares of APT, ESS_1 is error sum of squares of CAPM, N is number of observations, K_0 is number of independent variables of the APT and K_1 is number of independent variables of the CAPM. As according to the ratio when;

$R > 1$ means CAPM is more strongly supported by data under consideration than APT.

$R < 1$ means APT is more strongly supported by data under consideration than CAPM.

IV. RESULTS AND DISCUSSION

4.1 Results of Descriptive Statics of Study Variables

Table 4.1: Descriptive Statics

Variable	Minimum	Maximum	Mean	Std. Deviation	Jarque-Bera test	Sig
KSE-100 Index	-0.11	0.14	0.020	0.047	5.558	0.062
Inflation	-0.01	0.02	0.007	0.008	1.345	0.510
Exchange rate	-0.07	0.04	0.003	0.013	1.517	0.467
Oil Prices	-0.24	0.11	0.041	0.060	2.474	0.290
Interest rate	-0.13	0.05	0.047	0.029	1.745	0.418

Table 4.1 displayed mean, standard deviation, maximum minimum and jarque-bera test and its p value of the macroeconomic variables of the study. The descriptive statistics indicated that the mean values of variables (index, INF, EX, OilP and INT) were 0.020, 0.007, 0.003, 0.041 and 0.047 respectively. The maximum values of the variables between the study periods were 0.14, 0.02, 0.04, 0.41, 0.11 and 0.05 for the KSE- 100 Index, inflation, exchange rate, oil prices and interest rate.

The standard deviations for each variable indicated that data were widely spread around their respective means.

Column 6 in table 4.1 shows jarque bera test which is used to check the normality of data. The hypotheses of the normal distribution are given;

H_0 : The data is normally distributed.

H_1 : The data is not normally distributed.

Table 4.1 shows that at 5 % level of confidence, the null hypothesis of normality cannot be rejected. KSE-100 index and macroeconomic variables inflation, exchange rate, oil prices and interest rate are normally distributed.

The descriptive statistics from Table 4.1 showed that the values were normally distributed about their mean and variance. This indicated that aggregate stock prices on the KSE and the macroeconomic factors, inflation rate, oil prices, exchange rate, and interest rate are all not too much sensitive to periodic changes and speculation. To interpret, this study found that an individual investor could not earn higher rate of profit from the KSE. Additionally, individual investors and corporations could not earn higher profits and interest rates from the economy and foreign companies could not earn considerably higher returns in terms of exchange rate. The investor could only earn a normal profit from KSE.

Table 1 Table Type Styles

Table Head	TableColumnHead		
	Tablecolumnsubhead	Subhead	Subhead
copy	Moretablecopy ^a		

II. ACKNOWLEDGMENT

The preferred spelling of the word “acknowledgment” in American is without an “e” after the “g”. Avoid the stilted expression, “One of us (R.B.G.) thanks...”

Instead, try “R.B.G. thanks”. Put applicable sponsor acknowledgments here; DONOT place them on the first page of your paper or as a footnote.

REFERENCES

- [1] Ali, A. 2001. Macroeconomic variables as common pervasive risk factors and the empirical content of the Arbitrage Pricing Theory. *Journal of Empirical finance*, 5(3): 221–240.
- [2] Basu, S. 1997. The Investment Performance of Common Stocks in Relation to their Price to Earnings Ratio: A Test of the Efficient Markets Hypothesis. *Journal of Finance*, 33(3): 663-682.
- [3] Bhatti, U. and Hanif. M. 2010. Validity of Capital Assets Pricing Model. Evidence from KSE-Pakistan. *European Journal of Economics, Finance and Administrative Science*, 3 (20).



INTERNATIONAL JOURNAL OF VERSATILE RESEARCH AND ANALYSIS

IJVRA.ORG | ISSN: 2984-8903

An International Open Access, Peer - Reviewed, Refereed Journal

Ref No : IJVRA / Vol 4 / Issue 4 / 613

**To,
Savarala Thanmayee**

Subject: Publication of paper in International Journal of Versatile Research and Analysis (Transparent Peer Review Based on UGC CARE Approved Journal Publication and Global Indexing.)

Dear Author,

With Greetings we are informing you that your paper has been successfully published in the International Journal of Versatile Research and Analysis (ISSN: 2984-8903). Following are the details regarding the published paper.

About IJVRA : ISSN Approved - International Scholarly open access, Peer-reviewed, and Refereed Journal, Impact Factor: 9.12, (Calculate by google scholar and Semantic Scholar | AI-Powered Research Tool), Multidisciplinary, Monthly, Online, Print Journal, Indexing in all major database & Metadata, Citation Generator, Digital Object Identifier(DOI)

Registration ID : IJVRA_704074

Paper ID : IJVRA2604613

Title of Paper : Dual-Stage ANN-RNN Hybrid Model for Automated Diabetic Retinopathy Detection

Impact Factor : 9.12 (Calculate by Google Scholar) | License by Creative Common 3.0

DOI :

Published in : Volume 4 | Issue 4 | April-2026

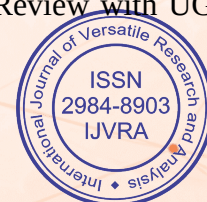
Page No : 608-612

Published URL : <https://ijpub.org/ijvra/viewpaperforall.php?paper=IJVRA2604613>

Authors : Savarala Thanmayee, P Keerthi, K Yohitha, Samudrala Kokila, Ms.M Deepika, Prof.M.E.Palanivel

Thank you very much for publishing your article in IJVRA Transparent Peer Review with UGC CARE Approved Journal Publication and global indexing systems.

Editor In Chief
International Journal of Versatile Research and Analysis
(ISSN: 2984-8903)



An International Scholarly, Open Access, Multi-disciplinary, Monthly, Indexing in all Major Database
Website: www.ijvra.org | Email ID: editor@ijvra.org



International Journal of Versatile Research and Analysis

IJVRA.ORG | ISSN: 2984-8903

An International Open Access, Peer - Reviewed, Refereed Journal

The Board of

International Journal of Versatile Research and Analysis

Is hereby awarding this certificate to

Savarala Thanmayee

In recognition of the publication of the paper entitled

Dual-Stage ANN-RNN Hybrid Model for Automated Diabetic Retinopathy Detection

Published in Volume 4 Issue 4, April-2026, 14-04-2026, | Impact Factor: 9.12 by Google Scholar

Co-Authors - P Keerthi, K Yohitha, Samudrala Kokila, Ms.M Deepika, Prof.M.E.Palanivel

Paper ID - IJVRA2604613



Registration ID - 704074

Editor-In Chief

Transparent Peer Review Based on UGC CARE Approved Journal Publication and Global Indexing

An International Scholarly, Open Access, Multi-disciplinary, Monthly, Indexing in all Major Database & Metadata, Citation Generator

IJVRA - International Journal of Versatile Research and Analysis

An International Scholarly, Open Access, Multi-disciplinary, Indexed Journal

Website: www.ijvra.org | Email: editor@ijvra.org | ESTD: 2023

Certificate of Publication

IJVRA | ISSN : 2984-8903



International Journal of Versatile Research and Analysis

IJVRA.ORG | ISSN: 2984-8903

An International Open Access, Peer - Reviewed, Refereed Journal

The Board of

International Journal of Versatile Research and Analysis

Is hereby awarding this certificate to

P Keerthi

In recognition of the publication of the paper entitled

Dual-Stage ANN-RNN Hybrid Model for Automated Diabetic Retinopathy Detection

Published in Volume 4 Issue 4, April-2026, 14-04-2026, | Impact Factor: 9.12 by Google Scholar

Co-Authors - Savarala Thanmayee, K Yohitha, Samudrala Kokila, Ms.M Deepika, Prof.M.E.Palanivel

Paper ID - IJVRA2604613



Registration ID - 704074

Editor-In Chief

Transparent Peer Review Based on UGC CARE Approved Journal Publication and Global Indexing

An International Scholarly, Open Access, Multi-disciplinary, Monthly, Indexing in all Major Database & Metadata, Citation Generator

IJVRA - International Journal of Versatile Research and Analysis

An International Scholarly, Open Access, Multi-disciplinary, Indexed Journal

Website: www.ijvra.org | Email: editor@ijvra.org | ESTD: 2023

Certificate of Publication

IJVRA | ISSN : 2984-8903



International Journal of Versatile Research and Analysis

IJVRA.ORG | ISSN: 2984-8903

An International Open Access, Peer - Reviewed, Refereed Journal

The Board of

International Journal of Versatile Research and Analysis

Is hereby awarding this certificate to

K Yohitha

In recognition of the publication of the paper entitled

Dual-Stage ANN-RNN Hybrid Model for Automated Diabetic Retinopathy Detection

Published in Volume 4 Issue 4, April-2026, 14-04-2026, | Impact Factor: 9.12 by Google Scholar

Co-Authors - Savarala Thanmayee, P Keerthi, Samudrala Kokila, Ms.M Deepika, Prof.M.E.Palanivel

Paper ID - IJVRA2604613



Registration ID - 704074

Editor-In Chief

Transparent Peer Review Based on UGC CARE Approved Journal Publication and Global Indexing

An International Scholarly, Open Access, Multi-disciplinary, Monthly, Indexing in all Major Database & Metadata, Citation Generator

IJVRA - International Journal of Versatile Research and Analysis

An International Scholarly, Open Access, Multi-disciplinary, Indexed Journal

Website: www.ijvra.org | Email: editor@ijvra.org | ESTD: 2023

Certificate of Publication

IJVRA | ISSN : 2984-8903



International Journal of Versatile Research and Analysis

IJVRA.ORG | ISSN: 2984-8903

An International Open Access, Peer - Reviewed, Refereed Journal

The Board of

International Journal of Versatile Research and Analysis

Is hereby awarding this certificate to

Samudrala Kokila

In recognition of the publication of the paper entitled

Dual-Stage ANN-RNN Hybrid Model for Automated Diabetic Retinopathy Detection

Published in Volume 4 Issue 4, April-2026, 14-04-2026, | Impact Factor: 9.12 by Google Scholar

Co-Authors - Savarala Thanmayee, P Keerthi, K Yohitha, Ms.M Deepika, Prof.M.E.Palanivel

Paper ID - IJVRA2604613



Registration ID - 704074

Editor-In Chief

Transparent Peer Review Based on UGC CARE Approved Journal Publication and Global Indexing

An International Scholarly, Open Access, Multi-disciplinary, Monthly, Indexing in all Major Database & Metadata, Citation Generator

IJVRA - International Journal of Versatile Research and Analysis

An International Scholarly, Open Access, Multi-disciplinary, Indexed Journal

Website: www.ijvra.org | Email: editor@ijvra.org | ESTD: 2023

Certificate of Publication

IJVRA | ISSN : 2984-8903



International Journal of Versatile Research and Analysis

IJVRA.ORG | ISSN: 2984-8903

An International Open Access, Peer - Reviewed, Refereed Journal

The Board of

International Journal of Versatile Research and Analysis

Is hereby awarding this certificate to

Ms.M Deepika,

In recognition of the publication of the paper entitled

**Dual-Stage ANN-RNN Hybrid Model for Automated Diabetic Retinopathy
Detection**

Published in Volume 4 Issue 4, April-2026, 14-04-2026, | Impact Factor: 9.12 by Google Scholar

Co-Authors - Savarala Thanmayee, P Keerthi, K Yohitha, Samudrala Kokila

Paper ID - IJVRA2604613



Registration ID - 704074

Editor-In Chief

Transparent Peer Review Based on UGC CARE Approved Journal Publication and Global Indexing

An International Scholarly, Open Access, Multi-disciplinary, Monthly, Indexing in all Major Database & Metadata, Citation Generator

IJVRA - International Journal of Versatile Research and Analysis

An International Scholarly, Open Access, Multi-disciplinary, Indexed Journal

Website: www.ijvra.org | Email: editor@ijvra.org | ESTD: 2023

Certificate of Publication

IJVRA | ISSN : 2984-8903



International Journal of Versatile Research and Analysis

IJVRA.ORG | ISSN: 2984-8903

An International Open Access, Peer - Reviewed, Refereed Journal

The Board of

International Journal of Versatile Research and Analysis

Is hereby awarding this certificate to

Prof.M.E.Palanivel

In recognition of the publication of the paper entitled

Dual-Stage ANN-RNN Hybrid Model for Automated Diabetic Retinopathy Detection

Published in Volume 4 Issue 4, April-2026, 14-04-2026, | Impact Factor: 9.12 by Google Scholar

Co-Authors - Savarala Thanmayee, P Keerthi, K Yohitha, Samudrala Kokila

Paper ID - IJVRA2604613



Registration ID - 704074

Editor-In Chief

Transparent Peer Review Based on UGC CARE Approved Journal Publication and Global Indexing

An International Scholarly, Open Access, Multi-disciplinary, Monthly, Indexing in all Major Database & Metadata, Citation Generator

IJVRA - International Journal of Versatile Research and Analysis

An International Scholarly, Open Access, Multi-disciplinary, Indexed Journal

Website: www.ijvra.org | Email: editor@ijvra.org | ESTD: 2023

Certificate of Publication

IJVRA | ISSN : 2984-8903



**SREENIVASA INSTITUTE OF TECHNOLOGY AND MANAGEMENT STUDIES
(AUTONOMOUS)**

Accredited by NBA (EEE and MCA)
MURUKAMBATTU, CHITTOOR - 517127, A.P.



5th INTERNATIONAL CONFERENCE ON
ADVANCES AND INNOVATION IN ENGINEERING AND SCIENCES
2nd APRIL - 2026

**CERTIFICATE
OF PARTICIPATION**

This certificate is proudly presented to

Mr / Ms. SAVARALA THANMAYEE

has participated and presented a paper titled DUAL-STAGE ANN-RNN HYBRID MODEL
FOR AUTOMATED DIABETIC RETINOPATHY DETECTION
in the 5th INTERNATIONAL CONFERENCE ON ADVANCES AND INNOVATION IN
ENGINEERING AND SCIENCES (ICAIES-2K26) held on 2nd APRIL 2026.



M.S.
CONVENOR

Neel
PRINCIPAL



**SREENIVASA INSTITUTE OF TECHNOLOGY AND MANAGEMENT STUDIES
(AUTONOMOUS)**

Accredited by NBA (EEE and MCA)
MURUKAMBATTU, CHITTOOR - 517127, A.P.



5th INTERNATIONAL CONFERENCE ON
ADVANCES AND INNOVATION IN ENGINEERING AND SCIENCES
2nd APRIL - 2026

**CERTIFICATE
OF PARTICIPATION**

This certificate is proudly presented to

Mr / Ms. P. KEERTHI
has participated and presented a paper titled DUAL-STAGE ANN-RNN HYBRID MODEL
FOR AUTOMATED DIABETIC RETINOPATHY DETECTION
in the 5th INTERNATIONAL CONFERENCE ON ADVANCES AND INNOVATION IN
ENGINEERING AND SCIENCES (ICAIES-2K26) held on 2nd APRIL 2026.



M.S.N

CONVENOR

N. S. S.

PRINCIPAL



**SREENIVASA INSTITUTE OF TECHNOLOGY AND MANAGEMENT STUDIES
(AUTONOMOUS)**

Accredited by NBA (EEE and MCA)
MURUKAMBATTU, CHITTOOR - 517127, A.P.



5th INTERNATIONAL CONFERENCE ON
ADVANCES AND INNOVATION IN ENGINEERING AND SCIENCES
2nd APRIL - 2026

**CERTIFICATE
OF PARTICIPATION**

This certificate is proudly presented to

Mr / Ms.

K. VOHITHA

has participated and presented a paper titled DUAL-STAGE ANN-RNN HYBRID MODEL
FOR AUTOMATED DIABETIC RETINOPATHY DETECTION
in the 5th INTERNATIONAL CONFERENCE ON ADVANCES AND INNOVATION IN
ENGINEERING AND SCIENCES (ICAIES-2K26) held on 2nd APRIL 2026.



MSI
CONVENOR

N. Sankar
PRINCIPAL



**SREENIVASA INSTITUTE OF TECHNOLOGY AND MANAGEMENT STUDIES
(AUTONOMOUS)**

Accredited by NBA (EEE and MCA)
MURUKAMBATTU, CHITTOOR - 517127, A.P.



5th INTERNATIONAL CONFERENCE ON
ADVANCES AND INNOVATION IN ENGINEERING AND SCIENCES
2nd APRIL - 2026

**CERTIFICATE
OF PARTICIPATION**

This certificate is proudly presented to

Mr. / Ms. SAMUDRALA KOKILA
has participated and presented a paper titled DUAL-STAGE ANN-RNN HYBRID MODEL
FOR AUTOMATED DIABETIC RETINOPATHY DETECTION
in the 5th INTERNATIONAL CONFERENCE ON ADVANCES AND INNOVATION IN
ENGINEERING AND SCIENCES (ICAIES-2K26) held on 2nd APRIL 2026.



M.S.
CONVENOR

[Signature]
PRINCIPAL